TABLES & HIGH-YIELD

Environment and Ecology

Teaching tables, Confused Pairs, Common Traps, Current Affairs, and other high-yield blocks.

UPPCS + UKPCS Study Library · chapter order follows Subject Notes.

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Topic 1 — Environment Basics

Current Affairs

Year	Fact	Why asked
2023	**Amrit Dharohar Scheme** — wetlands conservation; launched on WED 2023	Wetlands + governance
2023	**National Green Hydrogen Mission**	SD / climate policy overlap
2019	**NCAP** (राष्ट्रीय स्वच्छ वायु कार्यक्रम) — 132 cities; 20–30% PM2.5/PM10 reduction target by 2024	Urban (नगरीय) environment governance

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Environment	**Ecology**	Total biotic and abiotic surroundings influencing an organism vs the scientific *study* of interactions between organisms and environment	पर्यावरण / पारिस्थितिकी
Ecology	**Ecosystem**	Scientific discipline/field of study (Ernst Haeckel 1866) vs functional, structural ecological *unit* of nature (A.G. Tansley 1935)	पारिस्थितिकी / पारिस्थितिकी तंत्र
Habitat	**Ecological Niche**	Physical address / geographical place where an organism lives vs functional profession / ecological role of a species in an ecosystem	वास स्थान / पारिस्थितिक निकेत (Niche)
Biotic Components	**Abiotic Components**	Living organisms (producers, consumers, decomposers) vs non-living physical and chemical factors (light, temperature, soil, water)	जैविक घटक / अजैविक घटक
Lentic Ecosystem	**Lotic Ecosystem**	Standing or stagnant fresh water bodies (ponds, lakes, bogs, swamps) vs running or flowing fresh water bodies (rivers, streams, brooks)	स्थिर जल (लैंटिक) / प्रवाही जल (लोटिक)
Liebig's Law of the Minimum	**Shelford's Law of Tolerance**	Population growth is controlled by the scarcest limiting resource vs species survival requires environmental factors to remain between minimum and maximum tolerance limits	लाइबिग का न्यूनतम नियम / शेलफोर्ड का सहनशीलता नियम
Internal Environment	**External Environment**	Biochemical and physiological conditions within an organism (homeostasis) vs physical surroundings external to the organism	आंतरिक पर्यावरण / बाह्य पर्यावरण
Physical Environment	**Biological Environment**	Inanimate non-living components (lithosphere, hydrosphere, atmosphere) vs living communities of flora, fauna, and microbes	भौतिक पर्यावरण / जैविक पर्यावरण

1.1 Environment – Meaning

Source	Definition
NCERT (Class 12)	Sum of all **biotic + abiotic** factors influencing living organisms ← **most asked**
UNEP	Physical, chemical, and biotic factors acting on organism/community, determining form and survival
EPA 1986 (पर्यावरण संरक्षण 1986) (Legal)	Water, air, land + inter-relationship among them and with humans, other living beings, plants, and property
General	External conditions — physical, chemical, biological, social, economic, cultural — affecting life

1.2 Environmental Components

Abiotic factor category	Includes	Hindi
Edaphic	Soil texture, minerals, pH, moisture	मृदा संबंधी
Climatic	Temperature, rainfall, wind, humidity	जलवायु संबंधी
Topographic	Altitude, slope, aspect	स्थलाकृति संबंधी
Biotic group	Also called	Function
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Producers	Autotrophs	Convert solar energy → chemical energy
Consumers	Heterotrophs	Feed on other organisms
Decomposers	Saprotrophs	Break down dead matter; recycle nutrients
Detritivores	—	Feed on detritus (earthworms, termites)

1.3 Classification of Environment

Basis	Categories
Standard triad	**Physical** / **Biological** / **Cultural** (**Operational** is **not** a category)
Origin	Natural vs **Anthropogenic** (human-modified — cities, dams, cropland)
Components	Abiotic vs Biotic
Domain	**Terrestrial** (~29% surface) vs **Aquatic** (~71%)
Organism level	**Internal** (inside body — physiology) vs **External** (outside — ecology default)
Scale	Micro / Meso / Macro
Human use	Rural / Urban / Industrial

Physical, biological, and cultural environment

Aquatic type	Description	Indian example
Marine	Oceans/seas — high salinity	Arabian Sea, Bay of Bengal
Lentic	**Standing** freshwater	Loktak Lake (लोकटक), Dal Lake
Lotic	**Flowing** freshwater	Ganga, Yamuna
Estuarine	River meets sea — brackish	Sundarbans (सुंदरबन) delta (डेल्टा)
Wetland	Transitional — neither fully land nor water	Chilika, Keoladeo (केवलादेव)
Marine zone	Location	Key fact
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Neritic	Shallow shelf (0–200 m)	Fisheries, coral, mangroves
Pelagic	Open water column	Plankton, tuna
Benthic	Sea floor	Crabs, decomposers
Littoral	Intertidal shore	Barnacles, molluscs

India's 10 biogeographic zones

#	Zone	feature
1	**Trans-Himalaya**	Cold desert — Ladakh (लद्दाख), high-altitude sparse vegetation
2	**Himalaya** (हिमालय)	Alpine to subtropical belts along the main Himalayan range

#	Zone	feature
3	**Desert**	Thar (थार) — Rajasthan and adjoining Gujarat
4	**Semi-Arid**	Dry scrub/savanna (सवाना) fringe of Gujarat–Maharashtra–central India
5	**Western Ghats** (पश्चिमी घाट)	Biodiversity hotspot (जैव विविधता हॉटस्पॉट); high endemism
6	**Deccan (दक्कन) Peninsula**	Plateau forests and dry deciduous tracts
7	**Gangetic Plain**	Fertile alluvial (जलोढ़) plains — covers **most of UP**
8	**Coasts**	Mangroves, estuaries, coral reefs
9	**North-East India**	Indo-Malayan biodiversity corridor
10	**Islands**	Andaman (अंडमान) & Nicobar (निकोबार) and Lakshadweep (लक्षद्वीप)

1.4 Human–Environment Interaction

Model	Proponent	Idea	Status
Environmental Determinism	Ratzel	Environment **dictates** culture/society	Largely **rejected**
Possibilism	Vidal de la Blache	Environment offers **possibilities**; humans choose	**Accepted**
Neo-determinism	Griffith Taylor	Humans modify environment **within limits**	Balanced view
Concept	Definition / fact		
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Carrying capacity	Maximum load an environment can sustain indefinitely. Resource extraction must stay **below** regeneration. Waste must stay **within** absorption capacity		
IPAT	Impact = **P**opulation × **A**ffluence × **T**echnology		
Ecological footprint (पारिस्थितिक पदचिह्न)	Land/water area needed to support consumption — measured in **global hectares (gha)**		
Tragedy of the Commons	Shared resources overexploited when individual interest > collective good — **Garrett Hardin, 1968**		
Sustainable Development	"Meet present needs without compromising future generations' ability to meet theirs" — **Brundtland Report, 1987** (*Our Common Future*)		

Landmark Environmental Classics & Pioneer Thinkers

Thinker / Landmark	Year / Publication	Core Concept & Exam Significance
Rachel Carson	**1962** (*Silent Spring*)	Exposed environmental hazards of indiscriminate synthetic pesticides (**DDT** bioaccumulation and eggshell thinning); catalytic milestone that birthed the modern global environmental movement.
John W. Bennett	**1976** (*The Ecological Transition*)	Coined and developed the concept of **'Ecological Transition'**, analyzing human society's cultural shift from balanced ecological adaptation to intensive environmental exploitation.
Garrett Hardin	**1968** (*The Tragedy of the Commons*)	Postulated in *Science* that independent rational actors deplete shared public resources (pastures, fisheries, clean air) through unconstrained self-interest.
Club of Rome	**1972** (*The Limits to Growth*)	Commissioned MIT computer-simulation model (Meadows et al.) showing ecological collapse if exponential economic and population growth outpace planetary carrying capacity. *(Trap: Brundtland Report 1987 is Our Common Future, NOT Limits to Growth).*

Thinker / Landmark	Year / Publication	Core Concept & Exam Significance
Ernst Haeckel	**1866** (*Generelle Morphologie*)	Coined the term **'Ecology'** (*Oekologie*) for the study of organism-environment interactions.
A.G. Tansley	**1935**	Coined the term **'Ecosystem'** as the fundamental functional unit of nature.
Arne Naess	**1973**	Coined **'Deep Ecology'** (advocating inherent biocentric rights of all living species) versus utilitarian 'Shallow Ecology'.
Joseph Grinnell	**1917**	Coined and formalized the concept of **'Ecological Niche'** as the functional role and distributional unit of a species.
Mathis Wackernagel & William Rees	**1996**	Formulated the concept of **'Ecological Footprint'**, measuring human resource appropriation in global hectares (gha).

Common Traps — Don't Fall For These

Trap	Correct fact
Art **48** vs **48A**	Environment = **48A** (48 = agriculture/cows)
Humus = abiotic?	**No** — organic, biotic-origin
Mangrove = terrestrial?	**No** — coastal/transitional
MoEFCC est. 1972?	**No** — **1985** (1972 = Stockholm)
Lentic = flowing?	**No** — Lentic = **standing**
Wildlife Day = 22 March?	**No** — **3 March**
Agenda 21 = 1995?	**No** — **Rio 1992**
Brundtland = *Limits to Growth*?	**No** — Brundtland = *Our Common Future* (1987)

Topic 2 — Ecology & Ecosystem

Current Affairs (this topic)

- Ecosystem-services and productivity stems stay conceptual — refresh MEA / IPBES wording if a CA stem cites (साइट्स) them.
- Keep Tansley / Haeckel / Odum name traps stable; they are not CA-dependent.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Ecology (Haeckel 1866)	**Ecosystem (Tansley 1935)**	Scientific academic study of organism-environment interactions vs concrete structural-functional unit of nature combining biotic and abiotic components	पारिस्थितिकी / पारिस्थितिक तंत्र
Habitat	**Ecological Niche**	Physical place or environmental zone where a species resides vs unique functional role, trophic position, and resource utilization pattern (Grinnell)	वास स्थान / पारिस्थितिक निकेत
Population	**Ecological Community**	Group of interbreeding individuals of the *same* species in an area vs assemblage of populations of *different* interacting species in an area	जनसंख्या (समष्टि) / जैविक समुदाय
Gross Primary Productivity (GPP)	**Net Primary Productivity (NPP)**	Total rate of organic matter synthesized by producers via photosynthesis vs actual biomass available to consumers after plant respiration losses (NPP = GPP - R)	सकल प्राथमिक उत्पादकता (GPP) / शुद्ध प्राथमिक उत्पादकता (NPP)
Fundamental Niche	**Realized Niche**	Full theoretical potential range of environmental conditions and resources a species can use without competition vs actual restricted niche occupied due to biotic competition	मूल (मौलिक) निकेत / वास्तविक निकेत
Primary Succession	**Secondary Succession**	Ecological succession beginning on barren, lifeless substrate with no pre-existing soil (cooled lava, bare rock) vs succession on disturbed areas where soil already exists (after forest fire, flood)	प्राथमिक अनुक्रमण / द्वितीयक अनुक्रमण
r-Selected Species	**K-Selected Species**	High reproductive rate, small body, short lifespan, little parental care (insects, rodents, weeds) vs low reproductive rate, large body, long lifespan, high parental care (elephants, whales, humans)	r-रणनीतिकारी / K-रणनीतिकारी प्रजाति
Ecotone	**Ecotype**	Sharp transition boundary zone between two distinct ecological communities (mangroves, forest-grassland edge) vs genetically distinct geographic variety or sub-population within a species	इकोटोन (संक्रमण क्षेत्र) / इकोटाइप
Sere	**Climax Community**	Entire chronological transitional sequence of developmental communities during ecological succession vs final, stable, self-perpetuating terminal community in equilibrium with regional climate	क्रमक (Sere) / चरम समुदाय
Bionomics	**Ecology**	Historical 19th-century synonym of ecology meaning “laws of life” (UPPCS trap: bionomics is NOT management of life) vs modern ecology	बायोनॉमिक्स / पारिस्थितिकी

2.1 Ecology – Meaning

Level	example
Organism	Camel water conservation in desert
Population	Tiger population in Dudhwa (दुधवा)
Community	Fish, birds, and plants sharing a wetland
Ecosystem	Ganga stretch at Varanasi (वाराणसी) with its biota, water, and nutrient flows
Biome	Tropical deciduous forest of India
Biosphere	Global life-support zone

2.3 Types of Ecosystems

Basis	Categories
Origin	Natural vs **Anthropogenic** (human-modified)
Medium	**Terrestrial** (~29% Earth's surface) vs **Aquatic** (~71%)
Water flow (freshwater)	**Lentic** (standing) vs **Lotic** (flowing)
Salinity	Freshwater vs Brackish vs Marine

Pond and lake zonation (lentic)

Zone	Location	Key fact
Littoral	Shallow shore with rooted plants	Warmest, highest light
Limnetic	Open water away from shore	Phytoplankton zone
Profundal	Deep water below light penetration	Low light, decomposers dominate

2.4 Structure, Functions & Components

Four functions of an ecosystem

Function	What it means
Productivity	Rate at which biomass or energy is produced
Decomposition	Breakdown of dead organic matter
Nutrient cycling	Movement of chemical elements through biotic and abiotic pools
Energy flow	One-way transfer of energy through trophic levels

2.8 Ecological Succession

Primary vs secondary succession

- **Primary succession** starts on bare lifeless substrate where no soil yet exists. Examples include cooled lava, bare rock, and new sand dunes.
- **Secondary succession** starts where soil and some organisms already remain after a disturbance. Examples include abandoned farmland, forest after fire, and flooded land once water recedes.
- **Pioneer species** are the first colonisers. Lichens on rock and moss on thin soil are classic terrestrial pioneers. Phytoplankton may pioneer a new pond.

Phases of biotic succession (full sequence)

Phase	What happens
Nudation	Creation of a bare site free of organisms (erosion, fire, glaciation, human clearance)
Migration	Seeds, spores, or other propagules reach the bare or disturbed area
Ecesis	The newcomers successfully establish, germinate, and grow (colonisation)
Reaction	Established species modify the habitat (soil, shade, moisture) for the next community
Stabilisation	Change slows and the climax community dominates

2.9 Ecosystem Services

Category	What humans get	Example
Provisioning	Direct material products	Food, water, timber
Regulating	Processes that stabilise the environment	Climate/disease control
Supporting	Services that underpin all other services	Nutrient cycling, soil formation
Cultural	Non-material benefits	Recreation, spiritual value
Preserving (WDR/UPSC five-fold framing)	Long-term diversity maintenance	Maintenance of diversity

2.10 Ecosystem Productivity

Term	Meaning
GPP (Gross Primary Productivity)	Total energy or biomass fixed by producers through photosynthesis
R	Energy lost by producers through respiration
NPP (Net Primary Productivity)	**GPP − R** = biomass actually available to consumers
NSP	NPP remaining after heterotroph consumption at producer level
Standing crop	Biomass present in an ecosystem at a given instant

2.12 Genetic, Species & Community Ecology

Community level ecology

Interaction	What happens	Example
Competition	Two species vie for the same limited resource	Two grazers on one pasture
Predation	One organism kills and eats another	Tiger hunting deer
Parasitism	One benefits while the host is harmed	Tick on cattle
Mutualism	Both species benefit	**Azolla** with *Anabaena* in rice fields
Commensalism	One benefits and the other is unaffected	Cattle egret feeding near grazing cattle

Common Traps — Don’t Fall For These

Trap	Correct fact
Odum coined “ecosystem”	**Tansley 1935** coined ecosystem
Ecosystem is a closed system	Ecosystem is normally **open**

Trap	Correct fact
Ecesis before Migration	Full: **Nudation → Migration → Ecesis → Reaction → Stabilisation**
Sere = nutrient cycle	**Sere** = succession community **sequence**
Lichens thrive in polluted air	Lichens are **sensitive** indicators
1% and 10% cannot both be true	**Both true** in different stages
Energy increases up food chain	Energy **decreases** (~10% transfer)
Coral reefs = most stable ecosystem	Coral reefs = high **diversity**; marine = high **stability**
Primary succession has soil already	Primary = **no soil**; secondary = soil present
Bionomics = management of life	Bionomics = **ecology**
Niche coined by Odum	Niche concept linked to **Grinnell (1917)**
Ecological Transition = Ratzel	**John W. Bennett**
ESZ under WPA 1972 / total human ban	ESZ under **EPA 1986**; activities regulated, not all banned
Eucalyptus is eco-friendly	Often treated as **groundwater hazard**
Productivity: oceans first	Decreasing order **Mangroves > grasslands > lakes > oceans**

Topic 3 — Food Chain & Energy Flow

Current Affairs (this topic)

- Energy-flow and trophic-level traps recycle yearly — 10% rule and pyramid types stay core.
- Bioaccumulation (जैव संचय) / biomagnification (जैव आवर्धन) CA usually rides on pesticide or heavy-metal news; keep the mechanism pair clear.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Food Chain	**Food Web**	Single linear sequence of organisms where nutrients and energy transfer through feeding vs complex network of interconnected, interlocking food chains providing alternative feeding pathways	खाद्य श्रृंखला / खाद्य जाल
Grazing Food Chain (GFC)	**Detritus Food Chain (DFC)**	Begins with living green photosynthetic plants (producers) and drives terrestrial energy vs begins with dead organic matter (detritus) decomposed by fungi and bacteria	चराई खाद्य श्रृंखला / अपरद (अवशेष) खाद्य श्रृंखला
Upright Pyramid of Biomass	**Inverted Pyramid of Biomass**	Characteristic of terrestrial grasslands and forests where producer biomass far exceeds herbivores vs characteristic of aquatic/pond ecosystems where standing crop of phytoplankton is smaller than zooplankton/fish	सीधा जैवभार पिरामिड (स्थलीय) / उल्टा जैवभार पिरामिड (जलीय/तालाब)
10% Energy Transfer Law	**Bioaccumulation**	Raymond Lindeman (1942) principle that only ~10% of energy transfers from one trophic level to the next vs progressive buildup of non-biodegradable chemicals within an individual organism over time	10% ऊर्जा नियम (लिंडमैन) / जैव संचय (बायोएक्यूमुलेशन)
Bioaccumulation	**Biomagnification**	Increase in toxicant concentration within a single organism's tissues through direct absorption from water/food vs increase in toxic chemical concentration (e.g. DDT, mercury) across successive trophic levels of a food chain	जैव संचय (एक जीव में) / जैव आवर्धन (खाद्य श्रृंखला में)
Primary Consumer	**Secondary Consumer**	Herbivores feeding directly on autotrophic green plants (cow, deer, zooplankton) vs primary carnivores that hunt and feed on herbivores (frog, fox, small fish)	प्राथमिक उपभोक्ता (शाकाहारी) / द्वितीयक उपभोक्ता (मांसाहारी)
Pyramid of Energy	**Pyramid of Numbers**	Always universally upright in all ecosystems because energy is irretrievably lost as metabolic heat at each level vs can be upright (grassland) or inverted (single large tree supporting thousands of birds/parasites)	ऊर्जा पिरामिड (सदैव सीधा) / संख्या पिरामिड

3.1 Food Chain

Type	Starts from	UP / textbook example
Grazing	Living producers	Grass → deer → tiger (Dudhwa); Grass → grasshopper → frog → snake
Detritus	Dead organic matter	Leaf litter → earthworm → fungi
Marine	Phytoplankton/diatoms	Diatoms → crustaceans → herrings
Parasitic (special)	Living host tissue	Tree → insects → parasites

3.3 Trophic Levels

Bioaccumulation and Biomagnification

Level	Role	UP / India example
T1	Producer	Terai grass, wheat, Ganga algae
T2	Primary consumer (प्राथमिक उपभोक्ता)	Barasingha, chital, rohu (juvenile)
T3	Secondary consumer (द्वितीयक उपभोक्ता)	Frog, catfish
T4	Tertiary consumer	Python, fishing eagle
T5	Top carnivore	Tiger (Dudhwa), Gangetic dolphin (river apex)

3.4 Energy Flow in Ecosystem

Where energy goes at each level	Key fact
Respiration and heat	Largest share — why pyramid of energy (ऊर्जा पिरामिड) is upright
Movement and metabolism	Not passed to next consumer
Egestion and uneaten parts	Lost from food chain pathway
Passed to next trophic level	**~10%** (Lindeman 1942)

3.5 Ecological Pyramid (Ecological Pyramids)

Pyramid	Unit	Always upright?	Inverted example
Energy	kcal/m^2/yr	**Yes — always**	Never inverted
Biomass	g/m^2	Usually	Open ocean (phytoplankton turnover)
Numbers	Individual count	Usually	One tree → many insects/parasites

Common Traps — Don't Fall For These

Trap	Correct fact
Energy flow is cyclic	**Unidirectional** — nutrients cycle, energy does not
All pyramids always upright	Only **energy** pyramid is always upright
Grazing chain starts from dead matter	Starts from **living green plants**
Sun is Trophic Level 1	Sun = energy **source**; **Producer = T1**
10% rule by Darwin	**Lindeman 1942**
Food web makes energy cyclic	Each pathway is **one-way**
Biomass pyramid measures energy flow	Biomass = **g/m^2**; energy = **kcal/m^2/yr**
Open ocean = inverted number pyramid	Open ocean / pond = inverted **biomass** pyramid
Man is only secondary consumer	Man is **primary + secondary** consumer
Virus is a decomposer	Decomposers = **fungi + bacteria**
Food chain within one species	Food chains are **across** feeding links, not within one species
Highest DDT in grasshopper	Highest in top consumer (e.g. **snake**)

Topic 4 — Biodiversity

Current Affairs (this topic)

- Track new Ramsar (रामसर) and IUCN (आईयूसीएन) Red List updates against the Must-Score baselines (India Ramsar **~100**; UP **12**).
- Invasive-species and hotspot stems recycle yearly — keep Western Ghats (पश्चिमी घाट) / Indo-Burma trap pairs stable.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Genetic Diversity	**Species Diversity**	Variation of genes and alleles *within* a single species across its geographic range vs the variety and abundance of *different species* within a given ecological region	आनुवंशिक विविधता / प्रजाति विविधता
Species Diversity	**Ecosystem Diversity**	Number of species and their relative abundance (species richness + evenness) vs variety of distinct habitats, biotic communities, and ecological processes in a landscape	प्रजाति विविधता / पारिस्थितिक तंत्र विविधता
Alpha Diversity	**Beta Diversity**	Species richness measured *within* a particular local habitat or community vs rate of species turnover / change in species composition *between* two different habitats across an environmental gradient	अल्फा विविधता (स्थानीय) / बीटा विविधता (आवासीय परिवर्तन)
In-situ Conservation	**Ex-situ Conservation**	Protecting endangered species within their natural evolutionary habitats (National Parks, Sanctuaries, Biosphere Reserves) vs conserving species outside their natural habitats in artificial environments (Zoological Parks, Botanical Gardens, Gene Banks)	स्व-स्थाने संरक्षण (In-situ) / बाह्य-स्थाने संरक्षण (Ex-situ)
Endangered (EN)	**Critically Endangered (CR)**	Very high risk of extinction in the wild (e.g. Asiatic Lion, Tiger) vs extremely high imminent risk of extinction in the wild (e.g. Great Indian Bustard, Gharial, Hangul)	लुप्तप्राय (EN) / अति संकटग्रस्त (CR)
Endemic Species	**Invasive Alien Species**	Native species strictly restricted to a specific unique geographic territory and found nowhere else (e.g. Lion-tailed Macaque in Western Ghats) vs non-native species introduced from outside that outcompete and displace indigenous wildlife (e.g. Lantana camara, Water Hyacinth)	स्थानिक प्रजाति (Endemic) / आक्रामक विदेशी प्रजाति (Invasive)
Biodiversity Hotspot	**Hope Spot**	Terrestrial regions with exceptionally high endemic species richness under severe threat (≥1,500 endemic vascular plants and ≥70% primary vegetation lost; Norman Myers) vs marine protected areas critical to the health of global oceans (Sylvia Earle / Mission Blue)	जैव विविधता हॉटस्पॉट (स्थलीय) / होप स्पॉट (समुद्री)
People's Biodiversity Register (PBR)	**Red Data Book**	Mandated under Biological Diversity Act 2002 prepared by local Biodiversity Management Committees (BMCs) documenting traditional ecological knowledge vs official compendium published by IUCN documenting global extinction risks	जन जैव विविधता रजिस्टर (PBR) / रेड डेटा बुक (IUCN)

4.3 Global & Indian Biodiversity

Region	Megadiverse countries (17)
South America	Brazil, Colombia, Ecuador, Peru, Venezuela
North / Central America	Mexico, United States
Asia	China, India, Malaysia, Indonesia, Philippines

Region	Megadiverse countries (17)
Africa	Democratic Republic of Congo, South Africa, Madagascar
Oceania	Australia, Papua New Guinea

Group	India's rough share of global species
Mammals	~**7.6%**
Birds	~**12.6%**
Reptiles	~**6.2%**
Amphibians	~**4.4%**
Indian endemism centre	Note
----------	------
Western Ghats (पश्चिमी घाट)	Richest among common Indian options
Northeast India	High endemism belt
Himalaya (हिमालय)	Richness from confluence of biogeographic zones
Andaman & Nicobar Islands	Island endemism

4.5 Biodiversity Assessment & Registers

IUCN Red List categories

Category	Meaning
LC	Least Concern
NT	Near Threatened
VU	Vulnerable
EN	Endangered
CR	Critically Endangered
EW	Extinct in the Wild
EX	Extinct

4.8 Biodiversity Hotspots

Hotspot	Indian region	UP link
Himalaya (often called Eastern Himalaya in older stems)	Himalaya + Terai belt; Sikkim is a botanist's paradise here	Dudhwa, Pilibhit Terai
Indo-Burma / Indo-Myanmar	Northeast India + **Andaman Islands**	—
Western Ghats & Sri Lanka	West coast mountains	—
Sundaland	**Nicobar Islands** (not Andaman)	Not mainland

4.9 Species Categories

Category	One-line fact	India example
Endemic	Found **nowhere else** in the world	Asiatic lion (सिंह) (Gir (गीर)), lion-tailed macaque (Western Ghats)

Category	One-line fact	India example
Endangered	IUCN **EN** — very high extinction risk	Gangetic dolphin, Indian vulture
Threatened	**VU + EN + CR** together	Broader than endangered alone
Extinct	No surviving individuals (**EX**) or wild population (**EW**)	Cheetah in India (1952 wild extinction)
Invasive alien	Non-native + ecological or economic harm	Water hyacinth, lantana, Parthenium
Keystone	Disproportionately large ecosystem impact if removed	Sea otter (kelp forest), tiger (apex regulation)
Flagship	Charismatic species that mobilises public support	Tiger, elephant, one-horned rhino
Umbrella	Protecting its large range saves many other species	Tiger reserve network
Indicator	Reflects environmental quality or change	**Lichens** (air pollution), frog (wetland health)

Common Traps — Don't Fall For These

Trap	Correct fact
Biodiversity = species count only	**Genetic + species + ecosystem**
Wilson coined "biodiversity"	**Walter G. Rosen (1985)** coined; Wilson popularised
Alpha = regional diversity	Alpha = **local** mean; gamma = regional
Russia is megadiverse	**Not** in the standard 17 list
India has 6 hotspots	**4** — Himalaya, Indo-Burma, WG–Sri Lanka, Sundaland
Sundaland = Andaman + Nicobar	**Nicobar** under Sundaland; **Andaman** under Indo-Burma
Eastern Ghats / Vindhyas = hotspot	**Not** hotspots
Hotspots only in tropics	Temperate hotspots also exist
Threatened = Endangered	Threatened = **VU + EN + CR**
Botanical garden = in-situ	**Ex-situ**
Cryo-bank gas = oxygen / CO_2	**Liquid nitrogen**
Lichens thrive in pollution	**Sensitive** indicators — fail in pollution
Ex-situ preferred over in-situ	**In-situ preferred**
NBA headquarters Delhi	**Chennai**
BirdLife invented hotspots	**Norman Myers**; BirdLife = IBAs
TRAFFIC = UNEP bureau	**WWF–IUCN** NGO alliance
Coral reefs in Sundarbans	Sundarbans = **mangroves**; reefs = GoM, GoK, A&N, Lakshadweep
Vegetarianism threatens biodiversity	**Not** a threat
NCMB = Mumbai / Bhavnagar	**Jamnagar**
Silent Valley = Uttarakhand	**Kerala**; Valley of Flowers = Chamoli UK
India Ramsar ≈ 26 forever	About **80**; TN **16**, UP **10**, Punjab **6**
Gangetic Plain is a hotspot	**Not** a global hotspot

Topic 5 — Habitat, Flora & Fauna

Current Affairs (this topic)

- Habitat-loss and corridor news often refresh Terai (तराई) / Himalaya (हिमालय) colour for UP–UK.
- Do not swap endemic vs endangered wording when CA names a new Red List update.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Habitat	**Ecological Niche**	Physical geographic environment and specific spatial location where an organism resides vs functional status, behavioral role, and trophic interactions of that species in the community	आवास / पारिस्थितिक निकेत
Flora	**Fauna**	Total native plant species assemblages inhabiting a particular geographic region or geological era vs total native animal species inhabiting a region or era	वनस्पति (फ्लोरा) / प्राणीजात (फॉना)
Microhabitat	**Ecotone**	Small, localized physical niche within a broader habitat (e.g., under a rotting log, a tree cavity) vs macro-scale transitional boundary ecosystem between two adjacent distinct biomes (e.g., marsh between dry land and water)	सूक्ष्म आवास (माइक्रोहैबिटैट) / इकोटोन (संक्रमणिका)
Keystone Species	**Foundation Species**	Species having a disproportionately massive impact on ecosystem structure relative to its biomass (loss triggers ecological collapse; e.g. Fig tree, Sea Otter) vs dominant primary producer that physically creates and structures the habitat (e.g. Corals, Kelp, Mangroves)	कीस्टोन प्रजाति / आधारभूत (फाउंडेशन) प्रजाति
Umbrella Species	**Flagship Species**	Wide-ranging species whose habitat protection automatically shields hundreds of co-occurring species (e.g. Tiger, Asian Elephant) vs charismatic, culturally iconic species chosen as marketing symbols to rally conservation campaigns (e.g. Giant Panda, Polar Bear)	छत्र (अंब्रेला) प्रजाति / प्रतीक (फ्लैगशिप) प्रजाति
Indicator Species	**Invasive Species**	Species whose presence, absence, or abundance directly signals environmental stress or health (e.g. Lichens for SO_2 air pollution, Mayfly larvae for clean water) vs non-native species proliferating aggressively and threatening native ecology	जैव संकेतक प्रजाति / आक्रामक विदेशी प्रजाति
In-situ Conservation	**Ex-situ Conservation**	On-site conservation in natural ecosystems (Sacred Groves, Wildlife Sanctuaries) vs off-site conservation under artificial human custody (Seed Banks, Tissue Culture Labs)	स्व-स्थाने संरक्षण / बाह्य-स्थाने संरक्षण

5.1 Habitat

Major habitat types (classification)

Type	Meaning	India example
Terrestrial	Land-based	Terai sal forest, Thar (थार) desert
Aquatic freshwater	Ponds, lakes, rivers	Ganga, Loktak
Marine	Oceans and seas	Arabian Sea coral/ fisheries belt
Wetland	Seasonally or permanently waterlogged	Keoladeo (केवलादेव), UP Ramsar (रामसर) sites
Mangrove / coastal	Tidal saline (लवणीय) coast	Sundarbans (सुंदरबन)
Grassland / savanna (सवाना)	Grass-dominated open tract	Dudhwa grasslands
Alpine / montane	High-altitude cold zone	Himalayan snow leopard belt

Type	Meaning	India example
Desert / arid	Low rainfall extreme temperature	Thar
Arboreal	Tree-dwelling	Lion-tailed macaque
Fossorial	Burrowing underground	Indian pangolin
Cursorial	Open plain runners	Blackbuck
Xericole	Desert-adapted	Camel, desert fox
Amphibious	Both land and water phases	Crocodile, bullfrog

5.2 Habitat Conservation

Key schemes and policies

Scheme / policy	Year	Ministry	Fact
MISHTI	Budget **2023–24**	MoEFCC	Mangrove shoreline habitat restoration + community income
NUHHP	**2007**	MoHUA	Urban housing and sustainable urban habitat
JNNURM	**2005**	MoHUA	Urban infrastructure renewal
AMRUT	**2015**	MoHUA	Urban water, sewerage, green spaces
Jal Jeevan Mission	**2019**	Jal Shakti (शक्ति)	Rural tap water connectivity
CAMPA	2009 / Act 2016	MoEFCC	Compensatory afforestation for diverted forest land
Green India Mission	2014	MoEFCC	Forest cover (वन आवरण) and habitat quality improvement under NAPCC

5.3 Flora and Fauna

UP flora and fauna facts

Category	UP example	Note
State animal	**Barasingha** (swamp deer)	Terai grassland specialist
State bird	**Sarus crane**	Wetland and farmland mosaic
Dominant Terai flora	**Sal** (*Shorea robusta*)	Dudhwa forest tree
Aquatic fauna	**Gangetic dolphin**	Ganga–Yamuna stretch
Flagship habitat fauna	Tiger, rhino, elephant	Dudhwa and Pilibhit
Invasive flora	Parthenium, Lantana, water hyacinth	Degrade native habitats

Common Traps — Don't Fall For These

Trap	Correct fact
Habitat = niche	Habitat = **place**; niche = **role**
Mangrove = terrestrial	Mangrove = **transitional**
MISHTI launched 2019	**Budget 2023–24**
BSI surveys animals	BSI = **plants**; ZSI = **animals**
Masai in West Africa	Masai = **East Africa**
Zoos = in-situ	Zoos = **ex-situ**

Trap	Correct fact
NUHHP before JNNURM	JNNURM **2005** → NUHHP **2007**
Mangrove flora in UP Terai	UP has **no coastline**

Topic 6 — Protected Areas & Conservation

Current Affairs (this topic)

- Track new Ramsar notifications and tiger-reserve counts against the Must-Score totals.
- Keep UP Ramsar **12** and India **~100** as the teaching baseline unless a stem cites (साइट्स) a later official total.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
National Park (NP)	**Wildlife Sanctuary (WLS)**	Declared under WPA 1972 Sec 35; highest protection, zero human settlement, grazing banned, boundaries altered only by State Legislature on NBWL advice vs WPA Sec 18; regulated grazing and traditional rights may be permitted by Chief Wildlife Warden	राष्ट्रीय उद्यान (NP) / वन्यजीव अभयारण्य (WLS)
Conservation Reserve	**Community Reserve**	Established on government-owned lands acting as buffer corridors between PAs under WPA Sec 36A vs established on private or community-owned lands with voluntary community participation under WPA Sec 36C	संरक्षण आरक्षित क्षेत्र / सामुदायिक आरक्षित क्षेत्र
Tiger Reserve	**Elephant Reserve**	Statutory protected area notified under Wildlife Protection Act 1972 (amended 2006) with legally delineated Core and Buffer zones supervised by NTCA vs administrative/executive designation under Project Elephant without statutory backing under WPA	बाघ अभयारण्य (सांविधिक) / हाथी अभयारण्य (प्रशासनिक)
Biosphere Reserve (MAB)	**National Park**	Landscape-level conservation framework under UNESCO MAB with 3 distinct concentric zones (Core, Buffer, Transition) integrating human development vs strict statutory preservation excluding human commercial activity	जैवमंडल आरक्षित क्षेत्र / राष्ट्रीय उद्यान
Eco-Sensitive Zone (ESZ)	**Protected Area (PA)**	Buffer zone up to 10 km notified around National Parks and Sanctuaries under Section 3 of Environment (Protection) Act 1986 to regulate commercial activities vs statutory core area notified under Wildlife Protection Act 1972	पर्यावरण-संवेदनशील क्षेत्र (ESZ) / संरक्षित क्षेत्र (PA)
Ramsar Wetland Site	**Natural World Heritage Site**	International designation under 1971 Ramsar Convention specifically targeting ecological wise use of waterfowl/wetland habitats vs UNESCO World Heritage designation recognizing universal natural value under 1972 Convention	रामसर आर्द्रभूमि स्थल / प्राकृतिक विश्व धरोहर स्थल
Core Zone (Biosphere)	**Transition Zone (Biosphere)**	Legally protected untouched sanctum sanctorum where all human activity except non-destructive research is barred vs outermost open area where local sustainable agriculture, settlements, and economic activities are actively fostered	कोर क्षेत्र (पूर्ण सुरक्षित) / संक्रमण क्षेत्र (सहअस्तित्व)
### Protected-area hierarchy — one card			

Category	Legal home	Strictness / use	Classic trap
National Park	WPA Sec **35**	Strictest WPA PA	Boundary change needs **State Legislature**
Wildlife Sanctuary	WPA Sec **18**	Limited use with **CWLW** permission	Not as strict as NP
Tiger Reserve	WPA Sec **38V** + NTCA	Core + buffer mandatory	Statutory — unlike Elephant Reserve
Conservation Reserve	WPA Sec **36A**	Forest buffer near PA	Not private-land Community Reserve

Category	Legal home	Strictness / use	Classic trap
Community Reserve	WPA Sec **36B**	Private/community land + consent	Needs voluntary consent
Biosphere Reserve	UNESCO MAB (executive in India)	Core–buffer–transition	**No** WPA statutory basis
Elephant Reserve	Project Elephant 1992	Corridor / conflict focus	**No** WPA statutory basis
Ramsar site	Ramsar Convention 1971	Wise-use wetland	Not a forest or ozone treaty
Natural World Heritage	UNESCO WH Convention	Outstanding universal value	≠ Ramsar ≠ BR WNBR automatically
ESZ	EPA **1986**	Buffer restrictions around PA	**Not** a WPA category

- **UP teaching row:** Dudhwa NP / TR, Pilibhit TR, Ranipur TR, **12 Ramsar**, Terai corridors, **no marine PA**, **no biosphere reserve** inside present UP.
- **Three UNESCO instruments:** World Heritage · Biosphere (MAB / WNBR) · Intangible Cultural Heritage — do not force one site into all three without checking the list.

6.1 Wildlife Sanctuary

Feature	Wildlife Sanctuary	National Park
WPA section	Sec 18–27	Sec 35
Human activity	Allowed with CWLW permission	Largely prohibited
Grazing / forestry	May be permitted	Prohibited
Biomass collection	Limited use may be allowed	**Not** allowed as in sanctuary
Boundary change	State process	Needs **State Legislature** resolution
Strictness	Moderate (नरम दल)	**Highest** among WPA terrestrial PAs

6.3 Biosphere Reserve

#	Biosphere Reserve	State(s)
1	Nilgiri	Tamil Nadu, Karnataka, Kerala
2	Nanda Devi (नंदा देवी)	Uttarakhand
3	Nokrek	**Meghalaya**
4	Gulf of Mannar	Tamil Nadu
5	Sundarbans	West Bengal
6	Manas	Assam
7	Great Nicobar	Andaman & Nicobar
8	Simlipal	Odisha
9	Dibru-Saikhowa	Assam
10	Dihang-Dibang	Arunachal Pradesh
11	Pachmarhi	Madhya Pradesh
12	Khangchendzonga	Sikkim
13	Agasthyamalai	Kerala, Tamil Nadu
14	Achanakmar-Amarkantak	MP, Chhattisgarh

#	Biosphere Reserve	State(s)
15	Great Rann of Kutch	Gujarat
16	Cold Desert	Himachal (हिमाचल) Pradesh
17	Seshachalam Hills	Andhra Pradesh
18	Panna	Madhya Pradesh

Common Traps — Don’t Fall For These

Trap	Correct fact
No human activity in sanctuary	Activity **possible with CWLW permission**
Nokrek in Manipur	**Meghalaya**
Sultanpur in UP	**Haryana**
Rudrasagar in UP	**Tripura**
ESZ under WPA	**EPA 1986**
Elephant Reserve under WPA	**No** — Project Elephant only
Biosphere Reserve under WPA	**No** — executive/MAB
UP has marine PA	**No** — landlocked
Grazing allowed in NP	**Prohibited** in NP

Topic 7 — Wildlife Conservation

Current Affairs (this topic)

- Track Project Tiger / Project Elephant / species project headlines against the statutory vs scheme trap.
- NTCA tiger-reserve count drifts — teach the concept; verify the live total before a count stem.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Wildlife Conservation	**Wildlife Protection**	Holistic scientific management and preservation of species, populations, and habitats to prevent extinction vs legal punitive enforcement and regulatory prohibition of hunting/poaching under Wildlife Protection Act 1972	वन्यजीव संरक्षण (वैज्ञानिक प्रबंधन) / वन्यजीव सुरक्षा (कानूनी प्रवर्तन)
Umbrella Species	**Flagship Species**	Ecologically wide-ranging species whose expansive habitat protection indirectly shelters the broader biological community (e.g. Tiger, Rhino) vs popular charismatic animal used as an emotional figurehead in conservation marketing (e.g. Giant Panda, Snow Leopard)	छत्र प्रजाति / प्रतीक प्रजाति
Keystone Species	**Indicator Species**	Species whose functional role exerts disproportionate control on community structure (loss causes trophic cascade; e.g. Fig tree, Sea Otter) vs species whose physiological response or presence directly indicates environmental conditions (e.g. Lichen, Frogs)	मुख्य आधार (कीस्टोन) प्रजाति / जैव संकेतक प्रजाति
Project Tiger (1973)	**Project Elephant (1992)**	Centrally Sponsored Scheme given statutory backing in 2006 under WPA creating the National Tiger Conservation Authority (NTCA) vs Centrally Sponsored Scheme operating through administrative guidelines without statutory authority	प्रोजेक्ट टाइगर (1973, सांविधिक) / प्रोजेक्ट एलिफेंट (1992, गैर-सांविधिक)
Diclofenac	**Meloxicam**	Non-steroidal anti-inflammatory drug (NSAID) administered to cattle causing lethal visceral gout and acute renal failure in Gyps vultures (banned in 2006) vs safe, vulture-tested veterinary NSAID recommended as its non-toxic replacement	डाइक्लोफेनाक (गिद्ध नाशक) / मेलोक्सिकैम (सुरक्षित विकल्प)
Wildlife Protection Act 1972	**Biological Diversity Act 2002**	Focuses on species protection, hunting bans, protected area network, and trade schedules (MoEFCC/WCCB) vs focuses on sovereign rights over biological resources, fair & equitable benefit sharing (ABS), and traditional knowledge protection (NBA/SBB/BMC)	वन्यजीव संरक्षण अधिनियम 1972 / जैव विविधता अधिनियम 2002
Central Zoo Authority (CZA)	**National Board for Wildlife (NBWL)**	Statutory body under WPA Sec 38A overseeing standards, recognition, and animal exchanges in Indian zoos vs apex advisory body chaired by the Prime Minister framing national wildlife policies and approving PA boundary alterations	केंद्रीय चिड़ियाघर प्राधिकरण / राष्ट्रीय वन्यजीव बोर्ड (NBWL)

7.2 Wildlife Protection

Schedule	Protection	Examples
I	Maximum — hunting banned	Tiger, elephant, rhino, gharial, vulture
II	High	Himalayan black bear, cobra
III–IV	Moderate (नरम दल)/lower fauna (प्राणीजात)	Chital, sambar, hare
V	Vermin — may be hunted	Crow, rat
VI	Specified plants — licence needed	Pitcher plant and other listed taxa

Common Traps — Don’t Fall For These

Trap	Correct fact
WPA enacted in 1973	**1972** — Project Tiger = 1973
World Wildlife Day = 22 March	**3 March**; Wildlife Week = **2–8 October**
Operation Olivia nesting in August	**Nov–Feb**
Elephant Reserve under WPA	**No** — executive only; covers wild **and captive**
All rhinos in Assam	Also **WB and Dudhwa UP**
Asiatic lion in UP	**Gir, Gujarat only**
Schedule VI = absolute ban	Cultivation needs a **licence**
Schedule I tortoise ≠ tiger level	**Same** maximum protection band
Spider monkey in India	**Central/South America** only
Kharai cannot be domesticated	**Can** be domesticated; swims + mangrove graze
AWBI under EPA 1986	Under **PCA 1960** (est. 1962)
WWF logo = tiger	**Giant Panda**
Best gharial site = Pulicat	**Chambal**
Tiger Summit 2010 = Delhi	**Saint Petersburg**
Snow leopard in Terai/Gir	**High Himalaya only**
Crocodile project = gharial only	**3 species**
Diclofenac ban = human medicine ban	**Veterinary** use caused crisis

Topic 8 — Forests & Forest Management

Current Affairs (this topic)

- Keep ISFR / FSI function traps stable; new ISFR cycle is the usual forest CA refresh.
- FCA / forest-conservation wording may appear in CA — separate from WPA PA categories.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Forest Cover	**Tree Cover**	All land patches with tree canopy density ≥10% and area ≥1 hectare irrespective of ownership or legal status (FSI remote sensing) vs scattered tree patches, linear trees along canals/roads, and isolated trees <1 hectare outside recorded forest area	वन आवरण (≥1 हे., ≥10% छतरी) / वृक्ष आवरण (<1 हेक्टेयर)
Reserved Forest (RF)	**Protected Forest (PF)**	Highest degree of protection under Indian Forest Act 1927; all activities (grazing, timber felling) are strictly prohibited unless expressly permitted vs forest where all activities are permitted unless specifically prohibited by notification	आरक्षित वन (RF, पूर्णतः प्रतिबंधित) / संरक्षित वन (PF, सशर्त अनुमत)
Recorded Forest Area (RFA)	**Forest Cover**	Land legally recorded or notified as forest in government records (RF + PF + Unclassed) regardless of actual tree presence vs physical presence of tree canopy detected by satellite imagery (even on private orchards/tea estates)	अभिलिखित वन क्षेत्र (कानूनी अभिलेख) / वन आवरण (भौतिक वृक्ष आच्छादन)
Forest Conservation Act (FCA 1980)	**Forest Rights Act (FRA 2006)**	Central regulatory statute preventing de-reservation or diversion of forest land for non-forest purposes without prior central approval vs restorative rights law vesting individual and community forest rights in traditional forest-dwelling tribes and OTFDs	वन संरक्षण अधिनियम 1980 / वन अधिकार अधिनियम 2006
Social Forestry	**Agroforestry**	Growing trees on non-forest communal lands, roadside strips, and wastelands to meet rural fodder and fuel needs (National Commission on Agriculture 1976) vs deliberate integration of woody perennials (trees/shrubs) with agricultural crops and/or livestock on the same land unit	सामाजिक वानिकी / कृषि वानिकी
Joint Forest Management (JFM)	**Community Forest Resource (CFR)**	Collaborative institutional partnership between state forest department and village committees to protect and regenerate degraded forests vs statutory community ownership and governance right recognized under Section 3(1)(i) of FRA 2006	संयुक्त वन प्रबंधन (JFM) / सामुदायिक वन संसाधन अधिकार (CFR)
Tropical Wet Evergreen Forest	**Tropical Moist Deciduous Forest**	Dense multi-layered forest with rainfall >200 cm, no distinct leafless season (Ebony, Mahogany, Rosewood) vs most widespread forest in India with rainfall 100–200 cm, shedding leaves for 6–8 weeks in dry spring (Teak, Sal, Shisham)	उष्णकटिबंधीय आर्द्र सदाबहार / उष्णकटिबंधीय नम पर्णपाती वन

8.1 Forests and their Types

IFA class	Protection level
Reserved Forest	Strictest — hunting, grazing, felling banned unless permitted
Protected Forest	Moderate (नरम दल) — state controls but some rights allowed
Unclassed Forest	Not yet formally classified

8.2 Forest Types in India

#	Champion & Seth type	Key region
1	Tropical Wet Evergreen	Western Ghats, NE India
2	Tropical Semi-Evergreen	Transition to evergreen belt
3	Tropical Moist Deciduous	Terai, central India
4	Littoral & Swamp	Sundarbans, coastal Odisha
5	Tropical Dry Deciduous	**Largest area (~38%)**
6	Tropical Thorn	Rajasthan, Bundelkhand
7	Tropical Dry Evergreen	East coast Karnataka/AP
8	Subtropical Broadleaved Hill	Nilgiri (नीलगिरि), NE hills
9	Subtropical Pine	Chir pine — outer Himalaya
10	Subtropical Dry Evergreen	Siwalik, Jammu
11	Montane Wet Temperate	Nilgiri, eastern Himalaya
12	Himalayan Moist Temperate	Oak, deodar — Uttarakhand, HP
13	Himalayan Dry Temperate	Inner Himalaya conifers
14	Sub-Alpine	Rhododendron, fir — >3000 m
15	Moist Alpine Scrub	High Himalaya
16	Dry Alpine Scrub	Cold desert Himalaya

Common Traps — Don’t Fall For These

Trap	Correct fact
Bahraich >20% forest	**Chandauli + Shravasti** only
Forest cover = RFA only	Cover exists **outside** notified forest
FCA enacted 1972	**1980**
JFM = FRA same year	JFM **1990**, FRA **2006**
Social forestry = JFM	Social = **non-forest land**; JFM = degraded forest
FRA IFR cap 10 ha	**4 ha**
Agroforestry Policy 2006	**2014**
ISFR by CPCB	**FSI Dehradun**
NFP 1952 vs 1988	Current policy = **1988**; 1952 classes ≠ National Park
CAFRI location	**Jhansi**, not Banda
MP vs Mizoram	MP = **largest area**; Mizoram = **highest %**
RF vs Protected Forest	RF = strictest IFA class
Red Sanders = rainforest	**Dry deciduous** South India / Eastern Ghats
Teak = rainforest	**Tropical moist deciduous**; MP max teak
HP = tropical wet evergreen	**Pine/temperate** belts
Urbanisation = deforestation impact	Urbanisation is a **cause**
Mangrove safety = canopy shelter only	**Extensive roots**
Bhitarkanika in West Bengal	**Odisha** — Brahmani–Baitarni

Trap	Correct fact
Gujarat < A&N mangrove	**Gujarat 2nd** after WB (ISFR 2021 style)
Aluminium green = saves forests	Recyclability/eco-profile; wood-substitute R often false

Topic 9 — Pollution & Waste Management

Current Affairs (this topic)

- NCAP (राष्ट्रीय स्वच्छ वायु कार्यक्रम) / CAQM / solid-waste and e-waste rule updates are the living pollution CA layer.
- Keep primary pollutant vs secondary pollutant and BOD vs COD pairs stable.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Environmental Pollution	**Environmental Degradation**	Direct or indirect release of harmful substances, radiation, or thermal energy into air, water, or soil causing toxic injury to living organisms vs broader long-term deterioration of environmental quality, natural capital, and ecosystem productivity	पर्यावरणीय प्रदूषण / पर्यावरणीय क्षरण
Primary Air Pollutant	**Secondary Air Pollutant**	Emitted directly from identifiable domestic or industrial exhaust sources into the atmosphere (CO, SO_2, NO, PM, Lead) vs formed in the atmosphere through chemical or photochemical reactions among primary precursors (Tropospheric Ozone, PAN, Smog, Nitric acid)	प्राथमिक प्रदूषक (सीधे उत्सर्जित) / द्वितीयक प्रदूषक (रासायनिक क्रिया)
Biochemical Oxygen Demand (BOD)	**Chemical Oxygen Demand (COD)**	Amount of dissolved oxygen needed by aerobic microorganisms to decompose biodegradable organic matter in water at 20°C over 5 days vs total amount of oxygen required to chemically oxidize all organic matter (both biodegradable and non-biodegradable) using a strong chemical oxidant	जैव रासायनिक ऑक्सीजन मांग (BOD) / रासायनिक ऑक्सीजन मांग (COD)
Oligotrophic Lake	**Eutrophic Lake**	Deep, crystal-clear water body with low nutrient levels, low biological productivity, and high dissolved oxygen vs shallow, murky water body rich in nitrates/phosphates, choked with algal blooms, and experiencing severe hypolimnetic oxygen crashes	अल्पपोषी झील (स्वच्छ/कम पोषक) / सुपोषित झील (अल्गल ब्लूम/ऑक्सीजन ह्रास)
Extended Producer Responsibility (EPR)	**Polluter Pays Principle**	Policy mandate making manufacturers legally and financially responsible for post-consumer collection, recycling, and safe disposal of their products (E-waste, plastic packaging) vs international environmental law doctrine requiring polluters to bear the financial costs of pollution mitigation and cleanup	विस्तारित उत्पादक दायित्व (EPR) / प्रदूषक भुगतान सिद्धांत
Central Pollution Control Board (CPCB)	**Central Ground Water Board (CGWB)**	Statutory apex pollution regulator established under Water Act 1974 for framing air/water standards and national monitoring vs subordinate scientific office under Ministry of Jal Shakti assessing groundwater resources and aquifer dynamics	केंद्रीय प्रदूषण नियंत्रण बोर्ड (CPCB) / केंद्रीय भूजल बोर्ड (CGWB)
Classical (London) Smog	**Photochemical (Los Angeles) Smog**	Occurs in cool, humid winter mornings; high concentrations of SO_2 and coal soot; chemically reducing in nature vs occurs in hot, dry, sunny afternoons; vehicle exhaust NOx + volatile organic compounds (VOCs); chemically oxidizing in nature with high ozone and PAN	लंदन स्मॉग (सल्फ्यूरस/अपचायक) / प्रकाश रासायनिक स्मॉग (ऑक्सीकारक)

9.4 Water Pollution

Disease	Causative Pollutant / Dust	Industry / Occupational Source	Key feature / Nickname
Silicosis	Free silica (SiO_2) dust	Stone crushing, quarrying, mining, glass factories	Progressive lung fibrosis; Rajasthan first state to frame Silicosis Policy (2019)

Disease	Causative Pollutant / Dust	Industry / Occupational Source	Key feature / Nickname
Asbestosis	Asbestos fibers	Roofing sheets, pipe insulation, shipbuilding	Leads to pulmonary fibrosis and **Mesothelioma** (cancer of pleural lining)
Anthracosis	Coal dust	Coal mining (Jharia, Raniganj, Singrauli)	**Black Lung Disease** / Coal Workers' Pneumoconiosis
Byssinosis	Cotton fiber dust	Textile mills, ginning factories	**Brown Lung Disease**; called **"Monday Fever"** (symptoms peak after weekend return)
Bagassosis	Sugarcane bagasse dust	Sugar mills, paper & cardboard manufacturing	Fungal spores (*Thermoactinomyces*) on decomposing crushed sugarcane stalk
Siderosis	Iron / iron oxide dust	Iron foundries, arc welding, iron ore mining	Iron dust accumulation in lung parenchyma
Farmer's Lung	Moldy hay / straw dust	Agriculture, silos, threshing	Hypersensitivity reaction to thermophilic actinomycetes

9.12 Biomedical Waste

Colour	Waste type	Treatment
Yellow	Anatomical, soiled, medicines	Incineration/autoclave
Red	Contaminated recyclables (IV tubes, catheters, urine bags, syringes without needles)	Disinfection → recycle
White	Sharps (needles)	Autoclave → disposal
Blue	Glass vials, metal	Disinfection → recycle

Common Traps — Don't Fall For These

Trap	Correct fact
Lichens grow well in polluted air	They **die** in pollution
CPCB publishes groundwater report	**CGWB**
O_3 at ground level is beneficial	**Harmful** — unlike stratospheric ozone
Yellow BMW bag for sharps	**White** container
PWM Rules year = SUP ban year	**2016 vs 2022**
All marine pollution from ships	**~80% from land**
BOD measures heavy metals	**Organic pollution only**
Gomti flows through Kanpur	**Lucknow**
E-waste Rules still 2016	**2022** rules current
Thermal kills fish by heat toxicity alone	Mainly **DO reduction**
E-waste rank 1st globally	**3rd**
Complete decay in one half-life	**50% decay**
Pesticide = biodegradable	**Not**
AQI includes CO_2/CH_4	**Eight** pollutants — no
PAN = primary pollutant	**Secondary**; eye irritant
Knock-knee = mercury	**Fluoride**
Eutrophication = air pollution	**Water**

Trap	Correct fact
Cyclone divider = control device	**Not**
Bhopal = CO/methane	**MIC, Dec 1984**
Bioremediation cleans all Cd/Pb	**Not** readily
Green muffler = soil only	**Noise** tree belts
Most important indoor = outdoor SO_2	Often **radon**

Topic 10 — Climate Change

Current Affairs (this topic)

- Keep India NDC / Net Zero 2070 / LiFE as living climate CA; treaty years stay in the ladder below.
- Do not put **Montreal** into a climate-mitigation answer slot when Kyoto (क्योटो) / Paris are present.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Global Warming	**Climate Change**	Long-term monotonic increase in Earth's average surface and tropospheric temperature driven by greenhouse gas buildup vs broad spectrum of shifting weather patterns, rising sea levels, altering precipitation, and intensifying extreme climate anomalies	भूमंडलीय तापन (ग्लोबल वार्मिंग) / जलवायु परिवर्तन
Climate Mitigation	**Climate Adaptation**	Interventions designed to reduce greenhouse gas emissions at the source or enhance carbon sinks (renewable energy, afforestation, carbon capture) vs adjustments in natural or human systems to moderate harm, exploit beneficial opportunities, and cope with actual/expected climate impacts (flood barriers, drought-tolerant seeds)	जलवायु शमन (उत्सर्जन रोकथाम) / जलवायु अनुकूलन (सहनशीलता वृद्धि)
Natural Greenhouse Effect	**Enhanced Greenhouse Effect**	Essential natural phenomenon keeping Earth's surface habitable at an average ~15°C (without it Earth would freeze at -18°C) vs excessive anthropogenic heat-trapping caused by fossil fuel combustion and industrial emissions driving global warming	प्राकृतिक ग्रीनहाउस प्रभाव / मानवजनित संवर्धित प्रभाव
Global Warming Potential (GWP)	**Radiative Forcing**	Metric measuring the cumulative heat-trapping ability of 1 kg of a specific GHG over 100 years relative to CO_2 (=1.0) vs the net change in Earth's radiative energy balance (measured in Watts/m^2) caused by changes in atmospheric gas concentrations	वैश्विक तापन क्षमता (GWP) / विकिरण दबाव
Carbon Sink	**Carbon Sequestration**	Natural or artificial reservoir that absorbs and stores more carbon than it releases into the atmosphere (oceans, boreal forests, soil) vs the actual physical, biological, or chemical process of capturing and securing carbon dioxide in long-term storage	कार्बन सिंक (अवशोषक भंडार) / कार्बन पृथक्करण (भंडारण प्रक्रिया)
Carbon Credit	**Carbon Offset**	Formal tradable permit or certificate representing the reduction or removal of 1 metric tonne of CO_2 equivalent under a compliance cap-and-trade system vs voluntary purchase of emission reduction units to neutralize an individual's or company's carbon footprint	कार्बन क्रेडिट (1 टन CO_2e प्रमाण-पत्र) / कार्बन ऑफसेट
Kyoto Protocol (1997)	**Paris Agreement (2015)**	Top-down legally binding emission reduction targets imposed exclusively on developed nations (Annex-I) with Common But Differentiated Responsibilities vs bottom-up universal framework requiring all countries to submit voluntary Nationally Determined Contributions (NDCs) aiming to cap warming well below 2.0°C/1.5°C	क्योटो प्रोटोकॉल (बाध्यकारी/विकसित देश) / पेरिस समझौता (सार्वभौमिक NDCs)

10.3 Greenhouse Gases (GHGs)

GHG / group	Rough relative share (WMO-style stems)	GWP (100-yr)
CO_2	~64%	1
CH_4	~19%	28–36
N_2O	~6%	265
CFCs + others	~11%	Very high (F-gases)

10.8 Climate Adaptation

#	NAPCC Mission
1	National Solar Mission
2	Enhanced Energy Efficiency
3	Sustainable Habitat (वास स्थान)
4	National Water Mission
5	Sustaining Himalayan Ecosystem
6	Green India
7	Sustainable Agriculture
8	Strategic Knowledge for Climate Change

10.9 Climate Mitigation

Treaty ladder — climate vs ozone (read the stem)

Ladder	Instruments	What it controls	Classic trap
Climate	UNFCCC (1992) (जलवायु परिवर्तन संधि) → **Kyoto** (क्योटो) (1997) → **Paris** (2015)	GHG mitigation / NDCs	Montreal is **not** climate mitigation
Ozone	Vienna (1985) → **Montreal** (1987) → **Kigali** (HFCs, 2016)	ODS / HFC phase-down	Kyoto is **not** ozone

Common Traps — Don't Fall For These

Trap	Correct fact
Montreal = climate treaty	**Ozone** — Kyoto/Paris = climate
LiFE proposed at COP-25	**COP-26 Glasgow**
Eliminate greenhouse effect	**Natural effect essential** — problem is enhanced
Wilson/IPCC invented greenhouse effect	**Joseph Fourier (1820s)**
CH_4 highest total contribution	**CO_2** (~64% WMO-style / ~76% IPCC forcing)
Water vapour = main Kyoto inventory driver	Most abundant **natural** GHG / feedback; policy driver = **CO_2**
Argon / H_2 / propane / N_2 are GHGs	**Not** GHGs
SO_2 / NOx warm Earth directly	**Not direct** GHGs
India highest per capita emitter	**Low per capita**, ~3rd total; **China** #1
Net zero = absolute zero emissions	Emissions **balanced** by removals
Nuclear power is an NAPCC mission	**Not** among the eight
GCF founded at Durban	**Cancun COP-16 (2010)**
Earth Hour = UNEP/UNESCO	**WWF**
Carbon credit price fixed by UNEP	**Market-traded**
Adaptation = cut emissions	**Mitigation** cuts emissions
Kyoto bound India to cuts	Bound **Annex I developed** countries
Solar flares = primary recent climate cause	**Not** major (NASA/IPCC framing)

Trap	Correct fact
Milankovitch includes solar irradiance	Eccentricity, obliquity, precession only
Blue carbon = petroleum carbon	**Ocean/coastal** ecosystems

Topic 11 — Ozone Layer

Current Affairs (this topic)

- Montreal / Kigali compliance headlines stay ozone-family CA — do not park them under Paris NDC.
- ODS vs GHG trap is permanent; HFC phase-down is Kigali, not Kyoto (क्योटो).

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Stratospheric Ozone	**Tropospheric Ozone**	Essential atmospheric shield absorbing harmful solar UV-B radiation (20–30 km altitude) vs toxic secondary pollutant and greenhouse gas causing smog and respiratory ailments	समतापमंडलीय ओजोन (सुरक्षा कवच) / क्षोभमंडलीय ओजोन (हानिकारक प्रदूषक)
Ozone Depletion	**Ozone Hole**	Widespread gradual thinning of the stratospheric ozone layer globally caused by halogenated gases vs severe, seasonal Antarctic springtime drop where column ozone drops below 220 Dobson Units (DU)	ओजोन क्षरण (वैश्विक स्तर पर) / ओजोन छिद्र (अंटार्कटिक <220 DU)
Vienna Convention (1985)	**Montreal Protocol (1987)**	Non-binding multilateral framework agreement establishing international cooperation on ozone research and monitoring vs legally binding international treaty mandating precise, phase-out timetables for ozone-depleting substances (ODS)	वियना कन्वेंशन (ढांचागत समझौता) / मॉन्ट्रियल प्रोटोकॉल (बाध्यकारी प्रतिबंध)
Chlorofluorocarbons (CFCs)	**Hydrofluorocarbons (HFCs)**	Powerful ozone-depleting substances with high chlorine and high GWP (phased out under Montreal) vs zero-ODP fluorinated gases that do not harm ozone but are super greenhouse gases (phased down under 2016 Kigali Amendment)	सीएफसी (ओजोन विनाशक + GHG) / एचएफसी (केवल GHG, किगाली द्वारा नियंत्रित)
Dobson Unit (DU)	**Parts Per Million (ppm)**	Standard unit measuring the total vertical column abundance of atmospheric ozone (1 DU = 0.01 mm thickness at STP; normal baseline ~300 DU) vs standard volumetric concentration ratio of trace gases in ambient air	डॉबसन यूनिट (स्तंभ ओजोन मोटाई) / पीपीएम (सान्द्रता अनुपात)
Polar Stratospheric Clouds (PSCs)	**Cirrus Clouds**	High-altitude clouds forming in extreme Antarctic winter cold (-78°C / Type I and II) that catalyze chlorine release from reservoirs vs normal high-altitude wispy ice-crystal weather clouds in the upper troposphere	ध्रुवीय समतापमंडलीय बादल (PSCs) / पक्षाभ मेघ (ट्रोपोस्फेरिक)
Kigali Amendment (2016)	**Paris Agreement (2015)**	Legally binding amendment to Montreal Protocol committing countries to progressively phase down consumption of hydrofluorocarbons (HFCs) by 80–85% vs UNFCCC climate framework agreement targeting global warming mitigation	किगाली संशोधन (HFC कटौती) / पेरिस समझौता (जलवायु लक्ष्य)

11.2 Ozone Depletion

ODS	Main use
CFCs	Refrigerants, aerosols, foam
Halons	Fire extinguishers
Carbon tetrachloride	Solvent

ODS	Main use
Methyl bromide	Pesticide/fumigant
HCFCs	Transitional refrigerants

11.3 Ozone Hole

Feature	Antarctic	Arctic
Severity	Severe, recurring	Occasional, milder
Season	Sep–Oct (austral spring)	Mar–Apr if event occurs
Polar vortex	Strong, very cold	Weaker, warmer

Common Traps — Don't Fall For These

Trap	Correct fact
Montreal = climate treaty	**Ozone / ODS**
Ozone layer in troposphere	**Stratosphere**
Ozone hole = physical hole	**Low ozone column < 220 DU**
Ozone hole over Arctic always	**Antarctic spring**
Tropospheric ozone is good	**Pollutant**
CFCs controlled under Kyoto	**Montreal Protocol**
Ozone Day = 5 June	**16 September**
Kigali targets CFCs	**HFC phase-down**
UV-C reaches surface normally	**Blocked by O_3/O_2**
Depletion permanent with no recovery	Recovery expected **~2040/2066**

Topic 12 — Acid Rain

Current Affairs (this topic)

- Acid-rain CA is rare; keep SO_2 / NO_x chemistry and dry vs wet deposition stable.
- Do not confuse acid rain with ozone hole (ओजोन छिद्र) or greenhouse trapping.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Normal Rain	**Acid Rain**	Slightly acidic precipitation with pH ~5.6 caused by dissolved atmospheric carbon dioxide forming weak carbonic acid (H_2CO_3) vs precipitation with pH strictly below 5.6 caused by anthropogenic emissions of sulfur dioxide (SO_2) and nitrogen oxides (NOx)	सामान्य वर्षा (pH ~5.6) / अम्ल वर्षा (pH < 5.6)
Sulfuric Acid (H_2SO_4)	**Nitric Acid (HNO_3)**	Accounts for ~60–70% of total acid deposition acidity (primarily emitted by coal-fired thermal power plants and metal smelters) vs accounts for ~30–40% of acidity (primarily emitted by automobile exhausts and nitrogenous fertilizer use)	सल्फ्यूरिक अम्ल (60–70% भारांक) / नाइट्रिक अम्ल (30–40% भारांक)
Wet Acid Deposition	**Dry Acid Deposition**	Direct transfer of acidic pollutants to Earth's surface incorporated into falling rain, fog, sleet, or snow vs settling of acidic gases, sulfate aerosols, and nitrate dust directly onto soil, vegetation, and buildings without moisture	आर्द्र निक्षेपण (वर्षा/कोहरा) / शुष्क निक्षेपण (गैस व धूलिकण)
Marble Cancer	**Eutrophication**	Chemical weathering and yellowing/spalling of calcium carbonate structures (e.g. Taj Mahal) reacting with sulfuric acid to form soluble gypsum ($CaSO_4$) vs excessive nutrient enrichment of water bodies leading to dense algal blooms and dissolved oxygen crashes	संगमरमर कैंसर (ताजमहल क्षरण) / सुपोषण (जलीय पोषक संवर्धन)
Gothenburg Protocol (1999)	**Montreal Protocol (1987)**	UNECE agreement under CLRTAP targeting multi-pollutant abatement of acidification, eutrophication, and ground-level ozone (SO_2, NOx, VOCs, NH_3) vs UNEP treaty targeting the global phase-out of ozone-depleting substances	गोथेनबर्ग प्रोटोकॉल (अम्लीकरण नियंत्रण) / मॉन्ट्रियल प्रोटोकॉल (ओजोन)
Taj Trapezium Zone (TTZ)	**Eco-Sensitive Zone (ESZ)**	Specifically demarcated ~10,400 km^2 protective geographical zone covering Agra, Firozabad, Mathura, and Bharatpur to safeguard the Taj Mahal from industrial air pollution vs buffer belt up to 10 km declared around National Parks under EPA 1986	ताज ट्रपेज़ियम ज़ोन (10,400 वर्ग किमी) / पारिस्थितिक संवेदनशील क्षेत्र

12.4 Prevention

Measure	Target
FGD scrubbers	SO_2 from TPP chimneys
Catalytic converters	NOx from vehicles
BS-VI fuel	SO_2 + NOx precursors
CNG/LPG switch	Near-zero SO_2
Gothenburg Protocol	Transboundary SO_2/NOx
Taj Trapezium	Heritage protection near Agra

Common Traps — Don’t Fall For These

Trap	Correct fact
Normal rain pH = 7	**~5.6** from CO_2
CO_2 = main acid rain cause	**Weak acid only**; SO_2/NOx = strong acids
Coal acid rain explained by CO_2 alone	A true (coal → acid rain via **SO_2**); R true (CO_2 emitted) but R **does not explain** A
Acid rain = rain only	**Wet + dry deposition**
Acidic pH > 7	**Acidic < 7**
Montreal = acid rain	**Gothenburg / CLRTAP**
SO_2 lighter than air	**Heavier**
Lichens thrive in pollution	**Die in SO_2**
Tall chimney = prevention	**Dispersal only**
Taj damage from CO_2	**SO_2 + acid deposition**
Acid rain stays in source country	**Transboundary**

Topic 13 — Desertification & Land Degradation

Current Affairs (this topic)

- UNCCD / LDN and Indian desertification atlas updates are the CA refresh.
- Bundelkhand / Rajasthan land-degradation colour stays high-yield for UP.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Desertification	**Land Degradation**	Persistent reduction or loss of biological and economic productivity occurring specifically in Arid, Semi-Arid, and Dry Sub-Humid areas (drylands) vs general decline in the productive capacity and ecosystem health of any land anywhere	मरुस्थलीकरण (शुष्क भूमियों में) / भू-क्षरण (सामान्य भूमि अवनति)
Drought	**Desertification**	Temporary natural meteorological period of deficient precipitation over months or a few years from which ecosystems can recover vs persistent, semi-permanent ecological degradation and productivity loss driven by unsustainable human use and climate stress	सूखा (अस्थायी मौसमी संकट) / मरुस्थलीकरण (स्थायी भूमि ह्रास)
Sheet Erosion	**Gully Erosion**	Uniform, imperceptible removal of a thin layer of topsoil over large flat cultivated fields by sheetwash runoff vs cutting of deep, severe channels and ravines by concentrated runoff carving badland topography (e.g. Chambal ravines)	परत (चादर) अपरदन / अवनलिका अपरदन (बीहड़ निर्माण)
Splash Erosion	**Rill Erosion**	Initial detachment and airborne displacement of soil particles caused directly by the physical impact of falling raindrops vs intermediate erosion stage where tiny, shallow, visible finger-like channels are formed on sloped ground	आघात (स्पलैश) अपरदन / क्षुद्र सरिता (रिल) अपरदन
UNCCD (1994)	**UNFCCC (1992)**	Sole legally binding international treaty linking environment and development to sustainable land management, desertification, and drought in drylands vs framework convention dedicated to addressing anthropogenic climate change and greenhouse gases	मरुस्थलीकरण रोकथाम संधि (UNCCD) / जलवायु परिवर्तन संधि (UNFCCC)
Land Degradation Neutrality (LDN)	**Net Zero Emissions**	Target under UN SDG 15.3 ensuring that the quantity and quality of land resources necessary to support ecosystem functions remain stable or increase by 2030 vs balancing greenhouse gas emissions with equivalent removals	भूमि क्षरण तटस्थता (LDN, SDG 15.3) / नेट जीरो उत्सर्जन

13.2 Land Degradation

Process	Example region
Salinization	Punjab-Haryana canal belt
Mining degradation	Sonbhadra, Jharkhand
Waterlogging	Over-irrigated plains
Ravine formation	Chambal basin

Common Traps — Don’t Fall For These

Trap	Correct fact
Desertification = desert expansion only	**Any ASDM dryland losing productivity**

Trap	Correct fact
Land degradation = desertification	**Degradation is broader**
Afforestation causes degradation	**It prevents degradation**
Sheet erosion = deep ravines	**Sheet is first, mildest stage**
Chambal ravines in one state	**MP + UP + Rajasthan**
UNCCD = climate treaty	**UNCCD = desertification; UNFCCC = climate**
Normal rain pH 7 stops erosion	**Unrelated — erosion is topsoil removal**
Wind erosion in Chambal ravines	**Water/gully erosion**
LDN target 2025	**2030 (SDG 15.3)**
UNCCD signed at Rio 1992	**Adopted 1994, Paris**

Topic 14 — Environmental Impact Assessment

Current Affairs (this topic)

- EIA notification amendments and Category A/B traps are the living CA layer.
- Keep screening–scoping–appraisal–monitoring order stable.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Environmental Impact Assessment (EIA)	**Strategic Environmental Assessment (SEA)**	Formal analytical process evaluating the potential ecological impacts of a specific *project* (e.g. thermal plant, highway) before sanction vs higher-level systemic evaluation applied to government *policies, plans, and programmes*	पर्यावरण प्रभाव आकलन (EIA, परियोजना स्तर) / रणनीतिक पर्यावरण आकलन (SEA, नीति स्तर)
Category A Projects	**Category B Projects**	Large-scale developmental projects requiring mandatory full-scale EIA appraised centrally at Union level by MoEFCC on recommendations of EAC vs smaller projects appraised at State level by SEIAA on recommendations of SEAC	श्रेणी 'ए' (केंद्रीय मूल्यांकन) / श्रेणी 'ब' (राज्य मूल्यांकन)
Category B1 Projects	**Category B2 Projects**	State-level projects that undergo complete EIA process and mandatory public hearing due to significant ecological footprint vs smaller Category B projects exempted from EIA study and public consultation, appraised solely on application summary	श्रेणी 'ब1' (EIA अनिवार्य) / श्रेणी 'ब2' (EIA से छूट)
Terms of Reference (ToR)	**Environmental Management Plan (EMP)**	Comprehensive boundary document outlining all environmental baseline issues and studies that the project proponent must investigate vs actionable mitigation, monitoring, and institutional plan submitted in the final EIA report	संदर्भ की शर्तें (ToR, आरंभिक ढांचा) / पर्यावरण प्रबंधन योजना (EMP, शमन कार्ययोजना)
Public Hearing (Consultation)	**Appraisal**	Statutory public stage where local affected communities and citizens voice concerns and objections to the draft EIA vs detailed scientific scrutiny of final EIA and public hearing proceedings by EAC or SEAC	जन सुनवाई (स्थानीय सहभागिता) / वैज्ञानिक मूल्यांकन (EAC/SEAC)
National Green Tribunal (NGT)	**Appellate Authority under Air/Water Acts**	Specialized judicial tribunal established under NGT Act 2010 hearing original environmental disputes and appeals against environmental clearances (EC) vs administrative appellate authorities designated under Water Act 1974 / Air Act 1981	राष्ट्रीय हरित अधिकरण (न्यायिक अधिकरण) / राज्य अपीलीय प्राधिकरण

14.1 Environmental Impact Assessment (EIA)

Category	Authority	Screening?	Full EIA?
A	MoEFCC + EAC	No	Usually yes
B1	SEIAA + SEAC	Yes	Yes
B2	SEIAA/DEIAA	Yes	No full report
B + General Condition	MoEFCC (elevated)	—	As Category A
Stage	What happens		
-------	----------		
Screening	B → B1 or B2		
Scoping	ToR for EIA study		

Category	Authority	Screening?	Full EIA?
Public Consultation	Public hearing + comments		
Appraisal	Recommend grant/reject EC		

14.2 Strategic Environmental Assessment (SEA)

Feature	EIA	SEA
Level	Individual project	Policy/plan/programme
Timing	After project concept	Before projects chosen
India law	Mandatory (2006)	No standalone law
Cumulative impacts	Limited	Core focus
Example	EC for one thermal plant	National sector plan before siting
Instrument	Year	Scope
----------	------	-------
Rio Principle 17	1992	Project EIA norm
Espoo Convention	1991	Transboundary EIA
SEA Protocol (Kiev)	2003	Plans/programmes
EU SEA Directive	2001/42/EC	EU mandatory SEA
India EIA Notification	2006	Project EC only

Common Traps — Don’t Fall For These

Trap	Correct fact
CPCB grants EC	**MoEFCC/SEIAA grant EC**
Screening for Category A	**Screening = Category B only**
EIA after construction	**EC before construction**
EIA = SEA	**EIA = project; SEA = plan/policy**
India has mandatory SEA law	**Project EIA only**
EIA for all projects	**Listed Schedule projects only**
Water Act 1981 base for EIA	**EPA 1986**
NGT established 1986	**NGT Act 2010**
EAC grants final EC	**EAC recommends; MoEFCC/SEIAA grant**
Rio Principle 17 = SEA	**Principle 17 = project EIA**

Topic 15 — Sustainable Development & Environmental Governance

Current Affairs (this topic)

- SDG progress reports and LiFE / Mission LiFE headlines refresh this chapter's CA.
- Brundtland / Agenda 21 / Rio years stay fixed — do not relocate them into Paris-only answers.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Brundtland Report (1987)	**Limits to Growth (1972)**	UN World Commission on Environment & Development (*Our Common Future*) coining the definition of Sustainable Development vs Club of Rome computer-simulation study modeling impending resource depletion and economic collapse	ब्रंटलैंड रिपोर्ट (1987, सतत विकास) / लिमिट्स टू ग्रोथ (1972)
Millennium Development Goals (MDGs)	**Sustainable Development Goals (SDGs)**	8 anti-poverty targets for 2000–2015 focused primarily on developing nations funded by rich countries vs 17 universal, integrated economic-social-environmental goals (Agenda 2030) adopted in 2015 applying to all nations equally	सहस्राब्दी विकास लक्ष्य (MDGs) / सतत विकास लक्ष्य (SDGs 2030)
Green Economy	**Circular Economy**	Macro-economic development paradigm promoting low-carbon growth, resource efficiency, and social equity (UNEP) vs regenerative industrial design philosophy eliminating waste and keeping materials in closed continuous cycles	हरित अर्थव्यवस्था (व्यापक मॉडल) / चक्रीय अर्थव्यवस्था (पुनर्चक्रण लूप)
Linear Economy	**Circular Economy**	Traditional unsustainable industrial model based on "take, make, use, and dispose" vs restorative circular model based on "reduce, reuse, repair, refurbish, and recycle"	रैखिक अर्थव्यवस्था (कचरा उत्पादक) / चक्रीय अर्थव्यवस्था (शून्य अपशिष्ट)
Mission LiFE (2022)	**Agenda 21 (1992)**	Mass global citizen movement conceptualized by India to replace mindless consumption with mindful individual sustainable lifestyle actions vs Rio 1992 comprehensive non-binding intergovernmental action plan for sustainable governance	मिशन लाइफ (LiFE, व्यक्तिगत जीवनशैली) / एजेंडा 21 (सरकारी कार्ययोजना)
Natural Capital	**Manufactured Capital**	World's natural ecological assets including geology, soil, air, water, biodiversity, and ecosystem services vs human-created physical infrastructure, machinery, factories, and financial assets	प्राकृतिक पूंजी (पारिस्थितिक संपदा) / निर्मित पूंजी (मानव-निर्मित संपत्ति)

15.4 Green Economy & Circular Economy

Feature	Green economy	Circular economy
Focus	Overall low-carbon inclusive growth	Material/waste loop design
Scope	Broader	Narrower — subset of green
Example	Solar parks, green hydrogen	EPR, plastic recycling
SDG link	7, 8, 13 and more	Mainly SDG 12

15.5 Sustainable Agriculture & NMSA

Idea	Correct fact
Sustainable agriculture	Land quality remains intact

Idea	Correct fact
Best strategy set	Mixed cropping + organic + N-fixing + pest-resistant varieties
NMSA	NAPCC mission
Conservation Agriculture	Min tillage + residue cover + rotation

Common Traps — Don’t Fall For These

Trap	Correct fact
Agenda 21 = 1995	**Rio 1992**
Brundtland = Limits to Growth	***Our Common Future* 1987**
SD = environment only	**Three pillars balanced**
SD = no development	**Sustainable, not zero, development**
LiFE at COP-25	**COP26 2021; launched June 2022**
UP SDG report = Finance Dept	**Planning Department**
Circular = identical to green	**Circular ⊂ Green**
MDGs = 17 goals	**MDGs = 8; SDGs = 17**
SDGs only for developing countries	**Universal**
Bhopal → Stockholm 1972	**Bhopal 1984 → EPA 1986**

Topic 16 — Environmental Organizations (India)

Current Affairs (this topic)

- MoEFCC / CPCB / SPCB / NBA HQ and function traps stay stable; scheme rebrands are the CA layer.
- NEERI = Nagpur remains a classic organisation location trap.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Botanical Survey of India (BSI)	**Zoological Survey of India (ZSI)**	Apex national taxonomic survey of plant resources (founded 1890, HQ: Kolkata) publishing the Red Data Book of Indian Plants vs apex national survey of faunal resources (founded 1916, HQ: Kolkata) documenting animal biodiversity	भारतीय वनस्पति सर्वेक्षण (BSI, 1890) / भारतीय प्राणी सर्वेक्षण (ZSI, 1916)
Forest Survey of India (FSI)	**Forest Research Institute (FRI)**	National organization (HQ: Dehradun) responsible for periodic remote-sensing assessment of forest cover and publishing the biennial India State of Forest Report (ISFR) vs premier academic research and deemed university under ICFRE (HQ: Dehradun)	भारतीय वन सर्वेक्षण (FSI, ISFR प्रकाशक) / वन अनुसंधान संस्थान (FRI)
Wildlife Institute of India (WII)	**Wildlife Crime Control Bureau (WCCB)**	Autonomous research and training institute under MoEFCC (Dehradun) conducting wildlife ecology research and census vs statutory enforcement bureau created under WPA 1972 Sec 38Y (New Delhi) combating organized wildlife poaching and illegal trade	भारतीय वन्यजीव संस्थान (WII, अनुसंधान) / वन्यजीव अपराध नियंत्रण ब्यूरो (WCCB, प्रवर्तन)
Central Pollution Control Board (CPCB)	**State Pollution Control Board (SPCB)**	Statutory apex body established under Water Act 1974 advising the Union Government, setting national ambient standards, and coordinating inter-state river/air monitoring vs state-level statutory body granting Consent to Establish/Operate to industrial units	केंद्रीय प्रदूषण नियंत्रण बोर्ड (मानक निर्माता) / राज्य प्रदूषण नियंत्रण बोर्ड (अनुमति प्रदाता)
National Green Tribunal (NGT)	**Central Empowered Committee (CEC)**	Specialized statutory environmental court established under NGT Act 2010 with judicial and expert members deciding civil environmental disputes vs Supreme Court monitoring committee overseeing forest conservation and wildlife sanctuary clearances	राष्ट्रीय हरित अधिकरण (NGT, सांविधिक पीठ) / केंद्रीय अधिकार प्राप्त समिति (CEC)
National Environmental Engineering Research Institute (NEERI)	**National Institute of Oceanography (NIO)**	CSIR premier lab specializing in water/air treatment, environmental engineering, and monitoring (founded 1958, Nagpur) vs CSIR oceanographic lab researching marine resources and ecology (Dona Paula, Goa)	नीरी (NEERI, नागपुर) / राष्ट्रीय समुद्र विज्ञान संस्थान (NIO, गोवा)

16.1 Ministry of Environment, Forest and Climate Change (MoEFCC)

Function	Body
Category A EC	**MoEFCC**
Forest clearance	**MoEFCC**
NAAQS standards	**CPCB**
Category B EC	**SEIAA**
EC appeals	**NGT**

16.2 Central Pollution Control Board (CPCB) (केंद्रीय प्रदूषण नियंत्रण बोर्ड) & State Pollution Control Board (SPCB) (राज्य प्रदूषण नियंत्रण बोर्ड)

SPCB

Term	Body	Purpose
Consent (CTE/CTO)	**SPCB**	Pollution control for industry operation
Environmental Clearance	**MoEFCC/SEIAA**	Project impact assessment
Forest Clearance	**MoEFCC**	Forest land diversion

16.6 Zoological Survey of India (ZSI) (भारतीय प्राणी सर्वेक्षण)

Body	Est.	HQ	Kingdom
BSI	1890	Kolkata	**Plants**
ZSI	1916	Kolkata	**Animals**
WII	1982	Dehradun	Wildlife research
FSI	1981	Dehradun	Forest cover

16.8 National Environmental Engineering Research Institute (NEERI)

Body	HQ	Job
NEERI	**Nagpur**	Env. engineering research (CSIR)
CPCB	New Delhi	Pollution standards / monitoring
NGT	New Delhi	Environmental adjudication

Common Traps — Don’t Fall For These

Trap	Correct fact
CPCB grants EC	**MoEFCC/SEIAA**
CPCB publishes ISFR	**FSI**
CPCB prepares groundwater report	**CGWB**
MoEFCC created 1972	**1985**
BSI surveys animals	**Plants only**
ZSI surveys plants	**Animals only**
NGT grants EC	**Hears appeals**
ISFR is annual	**Biennial**
BSI/ZSI in Dehradun	**Kolkata**
SPCB grants Category B EC	**SEIAA**

Topic 17 — Environmental Laws & Policies

Current Affairs (this topic)

- New Acts / Rules / NGT orders are living CA; WPA / EPA / FCA year spine stays fixed.
- Do not confuse NGT with NBA (biodiversity) on acronym stems.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Environment (Protection) Act (EPA 1986)	**Wildlife (Protection) Act (WPA 1972)**	Umbrella legislation enacted under Article 253 post-Bhopal disaster enabling the Union to frame rules for EIA, CRZ, ESZ, and hazardous waste vs dedicated legislation protecting wild animal/plant species, regulating hunting, and establishing National Parks/Sanctuaries	पर्यावरण संरक्षण अधिनियम 1986 (छाता कानून) / वन्यजीव संरक्षण अधिनियम 1972
Forest (Conservation) Act (FCA 1980)	**Forest Rights Act (FRA 2006)**	Central law barring de-reservation or diversion of forest land for non-forest purposes without prior Union MoEFCC approval vs rights-based law administered by Ministry of Tribal Affairs restoring land tenure and customary rights to forest-dwelling STs and OTFDs	वन संरक्षण अधिनियम 1980 (विकास नियंत्रण) / वन अधिकार अधिनियम 2006 (जनजातीय अधिकार)
Water (Prevention and Control of Pollution) Act 1974	**Air (Prevention and Control of Pollution) Act 1981**	India's first dedicated environmental pollution statute that established CPCB and SPCBs under Article 252 vs legislation enacted under Article 253 to implement 1972 Stockholm decisions that empowered existing Water Boards to regulate air pollution and noise	जल अधिनियम 1974 (प्रदूषण बोर्डों का गठन) / वायु अधिनियम 1981 (स्टॉकहोम अनुपालन)
Biological Diversity Act (BDA 2002)	**Protection of Plant Varieties and Farmers' Rights Act (PPVFRA 2001)**	Implements 1992 CBD via three-tier structure (NBA, SBB, BMC) controlling commercial bio-resource access and benefit sharing (ABS) vs sui generis IPR regime granting legal rights to commercial plant breeders while safeguarding traditional farmers' seed-saving rights	जैव विविधता अधिनियम 2002 / पौधा किस्म व कृषक अधिकार संरक्षण 2001
Genetic Engineering Appraisal Committee (GEAC)	**Central Ground Water Authority (CGWA)**	Statutory apex regulatory body constituted under Rules 1989 of EPA 1986 approving environmental release of GMOs (e.g. Bt Cotton) vs regulatory authority constituted under Section 3(3) of EPA 1986 regulating commercial groundwater extraction NOCs	जीईएसी (GEAC, जीएमओ नियामक) / केंद्रीय भूजल प्राधिकरण (CGWA)
Coastal Regulation Zone (CRZ)	**Eco-Sensitive Zone (ESZ)**	Coastal belt up to 500 m from High Tide Line classified into CRZ-I to IV under EPA 1986 to regulate beach and marine development vs buffer belt around protected areas notified under Section 3 of EPA 1986 to curb polluting industrial operations	तटीय विनियमन क्षेत्र (CRZ) / पर्यावरण-संवेदनशील क्षेत्र (ESZ)

17.1 Environmental Law (Constitutional Framework)

Article	Type	Content
48A	DPSP	State protects environment
51A(g)	Fundamental Duty	Citizen protects environment
21	FR (expanded)	Healthy environment as part of life
48	DPSP	Agriculture — **not** environment

17.3 Wildlife (Protection) Act, 1972 & Wildlife Protection Laws

Schedule	Content
I	Highest protection — severe penalties
II–IV	Protected species — hunting restricted
V	**Vermin** — may be hunted
VI	Protected **plants**

17.4 Water (Prevention and Control of Pollution) Act, 1974 & Air Act, 1981

Air Act 1981

Act	Year	Key output
Water Act	**1974**	CPCB, SPCB, water consent
Air Act	**1981**	NAAQS, air pollution control

17.5 Forest (Conservation) Act, 1980 | Forest Rights Act, 2006 | National Forest Policy, 1952

National Forest Policy, 1952

Act	Purpose
FCA 1980	Stop forest **diversion** without Central approval
FRA 2006	Recognize forest **dweller rights**
NFP 1952	Early policy — revenue focus

17.6 Biological Diversity Act, 2002 & National Green Tribunal Act, 2010

NGT Act 2010

Act	Year	Key body
BDA	2002	NBA Chennai
NGT Act	2010	NGT Delhi

Common Traps — Don’t Fall For These

Trap	Correct fact
Water Act = 1981	**Water 1974; Air 1981**
EPA = 1972	**EPA 1986; WPA 1972**
FCA = FRA	**1980 diversion vs 2006 rights**
Art 48 = environment	**Art 48A**
Schedule V highest protection	**Schedule I**
EIA under Water Act	**EPA 1986**
NGT grants EC	**MoEFCC/SEIAA**

Trap	Correct fact
NBA HQ Delhi	**Chennai**
NFP 1988 in syllabus list	Syllabus says **1952** — know both
Consent = EC	**SPCB consent ≠ EC**

Topic 18 — International Environmental Agreements & Conferences

Current Affairs (this topic)

- COP outcomes and new listings refresh CA; treaty years and Montreal≠Kyoto stay fixed.
- Ramsar / CITES / CMS updates — verify live lists; UP Ramsar baseline remains **12**.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Stockholm Conference (1972)	**Rio Earth Summit (1992)**	First UN global conference on the Human Environment that placed environment on the global political agenda and created UNEP vs UN Conference on Environment and Development (UNCED) that produced Rio Declaration, Agenda 21, CBD, UNFCCC, and UNCCD	स्टॉकहोम सम्मेलन 1972 (UNEP उद्भव) / रियो पृथ्वी शिखर सम्मेलन 1992 (रियो संधियां)
Kyoto Protocol (1997)	**Paris Agreement (2015)**	Annex-I developed nations bound by legal emission targets with punitive compliance mechanisms vs universal bottom-up framework where all 196+ nations self-determine mitigation contributions (NDCs) with five-yearly global stocktakes	क्योटो प्रोटोकॉल (शीर्ष-से-नीचे बाध्यकारी) / पेरिस समझौता (स्वैच्छिक NDCs)
Basel Convention (1989)	**Rotterdam Convention (1998)**	Restricts and regulates transboundary movements and disposal of hazardous wastes (especially preventing dumping in developing countries) vs establishes legally binding Prior Informed Consent (PIC) procedure for hazardous international trade chemicals and pesticides	बासेल कन्वेंशन (खतरनाक अपशिष्ट आवागमन) / रॉटरडैम कन्वेंशन (पूर्व सूचित सहमति)
Stockholm Convention (2001)	**Minamata Convention (2013)**	Mandates the global elimination, restriction, and unintentional release reduction of Persistent Organic Pollutants (POPs - "Dirty Dozen") vs global treaty designed to protect human health and the environment from anthropogenic emissions and releases of toxic Mercury (Hg)	स्टॉकहोम संधि (POPs उन्मूलन) / मिनामाता संधि (पारा प्रदूषण नियंत्रण)
Ramsar Convention (1971)	**Convention on Migratory Species (CMS / Bonn 1979)**	Dedicated specifically to the ecological conservation and sustainable wise use of internationally significant wetlands and waterfowl habitats vs treaty dedicated to the conservation of terrestrial, aquatic, and avian migratory wild animal species across range states	रामसर कन्वेंशन (आर्द्रभूमि संरक्षण) / बॉन कन्वेंशन (प्रवासी प्रजाति संरक्षण)
CITES (1973)	**Convention on Biological Diversity (CBD 1992)**	International trade treaty strictly regulating commercial import/export of endangered wild flora and fauna through Appendices I, II, III vs comprehensive framework convention covering in-situ/ex-situ conservation, sustainable use, and fair benefit-sharing	साइट्स (CITES, वन्यजीव व्यापार) / जैव विविधता संधि (CBD, समग्र संरक्षण)

18.5 Chemicals & Waste Treaties — Basel, Rotterdam, Stockholm POPs & Minamata

Treaty	Targets
Basel	Hazardous **waste** shipment
Rotterdam	Hazardous **chemicals** trade (PIC)
Stockholm POPs	Persistent organic pollutants
Minamata	**Mercury**

Mains Answer Framework

UKPCS Prelims 2025

COP session	Location
1. COP 27	A. Belem
2. COP 28	B. Baku
3. COP 29	C. Dubai
4. COP 30	D. Sharm El-Sheikh

Common Traps — Don’t Fall For These

Trap	Correct fact
Montreal = climate treaty	**Ozone/ODS**
Agenda 21 signed 1995	**Rio 1992**
Brundtland = Limits to Growth	***Our Common Future* 1987**
CITES born at Rio 1992	**1973 Washington**
Ramsar = forests	**Wetlands**
CMS = CITES	**Migration vs trade**
Vienna = binding ODS phase-out	**Framework; Montreal = binding**
Kyoto = Paris model	**Annex binding vs universal NDCs**
Stockholm POPs = Stockholm 1972	**2001 POPs treaty — different**
UNEP HQ Geneva	**Nairobi**

Topic 19 — Climate & Environmental Institutions

Current Affairs (this topic)

- IPCC AR cycle and UNEP / WMO headlines are institutional CA.
- Keep UNFCCC (यूएनएफसीसीसी) Secretariat vs IPCC role split clear.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Intergovernmental Panel on Climate Change (IPCC)	**UN Framework Convention on Climate Change (UNFCCC)**	Scientific assessment panel established in 1988 by WMO and UNEP that reviews peer-reviewed literature (does not conduct research or negotiate policy) vs international legal treaty and negotiating secretariat that coordinates annual Conference of Parties (COPs)	आईपीसीसी (वैज्ञानिक समीक्षा निकाय) / यूएनएफसीसीसी (वार्ता व संधि सचिवालय)
United Nations Environment Programme (UNEP)	**United Nations Development Programme (UNDP)**	Apex global environmental authority coordinating UN environmental activities, hosting MEA secretariats (HQ: Nairobi, Kenya) vs UN global development network focusing on poverty reduction, sustainable growth, and publishing the Human Development Report (HQ: New York)	यूएनईपी (नैरोबी, पर्यावरण एजेंसी) / यूएनडीपी (न्यूयॉर्क, विकास एजेंसी)
International Union for Conservation of Nature (IUCN)	**World Wide Fund for Nature (WWF)**	World's oldest and largest global conservation union composed of governments and civil society (1948, Gland) publishing the IUCN Red List vs independent international private conservation NGO (1961, Gland) famous for the Giant Panda symbol and Earth Hour	आईयूसीएन (रेड लिस्ट प्रकाशक) / डब्ल्यूडब्ल्यूएफ (पांडा प्रतीक एनजीओ)
Global Environment Facility (GEF)	**Green Climate Fund (GCF)**	Multilateral financial partnership established in 1991 serving as financial mechanism for 5 MEAs (CBD, UNCCD, Stockholm, Minamata, UNFCCC) vs dedicated operating financial mechanism of the UNFCCC established at COP16 Cancun focusing specifically on climate projects in developing nations	जीईएफ (GEF, बहु-संधि कोष) / जीसीएफ (GCF, हरित जलवायु कोष)
Food and Agriculture Organization (FAO)	**World Meteorological Organization (WMO)**	Specialized UN agency leading international efforts to defeat hunger, maintain agricultural biodiversity, and soil health (HQ: Rome) vs specialized UN agency dedicated to atmospheric science, climatology, and operational hydrology (HQ: Geneva)	एफएओ (रोम, खाद्य व कृषि) / डब्लूएमओ (जिनेवा, मौसम विज्ञान)
International Solar Alliance (ISA)	**International Renewable Energy Agency (IRENA)**	Intergovernmental treaty-based organization launched jointly by India and France at COP21 Paris to promote solar energy deployment (HQ: Gurugram, India) vs intergovernmental organization supporting countries in their transition to all renewable energy types (HQ: Abu Dhabi, UAE)	अंतरराष्ट्रीय सौर गठबंधन (ISA, गुरुग्राम) / इरेना (IRENA, अबू धाबी)

19.1 IPCC

Body	Role
IPCC	**Science assessment**
UNFCCC	**Treaty negotiations**

19.5 WWF

Org	Output
IUCN	**Red List**
WWF	**Living Planet Report**

Common Traps — Don’t Fall For These

Trap	Correct fact
IPCC sets NDCs	**UNFCCC/COP**
IPCC = UNFCCC	**Science vs treaty**
UNEP HQ Geneva	**Nairobi**
Emissions Gap = IPCC/UNDP	**UNEP**
HDR = UNEP	**UNDP**
HDI team included Priesner	**Haq + Sen**
IUCN regulates trade	**CITES**
WWF = IUCN	**Same city, different roles**
FAO HQ Nairobi	**Rome**
GEF donors only developed	**Stmt 2 false in Q145**

Topic 20 — Biodiversity Conservation Methods

Current Affairs (this topic)

- 30×30 / Kunming–Montreal CA sits here with in-situ vs ex-situ method traps.
- Sacred grove and seed-bank examples stay teaching baselines.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
In-situ Conservation	**Ex-situ Conservation**	On-site conservation of wild populations within their natural ecosystems where evolutionary processes continue uninterrupted (National Parks, Wildlife Sanctuaries, Sacred Groves) vs off-site conservation of threatened components outside natural habitats in human custody (Botanical Gardens, Zoological Parks, Cryobanks)	स्व-स्थाने संरक्षण (प्राकृतिक आवास) / बाह्य-स्थाने संरक्षण (मानव-निर्मित व्यवस्था)
Botanical Garden	**Herbarium**	Ex-situ facility maintaining curated collections of *living* plant species for scientific research, propagation, and conservation display vs reference repository of *dried, pressed, and mounted* plant specimens preserved on archival sheets for taxonomic identification	वनस्पति उद्यान (जीवित पादप संग्रह) / पादप संग्रहालय (हर्बेरियम, शुष्क नमूने)
Zoological Park (Zoo)	**Safari / Wildlife Sanctuary**	Ex-situ facility where wild animals are maintained in artificial enclosures under human management for captive breeding and public education vs in-situ natural protected habitat where free-ranging wild animals live in untamed wild conditions	चिड़ियाघर (कृत्रिम बाड़ा) / वन्यजीव अभयारण्य / सफारी
Orthodox Seeds	**Recalcitrant Seeds**	Desiccation-tolerant seeds that can survive dehydration to low moisture content (≤5%) and sub-zero freezing, ideal for conventional long-term seed banking (wheat, rice, legumes) vs desiccation-sensitive seeds that lose viability rapidly if dried or frozen, requiring cryopreservation (mango, rubber, jackfruit, tea)	सामान्य (रूढ़िवादी) बीज / कठिन (असहनीय) बीज
Cryopreservation	**Seed Bank**	Ultra-low temperature preservation of living cells, gametes, embryos, and tissues suspended in liquid nitrogen at **-196°C** stopping all metabolic decay vs storage of dried orthodox seeds in vault freezers at standard temperatures of -18°C to -20°C	क्रायो-संरक्षण (-196°C द्रवित नाइट्रोजन) / बीज बैंक (-18°C शुष्क भंडारण)
Sacred Groves	**Biosphere Reserves**	Community-protected virgin forest patches preserved through traditional spiritual beliefs and religious taboos (e.g. Deorais, Orans, Kavu, Law Kyntang) vs modern scientific statutory conservation landscape demarcated by government under UNESCO MAB	पवित्र उपवन (पारंपरिक संरक्षण) / जैवमंडल आरक्षित क्षेत्र (वैज्ञानिक प्रारूप)
National Bureau of Plant Genetic Resources (NBPGR)	**National Bureau of Animal Genetic Resources (NBAGR)**	Premier ICAR institute managing the National Gene Bank for agricultural crop and wild relative germplasm (HQ: New Delhi) vs premier ICAR institute maintaining registrations and genetic databases of indigenous livestock breeds (HQ: Karnal, Haryana)	एनबीपीजीआर (नई दिल्ली, पादप जीन बैंक) / एनबीएजीआर (करनाल, पशु आनुवंशिकी)

20.3 Gene Bank

Institution	Location	Conserves
ICAR-NBPGR	New Delhi	Plant seeds, pollen, tissue, DNA
National Gene Bank	New Delhi (NBPGR)	Long-term base collection
NBAGR	Karnal, Haryana	Livestock semen, embryos, DNA
Svalbard Global Seed Vault	Norway	Duplicate global seed backup

20.4 Seed Bank

Type	Storage	Examples
Orthodox	Conventional seed bank (-18°C)	Wheat, rice, chickpea
Recalcitrant	Living collection or cryo	Mango, coconut, jackfruit
Feature	Seed Bank	Gene Bank
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Material	Seeds primarily	Seeds + DNA + tissue + sperm + embryos
Scope	Narrower	Broader
Both are	**Ex-situ**	**Ex-situ**

20.5 Botanical Garden

Garden	Location	Notable
Indian Botanic Garden	Howrah	1787; Great Banyan; largest in India
Lalbagh	Bengaluru	1760; heritage trees
NBRI Garden	Lucknow	Medicinal plant research

20.6 Zoological Park

Facility	Location	Type
National Zoological Park	New Delhi	Zoo (ex-situ)
CZA	New Delhi	Regulator (MoEFCC)
Kukrail Crocodile Centre	Lucknow, UP	Captive breeding (ex-situ)
Katerniaghat WLS	UP	Sanctuary (**in-situ** contrast)

Common Traps

Trap	Correct
Zoo = in-situ	**Ex-situ** — captive, off-site
Seed bank = in-situ	**Ex-situ** — cold storage off-site
Botanical garden = in-situ	**Ex-situ** — managed living collection
NBPGR = Kolkata	**New Delhi** — Kolkata = BSI/ZSI
NBAGR = Delhi or NBPGR	**Karnal** — animals only

Trap	Correct
NBPGR stores animal semen	**NBAGR Karnal**
BSI = botanical garden	BSI **surveys** flora; garden is a **facility**
Zoos are primary conservation	**In-situ PAs preferred**
Mango in standard seed bank	**Recalcitrant** — living collection/cryo
Svalbard vault in India	**Norway** — duplicate backup only
Sanctuary = zoo	Sanctuary = **wild in-situ**; zoo = **captive ex-situ**
Gene bank = in-situ farmer field	Formal gene bank = **ex-situ** storage

Topic 21 — Species & Ecology

Current Affairs (this topic)

- New IUCN category moves for flagship species (प्रतीक (फ्लैगशिप) प्रजाति) are the usual CA refresh.
- Keystone / umbrella / flagship / indicator definitions stay concept-fixed.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
IUCN Red List	**Red Data Book of India**	Global extinction risk assessments maintained by IUCN (HQ: Gland, Switzerland) categorizing global species from LC to EX vs national taxonomic inventories published by BSI (plants) and ZSI (animals) cataloguing Indian threatened flora/fauna	आईयूसीएन रेड लिस्ट (वैश्विक) / भारतीय रेड डेटा बुक (राष्ट्रीय - BSI/ZSI)
Threatened Species	**Endangered Species (EN)**	Broader collective umbrella category encompassing three distinct risk levels: Vulnerable (VU) + Endangered (EN) + Critically Endangered (CR) vs a specific single IUCN conservation status indicating very high risk of extinction	संकटग्रस्त प्रजातियां (VU + EN + CR समग्र) / लुप्तप्राय प्रजाति (EN विशिष्ट वर्ग)
Mangrove Ecosystem	**Coral Reef Ecosystem**	Salt-tolerant halophytic coastal wetland forests adapted to intertidal muddy sediments featuring pneumatophores (Rhizophora, Avicennia) vs marine calcium carbonate structures formed by symbiotic coral polyps and zooxanthellae in clear, shallow, warm waters	मैंग्रोव पारितंत्र (श्वसन मूल वन) / प्रवाल भित्ति पारितंत्र (उष्ण जलीय प्रवाल)
Pneumatophores	**Stilt Roots**	Specialized vertical negative geotropic aerial roots that emerge above mud to facilitate oxygen gas exchange in waterlogged swamp soils (Avicennia) vs adventitious support roots emerging from lower trunk/branches providing mechanical anchor against coastal tidal surges (Rhizophora)	वात-मूल (श्वसन मूल, न्यूमेटोफोर) / अवस्तंभ मूल (सहारा जड़ें)
Coral Bleaching	**Ocean Acidification**	Stress response where coral polyps expel their symbiotic photosynthetic zooxanthellae due to elevated sea surface temperatures turning white vs chemical lowering of ocean pH caused by oceanic absorption of excess atmospheric CO_2 dissolving calcium carbonate aragonite structures	प्रवाल विरंजन (ऊष्मा तनाव से शैवाल निष्कासन) / महासागरीय अम्लीकरण (pH ह्रास)
Neritic Marine Zone	**Oceanic (Pelagic) Zone**	Shallow coastal marine zone extending from low-tide mark to the edge of the continental shelf (depth up to 200 m) with abundant light and high marine productivity vs deep open marine waters beyond the continental shelf edge where nutrients and light drop precipitously	नेरिटिक मंडल (महाद्वीपीय मग्नतट) / खुला महासागरीय मंडल (पेलाजिक)
Zooxanthellae	**Coral Polyp**	Photosynthetic single-celled dinoflagellate microalgae living symbiotically within coral tissues providing up to 90% of their organic nutrients vs tiny, soft-bodied marine invertebrate animal belonging to Phylum Cnidaria that secretes a hard calcium carbonate skeleton	ज़ूजैंथेली (सहजीवी प्रकाश-संश्लेषी शैवाल) / प्रवाल पॉलीप (कैल्शियम स्रावी जंतु)

21.1 Red Data Book

Feature	Red Data Book (India)	IUCN Red List
Scope	National	Global
Publisher	BSI (plants), ZSI (animals)	IUCN (Gland)
Function	National threat inventory	Standardised risk categories

21.6 Grasslands

Type	Location	Key species
Terai grassland	UP foothills	Barasingha, tiger — Dudhwa
Floodplain grassland	Assam	Rhino — Kaziranga
Semi-arid savanna	Deccan	Blackbuck
Alpine meadow	High Himalaya	Medicinal herbs

21.8 Marine Ecosystem

Zone	Character	Productivity
Intertidal	Tide-exposed shore	Moderate (नरम दल)
Neritic	Continental shelf	**High**
Pelagic	Open ocean	Low–moderate
Benthic	Seafloor	Variable

Common Traps

Trap	Correct
Threatened = Endangered only	**VU + EN + CR**
Red Data Book = IUCN Red List	**National vs global**
BSI = animal Red Data Book	**BSI = plants; ZSI = animals**
IUCN = CITES	**Assessment vs trade**
Sultanpur Ramsar in UP	**Haryana (Gurugram)**
Rudrasagar Ramsar in UP	**Tripura**
MISHTI launched 2019	**2023–24 Budget**
Coral bleaching from cooling	**Heat (+2°C) stress**
Mangroves = freshwater wetland only	**Coastal saline**
Gangetic dolphin = marine	**Freshwater river**
Zoo/sanctuary same as wetland	Different ecosystems
EW = EX	**EW = captive survival remains**

Topic 22 — Renewable Energy

Current Affairs (this topic)

- India's renewable capacity and green-hydrogen headlines refresh energy CA.
- Keep renewable vs clean vs conventional classification traps stable.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Solar Photovoltaic (PV)	**Concentrated Solar Power (CSP)**	Semiconductor solar cells that convert incident sunlight photon energy directly into direct-current electricity via the photoelectric effect vs optical mirrors/lenses that concentrate solar thermal energy to heat a fluid and drive a steam turbine	सौर फोटोवोल्टिक (सीधा विद्युत उत्पादन) / संकेंद्रित सौर ताप (CSP, टरबाइन चालित)
Green Hydrogen	**Grey Hydrogen / Blue Hydrogen**	Produced by zero-carbon electrolysis of water powered entirely by renewable electricity (solar/wind) vs produced via steam methane reforming of natural gas releasing unabated CO_2 (Grey) or with carbon capture and storage (Blue)	हरित हाइड्रोजन (शून्य कार्बन जल अपघटन) / धूसर व नीला हाइड्रोजन (जीवाश्म स्रोत)
Biomass Energy	**Biogas**	Direct thermal or electrical energy produced by combusting solid organic matter (bagasse, rice husk, wood pellets) vs gaseous fuel mixture consisting primarily of methane (50–70%) and CO_2 generated through anaerobic microbial digestion of organic waste	बायोमास ऊर्जा (ठोस जैविक ईंधन) / बायोगैस (अवायवीय पाचन, मीथेन प्रधान)
Geothermal Energy	**Ocean Thermal Energy (OTEC)**	Harnesses natural high-temperature hydrothermal heat and steam from Earth's deep crustal magma reservoirs (e.g. Puga Valley, Ladakh; Manikaran, HP) vs exploits the temperature differential between warm tropical surface ocean water and freezing deep seawater (≥20°C gradient)	भू-तापीय ऊर्जा (पृथ्वी का आंतरिक ताप) / महासागरीय तापीय ऊर्जा (OTEC, सतही-गहन ताप भेद)
Small Hydro Power (SHP)	**Large Hydro Power**	Hydropower installations with individual installed capacity up to and including **25 MW** administered under the Ministry of New and Renewable Energy (MNRE) vs large dam-based hydropower plants >25 MW administered under the Ministry of Power	लघु जल विद्युत (≤25 MW, MNRE) / बड़ी जल विद्युत परियोजनाएं (>25 MW)
International Solar Alliance (ISA)	**International Renewable Energy Agency (IRENA)**	Treaty-based intergovernmental solar coalition initiated by India and France at COP21 Paris with global HQ in Gurugram, India vs global renewable energy intergovernmental agency covering all renewables founded in 2009 with HQ in Abu Dhabi, UAE	अंतरराष्ट्रीय सौर गठबंधन (ISA, गुरुग्राम) / इरेना (IRENA, अबू धाबी)

22.2 Wind Energy

Feature	Onshore	Offshore
Cost	Lower	Higher
Wind quality	Moderate (नरम दल)	Stronger, steadier
India status	Mature	Early policy stage

22.7 Green Hydrogen

Type	Source	CO_2
Green	Renewable electrolysis	Very low

Type	Source	CO_2
Grey	Natural gas	High
Blue	Gas + CCS	Moderate

Common Traps

Trap	Correct
Biogas = butane + CO_2	**Methane + CO_2**
ISA HQ Paris/New Delhi	**Gurugram**
ISA exactly 38 members / all UN	**Outdated / false**
Rajasthan leads wind	**Tamil Nadu wind; Rajasthan solar**
Geothermal from Sun	**Earth interior heat**
Geothermal in Ganga Delta	**Not** a listed province
Tidal = wave energy	**Moon gravity vs wind waves**
Tidal potential in Thar	**Khambhat/Kutch coast**
SHP = 50 MW	**≤25 MW**
Nepal border = zero Indian hydro	**False — Q31**
Grey hydrogen = green	**Grey = fossil gas**
Largest solar park = always Gujarat	**Bhadla, Rajasthan**
Fuel cell = AC	**DC**; heat + water
DCR = TV licences	**Domestic solar PV content**
India #3 silicon wafers	**Not** a wafer leader
Green H_2 Mission 2019	**January 2023**
Nuclear = renewable in MCQs	**Usually non-renewable category**
Singrauli = hydro	**Thermal power hub**

Topic 23 — Disaster & Environment

Current Affairs (this topic)

- Flood / landslide / heat-wave seasons refresh disaster–environment CA for UP–Himalaya (हिमालय).
- Sendai Framework year (**2015**) stays fixed — link hazards to land-use and climate stress.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Meteorological Drought	**Agricultural Drought**	Situation where actual seasonal rainfall over an area is significantly less than the long-term climatological mean (>25% deficient = moderate; >50% deficient = severe) vs condition where soil moisture deficiency severely impedes crop growth, triggering wilting and agrarian loss	मौसमी (मौसम विज्ञान) सूखा / कृषि सूखा (मृदा नमी संकट)
Hydrological Drought	**Socio-Economic Drought**	Prolonged meteorological drought leading to depletion of surface water reservoirs, dried-up streams, and falling groundwater aquifer tables vs severe water scarcity disrupting economic goods, public water supply, and livelihoods	जल-वैज्ञानिक सूखा (जलाशय ह्रास) / सामाजिक-आर्थिक सूखा (आजीविका संकट)
Cloudburst	**Urban Flash Flood**	Extreme, highly localized torrential rainfall event exceeding **100 mm per hour** over a small geographic area (~20–30 km^2) causing catastrophic flash torrents vs rapid inundation of built-up city areas caused by impermeable concrete surfaces and choked storm drains	बादल फटना (≥100 मिमी/घंटा अतिवृष्टि) / नगरीय आकस्मिक बाढ़
Storm Surge	**Tsunami**	Abnormal coastal sea-level rise and massive sea surge pushed onshore by the fierce winds and extreme low atmospheric pressure of a tropical cyclone vs series of immense sea waves generated by underwater seaquake, submarine volcanic eruption, or seafloor tectonic dislocation	तूफानी महोर्मि (चक्रवात जनित उभार) / सुनामी (अंतःसमुद्री भूकंप जनित तरंग)
National Disaster Management Authority (NDMA)	**National Institute of Disaster Management (NIDM)**	Apex statutory policy-making and executive body for disaster management established under DM Act 2005, chaired ex-officio by the Prime Minister vs premier national statutory institute responsible for capacity building, research, training, and documentation in disaster management	राष्ट्रीय आपदा प्रबंधन प्राधिकरण (NDMA, नीति पीठ) / राष्ट्रीय आपदा प्रबंधन संस्थान (NIDM, प्रशिक्षण)
Hazard	**Vulnerability**	Dangerous physical event, phenomenon, or human activity that has the potential to cause loss of life, injury, or property damage (e.g. earthquake, cyclone) vs conditions determined by physical, social, economic, and environmental factors increasing susceptibility to hazards	आपदा जोखिम/खतरा (Hazard) / सुभेद्यता (Vulnerability)

23.1 Floods

Type	Speed	UP example
Riverine	Days–weeks	Ganga at Prayagraj (प्रयागराज)/Varanasi (वाराणसी)
Flash	Hours	Terai foothills
Urban	Hours	Lucknow (लखनऊ), Kanpur (कानपुर) waterlogging

23.5 Heat Waves

Feature	Heat wave	Urban heat island
Scale	Regional weather event	Local city microclimate
Duration	Days–weeks	Persistent year-round bias
Cause	Atmospheric pattern	Built surfaces, lost greenery

Common Traps

Trap	Correct
DPAP = 1952	**1973** (CDP = 1952)
Strongest winds in cyclone eye	**Eyewall**
Landslides purely natural	**Deforestation/road cutting trigger**
Heat wave = urban heat island	**Different scale/cause**
All drought = only rainfall deficit	Also **agricultural/hydrological**
Piral Lao = UP scheme	**Uttarakhand**
India disaster-free	**False**
Arabian Sea > Bay for cyclones	**Bay of Bengal more**
Floods only coastal	**UP riverine/Terai major**
Mangroves worsen cyclones	**Reduce surge/wind damage**
NDMA Chair = Home Minister	**Prime Minister**
Sendai = ozone treaty	**Disaster risk reduction 2015**

Topic 24 — Current Environmental Issues

Current Affairs (this topic)

- This chapter is the CA layer — keep NbS, blue economy, and plastics stems current.
- UP is landlocked: do not invent statewide blue-economy projects.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Primary Microplastics	**Secondary Microplastics**	Microscopic plastic particles (<5 mm) intentionally manufactured for commercial applications (microbeads in facial scrubs, toothpaste, virgin industrial resin pellets) vs microplastics formed by physical weathering, photo-degradation, and fragmentation of macro-plastics in the environment	प्राथमिक माइक्रोप्लास्टिक (निर्मित मनके) / द्वितीयक माइक्रोप्लास्टिक (अपघटित कण)
Carbon Credit	**Green Credit (2023)**	Tradable compliance unit representing 1 metric tonne of CO_2 equivalent emission reduction or removal registered under carbon markets vs incentives under India's Green Credit Programme (2023) rewarding diverse eco-actions (afforestation, water harvesting, mangrove conservation) beyond carbon	कार्बन क्रेडिट (1 टन CO_2e मानक) / ग्रीन क्रेडिट (पर्यावरण कार्य प्रोत्साहन)
Net Zero Emissions	**Gross Zero Emissions**	Balancing residual greenhouse gas emissions across all sectors with equivalent anthropogenic carbon removals and sequestration by 2050/2070 vs absolute elimination of all greenhouse gas emissions to absolute zero without relying on offsets or sinks	नेट जीरो (उत्सर्जन-अवशोषण संतुलन) / सकल शून्य (पूर्ण उत्सर्जन शून्यता)
Blue Economy	**Blue Carbon**	Sustainable utilization of ocean, coastal, and marine resources for economic growth, improved livelihoods, and jobs while preserving marine ecosystem health vs organic carbon captured, sequestered, and stored over centuries by coastal marine ecosystems (mangroves, seagrasses, salt marshes)	नीली अर्थव्यवस्था (महासागरीय संधारणीय विकास) / नीला कार्बन (तटीय मैंग्रोव संचित कार्बन)
Green Finance	**Greenwashing**	Structured financial investments, bonds, and banking loans directed specifically towards environmentally sustainable development and clean energy projects vs deceptive marketing, publicity, or PR spin falsely exaggerating a company's environmental credentials and sustainability	हरित वित्त (टिकाऊ निवेश) / ग्रीनवॉशिंग (झूठा पर्यावरण प्रचार)
Mission LiFE (2022)	**Lifestyle for Environment**	Global citizen movement launched in October 2022 at Kevadia by PM Modi and UN Secretary-General mobilizing individuals to become 'Pro-Planet People' (P3) through 75 actionable behavioral changes vs abstract corporate environmental pledges	मिशन लाइफ (LiFE, P3 नागरिक आंदोलन) / जीवनशैली पर्यावरण

24.1 Microplastics

Type	Origin	Example
Primary	Manufactured small	Microbeads, nurdles
Secondary	Large plastic breaks down	Bag fragments, tyre particles

24.4 Net Zero Emissions

Country	Net zero year
EU / UK	2050

Country	Net zero year
China	2060
India	**2070**

24.5 Green Finance

Instrument	India milestone
Sovereign Green Bonds	FY **2022–23**
SEBI Green Bond Framework	**2023**
Loss and Damage Fund	**COP28 2023**

Common Traps

Trap	Correct
LiFE idea at COP-25 Madrid	**COP26 Glasgow 2021**
LiFE launched 2019	**June 2022**
MISHTI launched 2019	**2023–24 Budget**
India net zero 2050	**2070**
Microplastics <5 cm	**<5 mm**
Microplastics biodegrade quickly	**Persist decades**
Carbon credit = carbon trading	**Unit vs market**
Green Credit = carbon credit	**Different schemes**
Green finance = Green Credit	**Capital vs incentive certificate**
Montreal = climate treaty	**Ozone — Kyoto/Paris = climate**
PAT trades carbon credits directly	**ESCerts — energy efficiency**
Net zero = gross zero tomorrow	**Balanced with removals**
MISHTI for Ganga/UP	**Coastal mangroves**

Topic 25 — Global Environmental Geography

Current Affairs (this topic)

- Biome (जीवोम) and realm stems stay geographic; CA rarely moves the Köppen (कोपेन) or biome map.
- Bridge biome teaching with Ecology (पारिस्थितिकी) & Ecosystem (पारिस्थितिकी तंत्र) without duplicating full cards.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Biome	**Ecosystem**	Continental-scale geographical region characterized by distinct climate regimes, soil types, and climax vegetation communities (e.g. Tundra, Taiga, Savanna) vs localized structural and functional ecological unit where living organisms interact with abiotic components (e.g. a pond, forest patch)	जीवोम (बायोम, महाद्वीपीय स्तर) / पारिस्थितिकी तंत्र (स्थानीय कार्यात्मक इकाई)
Taiga (Boreal Forest)	**Tundra**	World's largest terrestrial biome dominated by evergreen needleleaf coniferous trees (Spruce, Fir, Pine, Larch) in subarctic latitudes south of the tree-line vs cold, treeless arctic and alpine biome dominated by mosses, lichens, dwarf shrubs, underlain by permafrost	टैगा (बोरियल शंकुधारी वन) / टुंड्रा (वृक्षविहीन शीत मरुस्थल)
Savanna	**Steppe / Prairie**	Tropical grassland biome characterized by continuous tall grass cover interspersed with scattered, drought-deciduous drought-resistant trees (e.g. Acacia, Baobab) vs temperate mid-latitude semi-arid grassland biome with treeless expanses of short to tall grasses	सवाना (उष्णकटिबंधीय सवृक्ष घास मैदान) / स्टेपी/ प्रेयरी (शीतोष्ण घास मैदान)
Selva	**Puszta**	Dense equatorial tropical rainforest biome of the Amazon River basin characterized by evergreen multi-tiered canopy and extreme biodiversity vs temperate flat grassland plain of the Pannonian Basin in Hungary characterized by continental climate and fertile chernozem soils	सेल्वा (अमेज़न वर्षावन) / पुस्ज़टा (हंगरी शीतोष्ण घास मैदान)
Permafrost	**Mollisols (Chernozem)**	Subsurface ground (soil, sediment, or rock) that remains continuously frozen below 0°C for at least two consecutive years, trapping massive ancient methane and organic carbon deposits vs deep, dark, humus-rich organic soils characteristic of temperate grasslands with highest agricultural fertility	स्थायी हिम मृदा (पर्माफ्रॉस्ट) / चेर्नोज़म (काली उपजाऊ घास मृदा)
Biodiversity Hotspot	**Ecological Island**	Large biogeographic region meeting Norman Myers' criteria (≥1,500 endemic plant species and ≥70% primary habitat lost) vs isolated terrestrial habitat surrounded by completely different ecosystems or human urbanization functioning like an oceanic island	जैव विविधता हॉटस्पॉट (ह्रासग्रस्त समृद्ध क्षेत्र) / पारिस्थितिक द्वीप (पृथक वास)
### Biome ↔ ecosystem bridge (read with Ecology chapter)			

- A **biome** is the large climate-defined vegetation–fauna belt (Amazon Selva, taiga, savanna).
- An **ecosystem** (पारिस्थितिकी तंत्र) is the local functional unit of energy flow and nutrient cycling inside that belt.
- Global textbook biome counts (often **~14**) are **not** India's **ten** biogeographic zones (Rodgers and Panwar).
- UP plains sit mainly in the **Gangetic Plain** biogeographic colour with Terai (तराई)–deciduous transitions — not a tropical rainforest biome answer.

25.2 Major Global Biomes

Biome	Climate	Global example
Tropical rainforest	Hot wet year-round	Amazon Selva
Desert	Arid <25 cm	Sahara, Gobi, Thar
Savanna	Wet–dry grassland + trees	East Africa
Temperate grassland	Moderate (नरम दल) rain	Pampas, Puszta
Taiga	Cold coniferous	Siberia, Canada
Tundra	Very cold, permafrost	Arctic

25.6 Biodiversity Hotspots & Degradation Belts

India hotspot	Region
Himalaya	Mountain endemism
Western Ghats	Peninsular rainforest
Indo-Burma	NE India
Sundaland	Nicobar

Common Traps

Trap	Correct
Northern Australia temperate	**Tropical**
Gobi in Russia/Kazakhstan as main	**Mongolia + China**
Masai = West Africa	**East Africa**
Selva false / Terra Roxa false	**Both TRUE**
Engler = Planting Hearth	**Sauer = Hearth; Engler = Floral**
Puszta = Argentina	**Hungary** (Pampas = Argentina)
Permafrost in savanna	**Taiga + Tundra**
Mangrove purely terrestrial	**Transitional coastal**
India has 14 biogeographic zones	**10 zones**
Hotspot = richness alone	**Endemism + habitat loss**

Topic 26 — Water Resources & Water Conservation

Current Affairs (this topic)

- Jal Jeevan / Namami Gange (नमामि गंगे) / wetland CA refresh water chapters.
- Keep watershed vs command-area and BOD water-quality traps stable.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Integrated Water Resources Management (IWRM)	**Watershed Management**	Coordinated macro-framework promoting the balanced development of water, land, and related resources across an entire river basin to maximize economic and social welfare equitably vs micro-level land and water conservation technique focusing on a ridge-to-valley catchment draining to a single outlet	एकीकृत जल संसाधन प्रबंधन (IWRM) / वाटरशेड (जलग्रहण) प्रबंधन
Unconfined Aquifer	**Confined (Artesian) Aquifer**	Permeable water-bearing geological formation whose upper boundary is the free atmospheric water table directly recharged by infiltrating rainfall vs deep aquifer bounded above and below by impermeable aquitards where groundwater is stored under hydrostatic artesian pressure	अप्रतिबंधित जलभृत (मुक्त भूजल स्तर) / प्रतिबंधित (आर्टेशियन) जलभृत
Sewage Treatment Plant (STP)	**Common Effluent Treatment Plant (CETP)**	Treatment facility designed specifically to clean domestic municipal sewage and blackwater from residential settlements vs collective facility built to treat hazardous industrial toxic effluents generated by a cluster of small- and medium-scale industrial factories	मलजल उपचार संयंत्र (STP, घरेलू सीवेज) / साझा उत्प्रवाह उपचार संयंत्र (CETP, औद्योगिक अपशिष्ट)
Rainwater Harvesting (RWH)	**Artificial Groundwater Recharge**	Collection, conveyance, and storage of runoff rainwater from rooftops or ground surfaces into surface storage tanks for immediate usage vs purposeful diversion and injection of harvested surface runoff deep into depleted subterranean aquifers through recharge shafts/borewells	वर्षा जल संचयन (सतही उपयोग) / कृत्रिम भूजल पुनर्भरण (भूजल रीचार्ज)
Central Ground Water Board (CGWB)	**Central Ground Water Authority (CGWA)**	Apex national scientific multidisciplinary organization under Ministry of Jal Shakti assessing groundwater resources, aquifer mapping, and monitoring vs statutory regulatory authority constituted under Section 3(3) of EPA 1986 issuing mandatory NOCs and regulating industrial groundwater extraction	केंद्रीय भूजल बोर्ड (CGWB, वैज्ञानिक आकलन) / केंद्रीय भूजल प्राधिकरण (CGWA, कानूनी नियमन)
Jal Jeevan Mission (JRM / Har Ghar Jal)	**Atal Bhujal Yojana (ABHY)**	Centrally sponsored flagship mission launched in August 2019 ensuring 55 lpcd functional household tap connections (FHTC) to all rural households by 2024 vs World Bank assisted community-led groundwater management scheme focused on water-stressed districts across 7 states	जल जीवन मिशन (हर घर जल, नल कनेक्शन) / अटल भूजल योजना (सामुदायिक भूजल प्रबंधन)

26.2 Integrated Water Resources Management (IWRM) (एकीकृत जल संसाधन प्रबंधन)

Traditional	IWRM
Departmental silos	River basin planning
Supply focus	Supply + demand + quality
Environment (पर्यावरण) ignored	Ecological flows included

26.3 Rainwater Harvesting

Method	Function
Rooftop tank	Direct use
Recharge pit/well	Aquifer recharge
Check dam (दाम -)	Slow runoff + percolation
Johad/taanka	Traditional arid storage

26.5 River Conservation

Programme	Year	Fact
Central Ganga Authority / GAP	**1985**	First Ganga STPs
NRCA rename	**1995**	Broader river mandate
YAP launch	**1993**	Yamuna bilateral Japan–India track
NGRBA	**2009**	PM-chaired basin authority
Namami Gange	**2014**	NMCG nodal
YAP-III	**2016**	Under Namami Gange
NRCP	1995	Other rivers

Common Traps

Trap	Correct
Rivers second after oceans	**Ice then GW then rivers**
Groundwater report = CPCB	**CGWB**
CGWA = CGWB	**EPA regulator ≠ assessment board**
JJM launched 2014	**2019**
Namami Gange = 1985	**2014** (GAP = 1985)
Freshwater = biotic resource	**Not** — abiotic
NWM = river linking + free borewells	**Recycling / coastal tech / +20% efficiency** yes; linking/reimbursement **no**
UV removes odour + turbidity	**Only microbial kill** among classic claims
Chilika under NLCP	**Not**
NGRBA chair = rotating CM	**Prime Minister**
Dolphin decline = more crocodiles	**Dams, nets, agrochemicals**
Lisbon = freshwater glaciers	**Oceans**
Hariyali = Jal Kranti	**Watershed programme**
Namami Gange = 1985	**2014** (GAP = 1985)
Gomti = Kanpur	**Lucknow**
Sultanpur in UP	**Haryana**
Watershed increases erosion	**Reduces erosion**
IWRM = dams only	**Basin + quality + ecology**
CWC = groundwater quality	**Surface water data**
Drinking water unlimited fluoride	**≤1.5 mg/L**

Topic 27 — Renewable & Non-Renewable Energy

Current Affairs (this topic)

- Capacity and fuel-mix CA overlaps the Renewable Energy chapter — keep classification traps consistent.
- Nuclear is low-carbon but conventionally non-renewable in many Prelims frames.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Renewable Energy	**Non-Renewable Energy**	Energy derived from natural regenerative sources that replenish naturally on a human timescale without depleting stocks (solar, wind, small hydro, geothermal, biomass) vs finite geological energy reserves that take millions of years to form and cannot be replenished once consumed (coal, petroleum, natural gas, uranium)	नवीकरणीय ऊर्जा (अक्षय स्रोत) / अनवीकरणीय ऊर्जा (परिमित जीवाश्म भंडार)
Conventional Energy	**Non-Conventional Energy**	Traditional energy sources that have been historically established and extensively utilized for commercial power generation for centuries (coal, oil, gas, large hydro, commercial nuclear) vs alternative, modern clean renewable sources developed to replace fossil fuels (solar, wind, tidal, ocean thermal, fuel cells)	पारंपरिक ऊर्जा स्रोत / गैर-पारंपरिक (वैकल्पिक) ऊर्जा स्रोत
Bioethanol	**Biodiesel**	Pure ethanol fuel produced through the biochemical fermentation of sugary and starchy feedstocks (sugarcane juice, molasses, damaged grains, corn) blended with petrol (E20 mandate) vs fatty acid methyl esters (FAME) produced via the transesterification of non-edible plant vegetable oils (Jatropha curcas, Pongamia, animal fats) blended with diesel	बायो-इथेनॉल (किण्वन, पेट्रोल संवर्धन) / बायो-डीजल (ट्रांसएस्टरीकरण, डीजल संवर्धन)
Perform, Achieve and Trade (PAT)	**Renewable Purchase Obligation (RPO)**	Market-based regulatory mechanism under the National Mission for Enhanced Energy Efficiency (BEE / EC Act 2001) issuing tradable Energy Saving Certificates (ESCerts) to designated energy-intensive industries vs statutory mandate under Electricity Act 2003 requiring power discoms to purchase a minimum percentage of electricity from renewable sources	निष्पादन, उपलब्धि और व्यापार (PAT, ऊर्जा बचत प्रमाण पत्र) / नवीकरणीय खरीद दायित्व (RPO)
Bureau of Energy Efficiency (BEE)	**Petroleum Conservation Research Association (PCRA)**	Statutory body established under the Energy Conservation Act 2001 (Ministry of Power) promoting star rating standards for electric appliances and energy conservation codes vs registered society under the Ministry of Petroleum and Natural Gas promoting conservation and fuel efficiency in the petroleum sector	ऊर्जा दक्षता ब्यूरो (BEE, स्टार लेबलिंग) / पेट्रोलियम संरक्षण अनुसंधान संघ (PCRA)
Biogas	**Compressed Biogas (CBG)**	Unrefined raw gaseous fuel mixture generated from anaerobic digestion of cattle dung and agro-waste containing ~55–60% methane and ~40% CO_2 vs purified, upgraded methane gas (≥90% CH_4) scrubbed of CO_2 and H_2S and compressed to ~250 bar for use as clean green automotive transport fuel (SATAT initiative)	बायोगैस (कच्ची गोबर गैस) / संपीडित बायोगैस (CBG, परिष्कृत 90%+ मीथेन)

27.3 Conventional Sources of Energy

Conventional	Non-conventional
Coal, oil, gas	Solar, wind, biomass
Large hydro	Small hydro ≤25 MW

Conventional	Non-conventional
Nuclear	Tidal, geothermal, ocean

27.8 Biofuels

Fuel	Feedstock	Blend use
Bioethanol	Molasses, maize, 2G waste	Petrol E10/E20
Biodiesel	Jatropha, Pongamia, UCO	Diesel B5/B10
Biogas/CBG	Dung, organic waste	Cooking / transport

27.14 Ocean Energy

Type	Driver
Tidal	Moon gravity
Wave	Wind on sea surface
OTEC	Warm–cold water ΔT

27.16 Renewable Energy Scenario in India

Source	Key fact
Solar	Fastest capacity growth
Wind	TN/Gujarat strongholds
SHP	Himalayan streams ≤25 MW
Biomass	Bagasse, biogas, waste-to-energy

27.19 Energy Conservation

Item	Year/body
Energy Conservation Act	**2001**
BEE	Under Power Ministry
Star labeling	Appliance efficiency

27.20 Energy Efficiency

Scheme	What it trades/does
PAT	ESCerts (energy efficiency)
CCTS	Carbon credits (CO_2e)
UJALA	LED bulbs at scale

Common Traps

Trap	Correct
Nuclear = renewable	Usually **non-renewable**
Non-conventional = non-renewable	**= renewable**
Coal = biomass	**Fossil, not biomass**
Uranium = fossil fuel	**Nuclear fuel**, not fossil
Biogas = butane	**CH_4 + CO_2**
Tidal = OTEC/wave	**Different drivers**
PAT = carbon credits	**ESCerts**
BEE = MNRE	**Power vs Renewables**
Large hydro = non-conventional	Usually **conventional**
E20 already forever since 2010	**Target ~2025–26**
SPR only in Delhi	**Vizag, Mangalore, Padur**
Energy crisis = only hydro shortage	**Fossil fuel exhaustion risk**
Biofuels always cheap	Often **not cost-effective**
Maize = biodiesel crop	**Ethanol**; Jatropha = biodiesel
Fuel cell = AC output	**DC**; by-products heat + water
Gasification gas = CO_2 only	**Producer gas** CO + H_2 (+ CH_4)

Topic 28 — Environmental Research & Institutions

Current Affairs (this topic)

- FRI / FSI / WII / IIRS function traps are evergreen Dehradun (देहरादून) CA colour.
- Do not swap ISFR (FSI) with FRI research mandate.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Botanical Survey of India (BSI)	**Zoological Survey of India (ZSI)**	Premier taxonomic institution (HQ: Kolkata, established 1890) undertaking floristic surveys, documenting plant species, and managing the Central National Herbarium vs premier institution (HQ: Kolkata, established 1916) surveying faunal diversity, animal taxonomy, and systematic collections	भारतीय वनस्पति सर्वेक्षण (BSI, 1890) / भारतीय प्राणी सर्वेक्षण (ZSI, 1916)
Forest Survey of India (FSI)	**Forest Research Institute (FRI)**	Subordinate organization under MoEFCC (HQ: Dehradun, established 1981) conducting national forest resource inventory via satellite imagery and publishing biennial ISFR vs deemed university and premier research institution under ICFRE (HQ: Dehradun, established 1906 as Imperial Forest Research Institute)	भारतीय वन सर्वेक्षण (FSI, द्विवार्षिक वन रिपोर्ट) / वन अनुसंधान संस्थान (FRI, वानिकी अनुसंधान)
National Biodiversity Authority (NBA)	**State Biodiversity Board (SBB)**	Autonomous statutory regulatory authority established in 2003 under Biological Diversity Act 2002 (HQ: Chennai) regulating foreign nationals' access to Indian bioresources and advising Union vs state-level statutory bodies regulating commercial utilization of bioresources by Indian citizens	राष्ट्रीय जैव विविधता प्राधिकरण (NBA, चेन्नई) / राज्य जैव विविधता बोर्ड (SBB)
National Bureau of Plant Genetic Resources (NBPGR)	**National Bureau of Agricultural Insect Resources (NBAIR)**	Nodal ICAR organization (HQ: New Delhi) managing the National Gene Bank preserving orthodox seeds and germplasm of food and agricultural crops vs premier ICAR research bureau (HQ: Bengaluru) dedicated to insect taxonomy, molecular characterization, and biocontrol insect resources	राष्ट्रीय पादप आनुवंशिक संसाधन ब्यूरो (NBPGR, दिल्ली) / कृषि कीट संसाधन ब्यूरो (NBAIR, बेंगलुरु)
Wildlife Institute of India (WII)	**Indira Gandhi National Forest Academy (IGNFA)**	Autonomous scientific research and training institute under MoEFCC (HQ: Dehradun) conducting wildlife ecological research and Tiger Census vs premier apex staff college (HQ: Dehradun) providing two-year professional induction training to directly recruited Indian Forest Service (IFS) officers	भारतीय वन्यजीव संस्थान (WII, अनुसंधान) / इंदिरा गांधी राष्ट्रीय वन अकादमी (IGNFA, IFS प्रशिक्षण)
India State of Forest Report (ISFR)	**Basic Animal Husbandry Statistics**	Biennial statutory publication by Forest Survey of India (FSI, Dehradun) reporting state-wise forest cover, tree cover, mangrove cover, and forest carbon stocks vs annual publication by Ministry of Fisheries, Animal Husbandry & Dairying reporting livestock census, milk, egg, and meat production	भारत वन स्थिति रिपोर्ट (ISFR, द्विवार्षिक) / बुनियादी पशुपालन सांख्यिकी (वार्षिक)

28.4 Forest Research Institute (FRI) (वन अनुसंधान संस्थान)

Feature	FRI	FSI
Year	1906	1981
Role	Research	Cover survey
Output	Silviculture protocols	**ISFR**

28.6 National Biodiversity Authority (NBA) (राष्ट्रीय जैव विविधता प्राधिकरण)

Level	Body	Seat	Role
National	NBA	**Chennai**	Foreign access, ABS
State	SBB	State capital	Domestic commercial access
Local	BMC	Panchayat/ward	**PBR**

Common Traps

Trap	Correct
BSI = animals	**Plants**
ZSI = plants	**Animals**
FRI publishes ISFR	**FSI publishes ISFR**
NBA HQ Delhi	**Chennai**
NBA prepares PBR	**BMC prepares PBR**
SBB = SPCB	**Biodiversity ≠ pollution**
NBPGR = Kolkata	**New Delhi**
NBPGR = animal bank	**NBAGR Karnal**
Bahraich >20% forest	**False**
ISFR annual	**Biennial (2 years)**

Topic 29 — Indian Environmental Movements

Current Affairs (this topic)

- Movement years and states are fixed history (इतिहास); CA rarely renames Chipko / Appiko / NBA.
- Keep NBA (Andolan) ≠ National Biodiversity Authority acronym trap ready.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Bishnoi Movement (1730)	**Chipko Movement (1973)**	Amrita Devi / Khejri tree sacrifice in Khejarli, Rajasthan vs Gaura Devi / Sunderlal Bahuguna anti-commercial logging in Chamoli (Uttarakhand)	बिश्नोई आंदोलन (1730) / चिपको आंदोलन (1973)
Chipko Movement (1973)	**Appiko Movement (1983)**	Himalayan forest conservation in Chamoli (Bahuguna/Chandi Prasad Bhatt) vs Western Ghats forest hugging in Uttara Kannada, Karnataka (Pandurang Hegde)	चिपको (उत्तराखंड) / अप्पिको (कर्नाटक)
Silent Valley Movement (1970s)	**Narmada Bachao Andolan (1985+)**	Kunthipuzha river anti-hydro dam protecting lion-tailed macaque in Kerala vs anti-large dam displacement on Narmada (Medha Patkar, Baba Amte)	साइलेंट वैली (केरल) / नर्मदा बचाओ आंदोलन
Jungle Bachao Andolan (1982)	**Chipko Movement (1973)**	Singhbhum (Jharkhand) tribals opposing replacement of natural sal with commercial teak ("Greed Game Political Populism") vs tree-hugging in UP hills	जंगल बचाओ (सिंहभूम) / चिपको आंदोलन
Appiko Movement (1983)	**Silent Valley (1970s)**	Forest conservation by tree embrace in Karnataka vs grassroots campaign stopping hydro-electric project in Kerala rainforest	अप्पिको (कर्नाटक) / शांत घाटी (केरल)
Tehri Dam Protest	**Narmada Bachao Andolan**	High-seismic zone anti-dam fast by Sunderlal Bahuguna on Bhagirathi vs multi-state anti-submergence movement across MP/Gujarat/Maharashtra	टिहरी बांध विरोध / नर्मदा बचाओ
Gadgil Commission (2011)	**Kasturirangan Committee (2013)**	WGEEP designated entire Western Ghats as ecologically sensitive (ESZ 1, 2, 3) vs HLWG recommending only 37% area as Eco-Sensitive Area (ESA)	गाडगिल समिति / कस्तूरीरंगन समिति

29.7 Save Western Ghats Movement

Cause, Course, Result

Report	Year	Tone
Gadgil WGEEP	**2011**	Stronger ESZ / conservation
Kasturirangan HLWG	**2013**	Narrower ESA / moderated

Common Traps

Trap	Correct
Appiko = Uttarakhand/UP	**Karnataka**
Chipko = Karnataka	**Uttarakhand 1973**
Silent Valley = Tamil Nadu	**Kerala**
Bishnoi = 1973 Chipko	**1730 Rajasthan**

Trap	Correct
Medha Patkar = Chipko	**Narmada Bachao**
Appiko = Save W Ghats	**1983 vs 2011+**
NBA = Biodiversity Authority	Context: movement = Narmada
Jungle Bachao = Garhwal	**Singhbhum Jharkhand**
Gaura Devi = Appiko	**Chipko Reni**
Hegde = Chipko	**Appiko**

Topic 30 — Environmental Literature & Awareness

Current Affairs (this topic)

- Book–author pairs (Carson, Fukuoka, etc.) stay fixed; new award themes are light CA.
- Silent Spring ≠ Silent Valley (साइलेंट वैली) (movement).

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Environmental Education	**Environmental Awareness**	Structured pedagogical curriculum in schools/colleges (Eco-Clubs, NGC) vs general public sensitization & behavioral change (Mission LiFE)	पर्यावरण शिक्षा / पर्यावरण जागरूकता
Silent Spring (1962)	**Tragedy of the Commons (1968)**	Rachel Carson's book exposing chemical pesticide (DDT) bioaccumulation vs Garrett Hardin's economic essay on shared resource degradation	साइलेंट स्प्रिंग / ट्रैजेडी ऑफ द कॉमन्स
Limits to Growth (1972)	**Brundtland Report (1987)**	Club of Rome computer model projecting economic/ecological collapse vs UN WCED report (*Our Common Future*) coining Sustainable Development	लिमिट्स टू ग्रोथ (1972) / ब्रंटलैंड रिपोर्ट (1987)
Small Is Beautiful (1973)	**Silent Spring (1962)**	E.F. Schumacher's critique of mass production ("economics as if people mattered") vs Carson's chemical toxicity expose	स्मॉल इज ब्यूटीफुल / साइलेंट स्प्रिंग
Eco-mark (1991)	**Greenwashing**	Official BIS earthen pot label for certified environment-friendly consumer goods vs deceptive corporate marketing claiming false eco-credentials	इको-मार्क (मिट्टी का घड़ा) / ग्रीनवॉशिंग
Greendex	**Ecological Footprint**	National Geographic consumer sustainability index vs Global Footprint Network measure of nature's demand vs biocapacity	ग्रीनडेक्स / पारिस्थितिक पदचिह्न
National Green Corps (NGC)	**Mission LiFE (2022)**	MoEFCC Eco-club programme in school networks vs international citizen-movement promoting lifestyle for environment	नेशनल ग्रीन कॉर्प्स / मिशन लाइफ (LiFE)

30.4 Environmental Communication

Channel	Example
Magazine	Down To Earth
Children's	Gobar Times
Scheme IEC	Namami Gange / Swachh Bharat
People's science	KSSP

30.5 Environment Awards (high-yield)

Identity: Indira Gandhi Paryavaran | Tyler | Global 500 → Champions of the Earth | Rajiv Gandhi | Borlaug

- **Indira Gandhi Paryavaran Puraskar (1987, MoEF)** honours significant measurable contribution to environment protection and improvement — not only wildlife research or wasteland alone.
- **Tyler Prize for Environmental Achievement** (University of Southern California) is often called the "Nobel of Environment" — environmental science, health, and energy. Notable awardees include Madhav Gadgil (2015 pairing), Partha Dasgupta (2016), Pavan Sukhdev & Gretchen Daily (2020).

- **Rajiv Gandhi Environment Award** recognises clean/new technology and industrial pollution-reduction practice (MoEFCC track).
- **Global 500 Award** (UNEP, from **1987**) honoured environmental achievement; later succeeded in spirit by **Champions of the Earth** (from **2005**). Narendra Modi received Champions of the Earth **2018**.
- **Borlaug Award** is for outstanding agricultural research/contribution. Norman Borlaug's Nobel was the **Peace** Prize (Green Revolution (हरित क्रांति)), not a medicine Nobel.

> **Logic:** Tyler ≠ drama/child-rights prize. Global 500 = environment, not anti-terror. AGMARK ≠ Eco-mark; AGMARK ≠ FAO mark.

Common Traps

Trap	Correct
Silent Spring = Hardin	**Carson 1962**
Brundtland = Limits to Growth	**Our Common Future 1987**
Limits to Growth = Brundtland	**Club of Rome 1972**
LiFE idea at COP-25	**COP26 2021**
Agenda 21 = only EE	**Primarily Sustainable Development**
ECO Club = CPCB	**MoEFCC NGC**
Awareness = only school EE	**Also campaigns + Art 51A(g)**
Bharat Nirman = NEP scheme	**Infrastructure, not NEP EE**
IPCC = MoEFCC awareness campaign	**Science assessment body**
Van Mahotsav = February	**July**

Topic 31 — Environmental Days

Current Affairs (this topic)

- Annual themes for WED / Earth Day / IDF refresh; calendar dates stay fixed (WED = **5 June**).
- Do not invent UP-only international day dates.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
World Wetlands Day	**World Wildlife Day**	**2 February** (signing of Ramsar Convention 1971) vs **3 March** (adoption of CITES 1973)	विश्व आर्द्रभूमि दिवस (2 फरवरी) / विश्व वन्यजीव दिवस (3 मार्च)
World Forestry Day	**World Water Day**	**21 March** (UN International Day of Forests) vs **22 March** (UN World Water Day)	विश्व वानिकी दिवस (21 मार्च) / विश्व जल दिवस (22 मार्च)
World Earth Day	**World Environment Day**	**22 April** (Gaylord Nelson, 1970) vs **5 June** (Stockholm Conference opening, 1972)	पृथ्वी दिवस (22 अप्रैल) / विश्व पर्यावरण दिवस (5 जून)
International Biodiversity Day	**World Earth Day**	**22 May** (Nairobi agreed text of CBD 1992) vs **22 April** (Earth Day)	जैव विविधता दिवस (22 मई) / पृथ्वी दिवस (22 अप्रैल)
World Ozone Day	**World Environment Day**	**16 September** (signing of Montreal Protocol 1987) vs **5 June** (Stockholm UNEP foundation)	विश्व ओजोन दिवस (16 सितंबर) / पर्यावरण दिवस (5 जून)
Earth Day	**Earth Hour**	Fixed annual civic observance on **22 April** vs WWF voluntary **last Saturday of March** 1-hour lights-off event	पृथ्वी दिवस (22 अप्रैल) / अर्थ आवर (मार्च अंतिम शनिवार)
World Soil Day	**World Day to Combat Desertification**	**5 December** (FAO celebration of King Bhumibol's birthday) vs **17 June** (UNCCD adoption anniversary)	विश्व मृदा दिवस (5 दिसंबर) / मरुस्थलीकरण रोकथाम दिवस (17 जून)

31.10 Other High-Yield Days

Identity: Toilet | No Tobacco | Habitat | Cities | Population (जनसंख्या) | Health | Overshoot

- **World Toilet Day = 19 November** (UNGA 2013; sanitation for all / end open defecation messaging).
- **World No Tobacco Day = 31 May** (WHO) — not 5 May. Themes rotate (e.g. tobacco–environment / grow food not tobacco).
- **World Habitat Day** = **first Monday of October** (UN; not a fixed calendar date like 5 October).
- **World Cities Day = 31 October**. **World Population Day = 11 July**. **World Health Day = 7 April**.
- **Earth Overshoot Day** marks when humanity's annual demand exceeds Earth's regenerative capacity — an environmental-degradation awareness marker (Global Footprint Network partnership framing).
- **Swachh Bharat Mission** launched **2 October 2014**. Sierra Leone's national cleanliness day is the last Saturday of each month.

> **Logic:** Anti-tobacco ≠ 5 May. Habitat Day ≠ fixed 5 Oct.

Common Traps

Trap	Correct
Wildlife Day = 22 March	**3 March** (Water is 22 March)
Wetlands Day = 2 / 3 March	**2 February**
Earth Day = 5 June	**22 April**; WED = 5 June
Earth Day = 22 March	**22 April**; Water = 22 March
Biodiversity Day = 22 March / April	**22 May**
Forests Day = 22 March	**21 March**
Ozone Day = 5 June	**16 September**
WED = Earth Day	Different dates and origins
Earth Hour = Earth Day	Hour ≠ Day
Population Day = 5 June	**11 July**; 5 June = WED

Topic 32 — National Parks & Protected Areas (Advanced)

Current Affairs (this topic)

- New NP / TR notifications refresh advanced PA CA — cross-check with the Protected Areas hierarchy.
- Location-match traps stay high-yield; UP has no marine PA.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
National Park (NP)	**Wildlife Sanctuary (WLS)**	WPA Sec 35; highest protection, no human settlement/grazing, boundaries altered only by State Legislature on NBWL advice vs WPA Sec 18; regulated grazing/traditional rights allowed	राष्ट्रीय उद्यान (NP) / वन्यजीव अभयारण्य (WLS)
Community Reserve	**Conservation Reserve**	Established on private or community-owned lands under WPA Sec 36C vs established on government-owned contiguous lands under WPA Sec 36A	सामुदायिक रिज़र्व / संरक्षण रिज़र्व
First Marine NP	**First National Park in India**	**Gulf of Kutch Marine NP** (1982, Gujarat) vs **Hailey / Jim Corbett NP** (1936, Uttarakhand)	प्रथम समुद्री उद्यान (कच्छ की खाड़ी) / प्रथम राष्ट्रीय उद्यान (कॉर्बेट 1936)
Dudhwa National Park	**Valmiki National Park**	Only National Park in **Uttar Pradesh** (Lakhimpur Kheri) vs only National Park in **Bihar** (West Champaran)	दुधवा राष्ट्रीय उद्यान (यूपी) / वाल्मीकि राष्ट्रीय उद्यान (बिहार)
Keibul Lamjao NP	**Khangchendzonga NP**	World's only floating park on Loktak Lake, Manipur (Sangai deer) vs India's only Mixed World Heritage Site in Sikkim	केइबुल लामजाओ (मणिपुर) / कंचनजंगा (सिक्किम)
Hemis National Park	**South Button Island NP**	India's largest National Park (Ladakh, snow leopard) vs India's smallest National Park (Andaman & Nicobar, 0.03 km^2)	हेमिस (लद्दाख, सबसे बड़ा) / साउथ बटन (अंडमान, सबसे छोटा)
Biosphere Reserve Core	**National Park**	Untouched strictly protected sanctuary within UNESCO MAB zoning vs statutory strictly protected area under Wildlife Protection Act, 1972	बायोस्फीयर कोर क्षेत्र / राष्ट्रीय उद्यान

32.2 Wildlife Sanctuaries of Uttar Pradesh

All 25 Wildlife Sanctuaries of UP

#	Sanctuary	District(s)	Key fact
1	Bakhira	Sant Kabir (कबीर) Nagar	Ramsar waterfowl
2	Chandraprabha	**Chandauli**	Early WLS (1957)
3	Hastinapur	Meerut–Muzaffarnagar–Ghaziabad–Bijnor–Amroha	**Largest**
4	Kaimur	Mirzapur (मिर्ज़ापुर), Sonbhadra	Vindhyan forest
5	Katarniaghat	Bahraich	**1975**; Terai Arc
6	Kishanpur	Lakhimpur Kheri	Dudhwa corridor
7	Lakh Bahosi	Kannauj (कन्नौज)	Lake birds
8	Mahavir Swami (स्वामी)	Lalitpur	Bundelkhand
9	National Chambal	Agra, Etawah (+ MP, RJ)	Gharial/dolphin
10	Nawabganj	Unnao	Ramsar birds
11	Okhla	GB Nagar, Ghaziabad	NCR wetland

#	Sanctuary	District(s)	Key fact
12	Parvati Arga	Gonda	Ramsar
13	Patna (पटना) Bird	Etah	Waterfowl
14	Ranipur	Chitrakoot (चित्रकूट), Banda	Ranipur TR 2023
15	Saman	Mainpuri	Ramsar
16	Samaspur	Rae Bareli	**IN UP** (Q58)
17	Sandi	Hardoi	Ramsar
18	Shekha Jheel	Aligarh (अलीगढ़)	Ramsar
19	Sohagibarwa	Maharajganj	Indo-Nepal
20	Sohelwa	Shravasti, Balrampur, Gonda	Bhabar (भाबर)–Terai
21	Sur Sarovar (Keetham)	Agra	Ramsar
22	Surha Tal	Ballia	Eastern wetland
23	Turtle	Varanasi (वाराणसी)	Gangetic turtles
24	Vijai Sagar	Mahoba	Bundelkhand lake
25	Sarsai Nawar	Etawah	**IN UP** (Q58)

32.3 National Parks of India

High-yield NP ↔ Leading states

National Park	State/UT	Extra fact
Jim Corbett	Uttarakhand	India first (1936); rivers **Ramganga + Kosi**
Dudhwa	Uttar Pradesh	Only UP NP; rhino rehab **1984+**
Valmiki	Bihar	NOT UP
Sultanpur / Kalesar	Haryana	Sultanpur trap
Dachigam / Salim Ali	J&K	Hangul; Salim Ali = ex–City Forest
Keoladeo Ghana	Rajasthan	Birds + WH — **not lion**
Kanha / Bandhavgarh / Panna / Van Vihar	Madhya Pradesh	Kanha hard-ground barasingha / **Bhoorsingh**; Bandhavgarh white tiger; Van Vihar = **Bhopal**
Gir	Gujarat	Asiatic lion
Kaziranga / Manas	Assam	Rhino / WH; Kaziranga rivers **Brahmaputra–Diphlu**
Periyar / Silent Valley / Eravikulam	Kerala	Elephant–tiger; **Kunthipuzha**; Nilgiri Tahr (“cloud goat”)
Bandipur / Nagarhole (Rajiv Gandhi)	Karnataka	Western Ghats
Hemis	Ladakh	Largest NP
Keibul Lamjao	Manipur	Loktak floating / Sangai
Mollem	Goa	—
Indravati	Chhattisgarh	Bastar; NP 1981 / TR 1983
Betla (Palamu)	Jharkhand	Project Tiger (प्रोजेक्ट टाइगर)
Simlipal / Bhitarkanika	Odisha	Simlipal ≠ MP; saltwater crocodile
Sundarbans / Buxa	West Bengal	Mangrove + WH; Buxa Jalpaiguri
Desert / Ranthambore (रणथंभौर) / Sariska	Rajasthan	GIB; both tiger reserves; Sariska Alwar

National Park	State/UT	Extra fact
Great Himalayan / Pin Valley	Himachal Pradesh	GHNP = **Kullu** WH 2014; Pin = Lahaul–Spiti (**not J&K**)
Rajaji / Govind / Valley of Flowers	Uttarakhand	Rajaji TR **2015**; Govind = **Tons** source; VoF alpine
Khangchendzonga	Sikkim	Mixed WH 2016
Namdapha	Arunachal Pradesh	Eastern Himalaya (हिमालय) hotspot; tropical→arctic climate belts
Gulf of Mannar	Tamil Nadu	Marine NP **1980**
Intanki / Ntangki	Nagaland	—
Guindy	Tamil Nadu	—
Papikonda	Andhra Pradesh	—
Phawngpui	Mizoram	Blue Mountain

River / species / wrong-pair keys

Stem	Correct lock
Corbett river	**Ramganga + Kosi** — not Ganga (गंगा)
Kaziranga river	**Brahmaputra / Diphlu** — not Manas
Silent Valley river	**Kunthipuzha** — not Kaveri (कावेरी)
Ghana / Bharatpur	Birds — **not lion**
Eravikulam	**Nilgiri Tahr** — not Hoolock
Hoolock Gibbon	**NE India / Assam**
Periyar	Elephant / tiger — **not Hangul**
Desert NP	GIB; **two districts**; habitation **exists**
Nokrek / Loktak / Namdapha	Garo (गारो) Hills / Manipur lake / Arunachal — only Nokrek–Garo matched among classic three-pair stems

32.4 Marine National Parks

#	Marine NP	State/UT	Fact
1	Gulf of Kutch	Gujarat	**First**
2	Gulf of Mannar	Tamil Nadu	Dugong / GoM BR
3	Mahatma Gandhi Marine	Andaman	Wandoor
4	Rani Jhansi Marine	Andaman	Ritchie's Archipelago

32.5 Important National Parks of the World

Park	Country	Fact
Yellowstone	USA	World first **1872**
Banff	Canada	**1885**
Serengeti	Tanzania	Savanna (सवाना) mega-fauna
Kruger	South Africa	Classic African NP
Kakadu	Australia	Wetland/rock art landscape

Park	Country	Fact
Galápagos	Ecuador	Evolution / WH

32.6 National Parks as Natural Heritage

Site	State	WH note
Kaziranga / Manas	Assam	1985 batch
Keoladeo	Rajasthan	Also Ramsar
Sundarbans	West Bengal	Mangrove WH
Nanda Devi + VoF	Uttarakhand	Alpine
Western Ghats	Multi-state	Serial **2012**
Great Himalayan	Himachal	**2014**
Khangchendzonga	Sikkim	Mixed **2016**

32.7 Coastal National Parks

Coastal / marine park	Coast	Fact
Sundarbans NP	East	Mangrove + WH
Bhitarkanika NP	East	Mangrove / croc
Gulf of Mannar Marine NP	East	Coral islands
Gulf of Kutch Marine NP	West	First Marine NP
MG / Rani Jhansi Marine	Andaman	Island marine

Common Traps

Trap	Correct
Valmiki / Sultanpur in UP	**Bihar / Haryana**
Dudhwa = Pilibhit district	**Lakhimpur Kheri**
First Marine NP = Sundarbans	**Gulf of Kutch**
World first NP = Corbett	**Yellowstone 1872**
Simlipal = MP	**Odisha**
Keoladeo = UP	**Rajasthan**
Samaspur / Sarsai Nawar not in UP	**Both are in UP**
Gahirmatha = Marine NP	**Marine sanctuary**
UP has UNESCO natural WH	**None**
Hangul = Gir	**Dachigam**
NP boundary needs Parliament	**State Legislature**
Corbett river = Ganga	**Ramganga + Kosi**
Silent Valley river = Kaveri	**Kunthipuzha**
Ghana = lion	**Birds** (Keoladeo)
Pin Valley = J&K	**Himachal (Lahaul–Spiti)**

Trap	Correct
Eravikulam = Hoolock	**Nilgiri Tahr**
Periyar = Hangul	**Elephant / tiger**
Desert NP = no people	Habitation **exists**; GIB yes
Nagarhole = Andhra	**Karnataka** (Rajiv Gandhi NP)
Bandipur = Tamil Nadu	**Karnataka**

Topic 33 — Biosphere Reserves

Current Affairs (this topic)

- WNBR new listings drift the UNESCO count; India national BR list stays the teaching spine.
- Biosphere ≠ WPA statutory category.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Biosphere Reserve (BR)	**National Park (NP)**	Landscape-level conservation integrating human livelihoods (MAB 3-tier zoning) vs species/habitat preservation strictly excluding human activity	जैवमंडल आरक्षित क्षेत्र / राष्ट्रीय उद्यान
National BRs in India	**WNBR (UNESCO) Listed**	Total **18** Biosphere Reserves designated by MoEFCC vs **12** inducted into UNESCO's World Network of Biosphere Reserves	18 राष्ट्रीय बायोस्फीयर / 12 यूनेस्को WNBR सूची
First BR in India	**Latest BR added to WNBR**	**Nilgiri** (1986, TN/KER/KAR) vs **Panna** (2020, Madhya Pradesh)	प्रथम बायोस्फीयर (नीलगिरि 1986) / नवीनतम यूनेस्को (पन्ना 2020)
Core Zone	**Buffer Zone**	Legally protected untouched sanctum sanctorum vs zone surrounding core where research, education, and limited tourism are permitted	कोर क्षेत्र (पूर्णतः सुरक्षित) / बफर क्षेत्र (सीमित गतिविधि)
Buffer Zone	**Transition Zone**	Managed area buffering the core vs outermost area where local communities, agriculture, and settlements sustainably coexist	बफर जोन / संक्रमण क्षेत्र (बस्तियां/खेती)
Nokrek Biosphere Reserve	**Loktak Lake**	Located in Garo Hills, **Meghalaya** (Citrus indica) vs freshwater wetland in **Manipur**	नोकरेक (मेघालय) / लोकटक (मणिपुर)
Uttar Pradesh BR count	**Madhya Pradesh BR count**	**Zero** Biosphere Reserves in UP vs **3** Biosphere Reserves in MP (Pachmarhi, Achanakmar-Amarkantak, Panna)	यूपी (शून्य बायोस्फीयर) / म.प्र. (3 बायोस्फीयर)

33.2 Major Biosphere Reserves of India

All 18 Biosphere Reserves of India

#	Biosphere Reserve	State(s)	Notified	WNBR note
1	Nilgiri	TN, KA, KL	**1986**	First; WNBR 2000
2	Nanda Devi	**Uttarakhand**	1988	WNBR 2004
3	Nokrek	**Meghalaya**	1988	Smallest; WNBR 2009
4	Gulf of Mannar	Tamil Nadu	1989	Marine; WNBR 2001
5	Sundarbans	West Bengal	1989	WNBR 2001
6	Manas	Assam	1989	**Removed UNESCO 2011**
7	Great Nicobar	A&N	1989	WNBR 2013
8	Simlipal	**Odisha** (ओडिशा)	1994	WNBR 2009
9	Dibru-Saikhowa	Assam	1997	National only
10	Dihang-Dibang	Arunachal Pradesh	1998	National only
11	Pachmarhi	Madhya Pradesh	1999	WNBR 2009

#	Biosphere Reserve	State(s)	Notified	WNBR note
12	Khangchendzonga	Sikkim	2000	WNBR 2018
13	Agasthyamalai	Kerala, Tamil Nadu	2001	WNBR 2016
14	Achanakmar–Amarkantak	MP, Chhattisgarh	2005	WNBR 2012
15	Great Rann of Kutch	Gujarat	2008	**Largest**
16	Cold Desert	Himachal Pradesh	2009	Recent WNBR addition
17	Seshachalam Hills	Andhra Pradesh	2010	National only
18	Panna	Madhya Pradesh	**2011**	Newest; WNBR 2020

33.3 Biosphere Reserves of Uttar Pradesh

State	BR count	Fact
Uttar Pradesh (उत्तर प्रदेश)	**0**	Any "UP BR" option is false
Uttarakhand	1	Nanda Devi
Madhya Pradesh	3	Pachmarhi, Achanakmar–Amarkantak, Panna
Bihar	0	Valmiki = NP, not BR

33.4 Biosphere Reserves of the World

World BR / landscape	Country	Fact
Yellowstone	USA	Iconic (साकार) MAB / NP landscape
Serengeti–Ngorongoro	Tanzania	Savanna (सवाना) mega-fauna
Galápagos	Ecuador	Evolution / WH overlap
Lake Baikal	Russia	Freshwater biodiversity
Doñana	Spain	Wetland BR classic
Mount Kenya	Kenya	Afro-alpine BR

Common Traps

Trap	Correct
Nokrek = Manipur	**Meghalaya**
Simlipal = Madhya Pradesh	**Odisha**
UP has a Biosphere Reserve	**Zero**
Nanda Devi = UP	**Uttarakhand**
Manas no longer a BR	Still **national** BR; UNESCO removed 2011
BR = WPA category	**Executive / MAB**, not WPA Sec 35
Core allows farming	**No** exploitative use in core
Dudhwa / Valmiki = UP BR	NP / Bihar NP — not UP BR
Agasthyamalai = Assam	**Kerala (+ TN)**
First BR = Nanda Devi	**Nilgiri 1986**
BR only for a few species	Conserves **all forms of life** + sustainable use

Trap	Correct
Largest BR = Nokrek	**Great Rann of Kutch** (national); Gulf of Mannar often largest UNESCO-certified
Khangchendzonga = HP	**Sikkim**; Cold Desert = HP
Bundala = India	**Sri Lanka**
Nallamalai = Indian BR	**Not** a notified BR
Transition = no people	Transition = **settlements / livelihoods**

Topic 34 — Climate Change (Advanced)

Current Affairs (this topic)

- Advanced climate CA mirrors Climate Change — NDC / net zero / extreme-event science updates.
- Keep adaptation vs mitigation and Montreal≠Paris traps identical to the core climate chapter.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Global Warming	**Climate Change**	Long-term rise in Earth's average surface temperature vs broader spectrum of changes including shifts in precipitation, weather extremes, and sea levels	भूमंडलीय तापन / जलवायु परिवर्तन
Climate Mitigation	**Climate Adaptation**	Actions to reduce or prevent GHG emissions (renewable energy, carbon taxes) vs actions to adjust and build resilience to actual or expected climate impacts (seawalls, drought crops)	शमन (उत्सर्जन कटौती) / अनुकूलन (अनुकूलनशीलता)
Carbon Sink	**Carbon Sequestration**	Natural or artificial reservoir absorbing more carbon than it releases (oceans, forests) vs the process/technique of capturing and storing carbon dioxide	कार्बन सिंक (अवशोषक) / कार्बन पृथक्करण (भंडारण प्रक्रिया)
Carbon Credit	**Carbon Tax**	Tradable permit allowing emission of 1 tonne of CO_2 equivalent under cap-and-trade vs direct price levied by government per tonne of GHG emitted	कार्बन क्रेडिट (1 टन CO_2e) / कार्बन कर
IPCC	**UNFCCC**	Scientific assessment body established in 1988 by WMO/UNEP (doesn't conduct research) vs multilateral legal treaty framework adopted at Rio 1992	आईपीसीसी (वैज्ञानिक निकाय) / यूएनएफसीसीसी (कानूनी संधि)
Emissions Gap Report	**Adaptation Gap Report**	Annual UNEP report comparing actual NDC pledges vs Paris 1.5°C targets vs annual UNEP assessment of adaptation finance and planning shortfalls	उत्सर्जन अंतराल रिपोर्ट / अनुकूलन अंतराल रिपोर्ट
Kyoto Protocol (1997)	**Paris Agreement (2015)**	Top-down binding targets on developed countries (Annex-I) vs bottom-up voluntary Nationally Determined Contributions (NDCs) applicable to all	क्योटो प्रोटोकॉल (1997) / पेरिस समझौता (2015)

34.16 IPCC Assessment Reports

Report	Year	Fact
AR1	1990	Helped lead to UNFCCC
AR4	2007	Nobel year
AR5	2014	Informed Paris
AR6 SYR	**2023**	Latest synthesis

Common Traps

Trap	Correct
Emissions Gap = IPCC	**UNEP**
Montreal = climate mitigation	**Ozone**; Kyoto = climate
Carbon credit = Montreal	**Kyoto**

Trap	Correct
Green Credit = carbon credit	**Not identical**
IPCC sets emission targets	**Science only**
Absolute emitter rank = 10th	~**3rd** absolute; 2023 Q109 keyed **10th**
Argon is a GHG	**Not**
Fourier missing / modern invent	**Joseph Fourier (1820s)**
GCF = Durban	**Cancun 2010**
Earth Hour = UNEP	**WWF**
Net zero = absolute zero	Balance with **removals**
Nuclear = NAPCC mission	**Not** listed
Milankovitch includes solar irradiance	**No** — eccentricity/obliquity/precession
CDM projects only in Annex-I	Projects in **developing** countries
Argon is a GHG	**Not**
India net zero 2050	**2070**
LiFE idea at COP-25	**COP-26**
PAT = carbon credit trading	**ESCerts** (efficiency)

Topic 35 — Atmosphere

Current Affairs (this topic)

- Atmosphere layer and composition facts are stable; WMO GHG / ozone bulletins are light CA.
- Troposphere weather vs stratosphere ozone remains the classic layer trap.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Troposphere	**Stratosphere**	Lowest weather layer, temperature decreases with altitude (normal lapse rate 6.5°C/km) vs calm dry layer where temperature increases with height due to UV absorption by ozone	क्षोभमंडल (मौसम परत) / समतापमंडल (ओजोन परत)
Stratosphere	**Mesosphere**	Contains protective ozone layer; stable air suited for jet aircraft vs coldest atmospheric layer (-90°C) where incoming meteors burn upon entry	समतापमंडल / मध्यमंडल (उल्कापिंड दहन)
Mesosphere	**Thermosphere (Ionosphere)**	Temperature plummets to lowest levels vs temperature surges past 1500°C; contains ionized gas reflecting terrestrial radio waves and hosting auroras	मध्यमंडल / तापमंडल (आयनमंडल)
Homosphere	**Heterosphere**	Lower ~85 km where atmospheric gases remain uniformly mixed in proportion vs upper layer where gases stratify by molecular weight (N_2, O, He, H)	सममंडल (समान मिश्रण) / विषममंडल (गुरुत्वाकर्षण परतें)
Insolation	**Terrestrial Radiation**	Shortwave electromagnetic solar radiation received by Earth vs outgoing longwave infrared radiation re-radiated by Earth's heated surface	सूर्यताप (लघुतरंग) / भौमिक विकिरण (दीर्घतरंग)
Good Ozone	**Bad Ozone**	Naturally occurring stratospheric ozone shielding Earth from solar UV-B vs ground-level tropospheric ozone acting as secondary pollutant and respiratory irritant	अच्छा ओजोन (समतापमंडल) / बुरा ओजोन (क्षोभमंडल)

35.3 Atmospheric Layers

Layer	Approx. height	Temp trend	Signature
Troposphere	0–8/18 km	↓ upward	Weather
Stratosphere	~12–50 km	↑ upward	Ozone
Mesosphere	~50–85 km	↓ upward	Meteors; coldest
Thermosphere	~85–600 km	↑ upward	Aurora / ISS
Exosphere	~600 km+	Transition	Space merge

Common Traps

Trap	Correct
Ozone layer in troposphere	**Stratosphere**
Atmosphere heated mainly by shortwave	**Longwave from surface**
N_2 is a GHG	**Not**; N_2O is
Temp falls in every layer	Tropo↓ Strato↑ Meso↓ Thermo↑

Trap	Correct
Weather in stratosphere	**Troposphere only**
Meteors burn in troposphere	**Mesosphere**
Max ozone in mesosphere/exosphere	**Stratosphere**
Argon is a GHG	**Not**
Thermosphere “feels” very hot	High T, **low density**
Composition % = climate importance	Trace CO_2 still critical

Topic 36 — Ozone Layer

Current Affairs (this topic)

- Twin of Ozone Layer core — keep Montreal / Kigali as ozone-family CA only.
- Prefer one consistent ODS story across both ozone chapters.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Stratospheric Ozone	**Tropospheric Ozone**	Beneficial atmospheric filter blocking lethal solar UV radiation vs harmful photochemical smog component damaging human lungs and crop yields	समतापमंडलीय ओजोन (सुरक्षा कवच) / क्षोभमंडलीय ओजोन (प्रदूषक)
Ozone Depletion	**Ozone Hole**	Global thinning of the stratospheric ozone layer by ODS vs severe seasonal Antarctic drop in total column ozone below 220 Dobson Units (DU)	ओजोन क्षरण / ओजोन छिद्र (<220 DU)
Vienna Convention (1985)	**Montreal Protocol (1987)**	Non-binding framework agreement establishing ozone research/cooperation vs binding international protocol mandating specific phase-out schedules for ODS	वियना कन्वेंशन (1985) / मॉन्ट्रियल प्रोटोकॉल (1987)
Chlorofluorocarbons (CFCs)	**Hydrofluorocarbons (HFCs)**	Chlorine-bearing potent ozone depleters + GHGs (phased out under Montreal) vs zero ODP non-ozone-depleting refrigerants that are super greenhouse gases (phased down under Kigali)	सीएफसी (ओजोन क्षयकारी) / एचएफसी (ग्रीनहाउस गैस)
Ozone Depletion Potential (ODP)	**Global Warming Potential (GWP)**	Measure of substance's ability to destroy O_3 relative to CFC-11 (=1.0) vs measure of heat-trapping efficiency relative to CO_2 (=1.0) over 100 years	ओजोन रिक्तीकरण क्षमता (ODP) / वैश्विक तापन क्षमता (GWP)
Dobson Unit (DU)	**Parts Per Million (ppm)**	Standard measure of atmospheric column ozone thickness (1 DU = 0.01 mm at STP; normal = ~300 DU) vs concentration ratio of gases in volume	डॉबसन यूनिट (DU) / पीपीएम (ppm)

36.4 Ozone Depleting Substances (ODS)

ODS family	Typical use	Fact
CFCs	Cooling, aerosols, foam, electronics cleaners	Main Cl source
Halons (1211, 1301)	Fire fighting	Br; high ODP
CCl_4 / methyl chloroform / CH_3Br	Solvent / fumigant	Controlled ODS
HCFCs	Transitional coolants	Lower ODP; being phased out
HFCs	Coolants	Not ODS; Kigali / GWP

36.6 Vienna Convention

Treaty	Year	Role
Vienna Convention	**1985**	Framework / cooperation
Montreal Protocol	**1987**	Binding ODS controls
Kigali Amendment	**2016** (force **1 Jan 2019**)	HFC phase-down under Montreal

Common Traps

Trap	Correct
Montreal = climate mitigation	**Ozone / ODS**; Kyoto = climate
Ozone layer in troposphere	**Stratosphere** (~90% of O_3)
Ozone hole over India like Antarctica	**Antarctic spring** feature
Vienna = detailed CFC ban alone	**Framework**; Montreal = phase-out
HFC = classic ODS	**High GWP**; Kigali track
Ozone Day = 5 June	**16 September**
Hole = physical hole	Column O_3 **< 220 DU**
Good ozone at ground level	Ground O_3 = **bad** pollutant
CFC not a greenhouse gas	CFCs **are** GHGs and ODS
CFC main source = automobiles	Refrigerants / foam / aerosols / electronics cleaners
CFC use = tubeless tyres	**Not** a classic CFC use
Fridge gas = methane	**Mafron**/halocarbon; large plants may use **ammonia**
Ozone absorbs IR as UV-shield role	Protective role = **UV**
UV harm key = liver/brain cancer	Classic key = **skin cancer**
Carbon trading = Montreal	**Kyoto**
Agenda 21 = ozone treaty	**Sustainable Development**
Kyoto = save water / ozone	**GHG / climate**
Ionosphere in homosphere	Homosphere ≈ tropo + strato + meso; **ionosphere not included**
Methyl chloroform not an ODS	It **is** Montreal-controlled ODS

Topic 37 — Greenhouse Gases

Current Affairs (this topic)

- WMO GHG Bulletin and methane CA refresh this twin of Climate Change (जलवायु परिवर्तन).
- GWP order traps (CO_2 / CH_4 / N_2O / fluorinated) stay core.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Global Warming Potential (GWP)	**Radiative Forcing**	Heat absorption efficiency of 1 kg of a gas over 100 years relative to CO_2 (=1) vs net change in energy balance of Earth (W/m^2) caused by that gas	वैश्विक तापन क्षमता (GWP) / विकिरण दबाव
Carbon Dioxide (CO_2)	**Water Vapour (H_2O)**	Primary anthropogenic driver of long-term climate change (atmospheric residence 100+ yrs) vs dominant natural greenhouse gas acting as a fast temperature feedback	कार्बन डाइऑक्साइड (मानवजनित) / जलवाष्प (प्राकृतिक फीडबैक)
Methane (CH_4)	**Nitrous Oxide (N_2O)**	GWP ~28–30 (residence ~12 yrs; rice paddies, enteric livestock, wetlands) vs GWP ~265–273 (residence ~114 yrs; chemical fertilizers, adipic acid)	मीथेन (धान/पशुपालन) / नाइट्रस ऑक्साइड (उर्वरक)
Sulfur Hexafluoride (SF_6)	**Carbon Dioxide (CO_2)**	Most potent GHG known with GWP >23,500 (used in electrical switchgear) vs benchmark baseline GHG with GWP = 1	सल्फर हेक्साफ्लोराइड (GWP 23,500) / CO_2 (मानक 1)
Kyoto Six GHGs	**Montreal Phase-Out Gases**	CO_2, CH_4, N_2O, HFCs, PFCs, SF_6 (later NF_3 added) targeted for climate reduction vs CFCs, Halons, CCl_4, HCFCs targeted for stratospheric ozone protection	क्योटो 6 ग्रीनहाउस गैसें / मॉन्ट्रियल ओजोन गैसें

37.7 Hydrofluorocarbons (HFCs)

Gas	ODS?	Climate role	Treaty home
CFC	Yes	High GWP GHG	Montreal
HFC	No	High GWP GHG	Kyoto + Kigali
HCFC	Yes (lower)	Transitional	Montreal phase-out

Common Traps

Trap	Correct
Argon / N_2 / O_2 are GHGs	**Not**
Hydrogen / propane are GHGs	**Not**
Paddy gas = CO_2 or CO	**Methane**
Laughing gas = ammonia / CFC	**N_2O**
Biogas = butane / CO	**CH_4 + CO_2**
Highest GWP = most total warming	**CO_2 dominates by volume**
Water vapour = primary human driver	**Feedback** / most abundant **natural** GHG
CFC is not a GHG	**It is** (and ODS)
SO_2 warms Earth directly	**Indirect** / acid-rain role

Trap	Correct
Fourier missing	**Joseph Fourier (1820s)**
HFC = ODS	**GHG only**; Kigali
Stratospheric O_3 = smog pollutant	**UV shield**; bad O_3 is tropospheric
CNG = butane	**Methane**

Topic 38 — Pollution (Advanced)

Current Affairs (this topic)

- Advanced pollution CA mirrors Pollution & Waste — AQI / CAQM / waste-rule updates.
- Keep criteria-pollutant chemistry pairs consistent with the core pollution chapter.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Primary Air Pollutant	**Secondary Air Pollutant**	Emitted directly from identifiable sources (CO, SO_2, NO, PM) vs formed in atmosphere through photochemical reactions (Ozone, PAN, Smog, Acid Rain)	प्राथमिक प्रदूषक (सीधे उत्सर्जित) / द्वितीयक प्रदूषक (रासायनिक क्रिया)
Biological Oxygen Demand (BOD)	**Chemical Oxygen Demand (COD)**	Amount of dissolved O_2 required by aerobic bacteria to decompose organic matter vs amount of oxygen required to chemically oxidize all biodegradable and non-biodegradable matter	जैव रासायनिक ऑक्सीजन मांग (BOD) / रासायनिक ऑक्सीजन मांग (COD)
Classical (London) Smog	**Photochemical (Los Angeles) Smog**	Occurs in cool, humid winter mornings; sulfurous, reducing, coal-smoke based vs occurs in dry, sunny afternoons; oxidizing, auto-exhaust NOx + VOCs generating ground ozone and PAN	लंदन स्मॉग (सल्फ्यूरस, अपचायक) / प्रकाश रासायनिक स्मॉग (ऑक्सीकारक)
Central Pollution Control Board (CPCB)	**Central Ground Water Board (CGWB)**	Statutory body under Water Act 1974 for overall air/water quality monitoring vs apex subordinate office under Jal Shakti Ministry for groundwater management	सीपीसीबी (प्रदूषण नियंत्रण) / सीजीडब्ल्यूबी (भूजल बोर्ड)
SAFAR	**NCAP**	Ministry of Earth Sciences real-time metropolitan air quality forecasting system vs MoEFCC national targeted programme to cut PM2.5/PM10 by 20–30% across 131 non-attainment cities	सफर (सटीक पूर्वानुमान प्रणाली) / राष्ट्रीय स्वच्छ वायु कार्यक्रम

38.3 Air Quality Index (AQI)

AQI	Category
0–50	Good
51–100	Satisfactory
101–200	Moderate
201–300	Poor
301–400	Very Poor
401–500	Severe

38.16 Heavy Metal Pollution

Metal / pollutant	Classic disease / effect
Cr	Kanpur tanneries
Hg	Minamata / neurological
Pb	Neurotoxin; IQ / CNS
Cd	Itai-itai / bone–kidney

Metal / pollutant	Classic disease / effect
As	Black Foot / cancers
F	Knock-knee / fluorosis
Asbestos	Lung scarring / cancer

Common Traps

Trap	Correct
Lichens thrive in polluted air	**Sensitive; die back**
Fly ash = secondary pollutant	**Primary**; smog is secondary
Fly ash = SiO_2/CaO only, no toxics	Also **Pb/As/Cu**-type metals
BOD = blood oxygen test	**Aquatic pollution assay**
Groundwater report = CPCB	**CGWB**
Pesticide = biodegradable	**Not**
Photochemical smog without O_3	**Always contains ozone**
PAN = primary stack gas	**Secondary**; eye irritant
AQI includes CO_2/CH_4	**Eight** pollutants — **not** those GHGs
Most important indoor = SO_2	Classic answer often **radon**
Knock-knee = mercury	**Fluoride**; Minamata = **Hg**
Eutrophication = air pollution	**Water** nutrient overload
Cyclone divider = control device	**Not** a standard name
CCAC = unique G20 initiative	Voluntary partnership; focus **CH_4 / BC / HFCs**
Bioremediation cleans all Cd/Pb	**Not** readily/completely
Noisiest UP city distractors	**Moradabad** (Frontier 2022)
Arsenic only = India exclusive	Classic **Bangladesh** fact
NCAP = SAFAR	Programme vs forecast system
Industrial noise day = 55 dB	**75 dB** (CPCB day)
Green muffler = soil fix	**Noise** (tree belts)
Bhopal gas = CO/SO_2	**MIC**, Dec **1984**

Topic 39 — Acid Rain

Current Affairs (this topic)

- Twin of Acid Rain core — chemistry fixed; rare CA only.
- Acid rain ≠ photochemical smog ≠ ozone depletion (ओजोन क्षरण).

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Normal Rain	**Acid Rain**	Slightly acidic with pH ~5.6 due to dissolved atmospheric CO_2 forming weak carbonic acid (H_2CO_3) vs precipitation with pH < 5.6 caused by anthropogenic SO_2 and NOx	सामान्य वर्षा (pH 5.6) / अम्ल वर्षा (pH < 5.6)
Sulfuric Acid (H_2SO_4)	**Nitric Acid (HNO_3)**	Contributes ~60–70% of total acid precipitation (emitted mainly by coal thermal power plants) vs contributes ~30–40% (emitted by automobile engines and fertilizers)	सल्फ्यूरिक अम्ल (60-70%) / नाइट्रिक अम्ल (30-40%)
Wet Acid Deposition	**Dry Acid Deposition**	Acidic compounds deposited on Earth via rain, fog, sleet, or snow vs acidic gases, sulfate, and nitrate particles settling directly on soil and buildings without precipitation	आर्द्र निक्षेपण (वर्षा/कोहरा) / शुष्क निक्षेपण (धूल/गैस कण)
Marble Cancer	**Eutrophication**	Yellowing and pitting of calcium carbonate monuments (Taj Mahal) reacting with sulfuric acid to form soluble gypsum vs excess nutrient loading in water bodies causing algal blooms	संगमरमर कैंसर (ताजमहल क्षरण) / सुपोषण (शैवाल प्रस्फुटन)
Gothenburg Protocol (1999)	**Montreal Protocol (1987)**	UNECE protocol to abate acidification, eutrophication, and ground-level ozone (SO_2, NOx, VOCs, NH_3) vs UNEP protocol phasing out ozone-depleting chlorofluorocarbons	गोथेनबर्ग प्रोटोकॉल (अम्लीकरण) / मॉन्ट्रियल प्रोटोकॉल (ओजोन)

39.3 Effects

Receptor	Effect
Lakes/rivers	Fish kills below ~pH 5
Soil	Nutrient leach + Al^{3+}
Forests	Conifer damage
Marble	Gypsum crust (Taj)
Lichens	Die in high SO_2

39.4 Prevention

Tool	Cuts
FGD	SO_2
Catalytic converter / BS-VI	NOx + fuel S
Fuel switch	Both
Taj Trapezium	Local heritage protection
Gothenburg/CLRTAP	Transboundary acidification

Common Traps

Trap	Correct
Normal rain pH = 7	**~5.6**
Acidic pH > 7	**Acidic < 7**
CO_2 = main severe acid rain gas	**Weak only**; SO_2/NOx drive strong acids
Montreal = acid rain	**Ozone**; Gothenburg/CLRTAP for acidifying
Acid rain = only liquid rain	Includes **dry deposition**
SO_2 lighter than air	**Heavier**
Taj damage from CO_2 alone	**SO_2 + acid → gypsum**
Lichens love polluted SO_2 air	**Die / indicate pollution**
FGD removes only NOx	Mainly **SO_2**
Neutral rain = natural rain	Natural rain already **slightly acidic**

Topic 40 — Desertification

Current Affairs (this topic)

- Twin of Desertification & Land Degradation — UNCCD / LDN CA refresh.
- Keep desertification ≠ drought-only wording clear.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Desertification	**Land Degradation**	Land degradation specifically occurring in arid, semi-arid, and dry sub-humid areas (drylands) from climatic variations and human activities vs any reduction in productive potential of land everywhere	मरुस्थलीकरण (शुष्क क्षेत्र) / भूमि क्षरण (सामान्य भूमि)
Drought	**Desertification**	Temporary natural period of deficient precipitation over months/years vs persistent, long-term degradation and loss of biological/economic productivity in dryland soils	सूखा (अस्थायी जल संकट) / मरुस्थलीकरण (स्थायी भूमि ह्रास)
Sheet Erosion	**Gully Erosion**	Uniform, imperceptible removal of thin topsoil layer by surface runoff on gentle slopes vs formation of deep pronounced ravines and gorges rendering land unfit for cultivation (Chambal Badlands)	परत अपरदन (अदृश्य मिट्टी ह्रास) / अवनलिका अपरदन (बीहड़ निर्माण)
UNCCD (1994)	**UNFCCC (1992)**	Only legally binding international agreement linking environment and development to sustainable land management in drylands vs framework treaty for combating greenhouse gases	मरुस्थलीकरण संधि (UNCCD) / जलवायु संधि (UNFCCC)
Land Degradation Neutrality (LDN)	**Net Zero Emissions**	Target under SDG 15.3 ensuring no net loss of healthy land resources by 2030 vs balancing anthropogenic GHG emissions with equivalent carbon removals	भूमि क्षरण तटस्थता (LDN) / नेट जीरो उत्सर्जन

40.2 Land Degradation

Process	Meaning
Water/wind erosion	Topsoil loss
Salinization	Excess irrigation salts
Waterlogging	Poor drainage
Mining / pollution	Direct land damage

Common Traps

Trap	Correct
Desertification = only desert expansion	**ASDM productivity loss**
UNCCD = climate treaty	**Desertification**; UNFCCC = climate
UNCCD signed at Rio 1992	**Adopted 1994**
Afforestation causes degradation	**Prevents** it
Gully = sheet	Different stages
Drought = desertification	Related, **not identical**

Trap	Correct
LDN year 2025/2040	**2030**
Chambal only in one state	**MP–UP–Rajasthan**
DPAP = 1994	**1973–74**
Normal erosion = focus	**Accelerated** erosion is

Topic 41 — Environmental Monitoring

Current Affairs (this topic)

- CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड) / SPCB monitoring network and satellite products are living CA.
- Keep ambient vs emission monitoring distinction clear.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Environmental Indicator	**Environmental Monitoring**	Specific quantitative metric reflecting state of an ecosystem (e.g., Lichens for SO_2, E. coli for fecal water contamination) vs continuous systematic collection and analysis of environmental data over time	पर्यावरणीय संकेतक / पर्यावरणीय निगरानी
Environmental Impact Assessment (EIA)	**Environmental Audit**	Prospective preventive study predicting ecological impacts of a project *before* environmental clearance vs retrospective verification of legal compliance and emissions *after* commissioning	पर्यावरण प्रभाव आकलन (EIA) / पर्यावरण लेखापरीक्षा
National Air Monitoring Programme (NAMP)	**Continuous Ambient Air Quality (CAAQMS)**	Nationwide manual sampling network monitoring SO_2, NO_2, PM10, PM2.5 twice weekly vs automated sensors measuring 12 criteria pollutants with real-time online data	राष्ट्रीय वायु निगरानी (NAMP) / सतत परिवेशी वायु निगरानी (CAAQMS)
Bioindicator	**Biomarker**	Organism whose presence/absence indicates environmental quality (Lichens for air, Diatoms/Mayfly for fresh water) vs cellular/biochemical alteration measured in an organism due to chemical exposure	जैव संकेतक (लाइकेन/ मेफ्लाई) / जैव मार्कर
Ringelmann Scale	**Decibel (dB)**	Visual chart scale (0 to 5) used to measure the apparent density/opacity of industrial smoke emissions vs logarithmic acoustic unit measuring environmental sound levels	रिंगेलमैन पैमाना (धुआं घनत्व) / डेसिबल (ध्वनि तीव्रता)

41.3 Environmental Audit

Tool	Role
Form V statement	Annual statutory disclosure to SPCB
ISO 14001	Voluntary EMS certification
EC compliance report	Project clearance conditions
SPCB inspection	Enforcement audit on ground

Common Traps

Trap	Correct
BOD = blood/altitude oxygen	**Aquatic pollution assay**
Ringelmann = fog/noise/water	**Smoke**
GW Quality Report = CPCB	**CGWB**
Monitoring = audit	Measure ≠ verify compliance
Lichens thrive in dirty air	**Sensitive indicators**

Trap	Correct
SAFAR = IMD alone / CPCB alone	**IITM** (+ MoEFCC support)
Form V = ISO certificate	Statutory **statement** to SPCB
COD < BOD always	**COD ≥ BOD**
NAMP = real-time only	Mostly **manual**; CAAQMS = continuous
High BOD = clean water	**Dirty** organics

Topic 42 — International Environmental Organizations

Current Affairs (this topic)

- UNEP / IUCN / IPCC / GEF role headlines refresh institutional CA.
- HQ and mandate traps stay fixed teaching.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
UNEP	**UNDP**	UN environmental coordinator established post-1972 Stockholm Conference (HQ: Nairobi, Kenya) vs UN development network producing the Human Development Index (HQ: New York)	यूएनईपी (नैरोबी, पर्यावरण) / यूएनडीपी (न्यूयॉर्क, विकास)
IPCC	**UNFCCC**	Scientific review panel established in 1988 by WMO/UNEP to assess peer-reviewed climate data (HQ: Geneva) vs legally binding treaty secretariat coordinating annual COPs (HQ: Bonn, Germany)	आईपीसीसी (वैज्ञानिक समीक्षा) / यूएनएफसीसीसी (वार्ता सचिवालय)
IUCN	**WWF**	Oldest global conservation union (1948, Gland) publishing the Red List of Threatened Species vs private international conservation NGO (1961, Gland) famous for Giant Panda logo and Earth Hour	आईयूसीएन (रेड लिस्ट) / डब्ल्यूडब्ल्यूएफ (पांडा प्रतीक)
CITES	**CMS (Bonn Convention)**	Regulates commercial cross-border trade in endangered species of wild flora and fauna (1973) vs promotes conservation and habitat protection for migratory wild animal species (1979)	साइट्स (वन्यजीव व्यापार) / बॉन कन्वेंशन (प्रवासी प्रजातियां)
Global Environment Facility (GEF)	**Green Climate Fund (GCF)**	Multilateral financial mechanism (World Bank 1991) serving CBD, UNCCD, Stockholm, Minamata, and UNFCCC vs dedicated operating financial entity of UNFCCC created at COP16 Cancun	जीईएफ (वैश्विक पर्यावरण कोष) / जीसीएफ (हरित जलवायु कोष)

42.8 WMO, Greenpeace & US EPA

Body	HQ / year	Job
WMO	Geneva	Weather/climate cooperation; IPCC parent
Greenpeace	Amsterdam	Environmental NGO
US EPA	Est. 1970	US federal environment agency
UNEP	Nairobi	UN environment programme

Org ↔ Report ↔ HQ Card

Org	HQ	Flagship fact
UNEP	Nairobi	Emissions Gap / GEO
UNDP	New York	HDR / HDI
FAO	Rome	SOFO; Rotterdam (+UNEP)
UNESCO	Paris	World Heritage; MAB
IUCN	Gland	Red List
WWF	Gland	Living Planet

Org	HQ	Flagship fact
IPCC	Geneva	Assessment Reports
WMO	Geneva	Meteorology; IPCC co-parent
Greenpeace	Amsterdam	Environmental NGO

Common Traps

Trap	Correct
Emissions Gap = IPCC/UNDP	**UNEP**
HDR = UNEP	**UNDP**
UNEP HQ Geneva/NY	**Nairobi**
IPCC sets NDCs	**UNFCCC/COP**
IUCN regulates trade	**CITES**
WWF = IUCN	Same city, **different roles**
FAO HQ Nairobi	**Rome**
GEF donors = developed only	**False**
HDI team = Priesner	**Haq–Sen** lineage
Limits to Growth = UNESCO	**Club of Rome**

Topic 43 — International Environmental Agreements

Current Affairs (this topic)

- Twin of Agreements & Conferences — COP / listing CA; treaty ladder identical.
- Ramsar India **~100** / UP **12** baseline unless a stem cites a later official total.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Kyoto Protocol (1997)	**Montreal Protocol (1987)**	Legally binding treaty targeting reduction of 6 greenhouse gases causing climate change vs universally ratified protocol phasing out ozone-depleting chlorofluorocarbons	क्योटो प्रोटोकॉल (जलवायु परिवर्तन) / मॉन्ट्रियल (ओजोन क्षरण)
Kyoto Protocol (1997)	**Paris Agreement (2015)**	Top-down legally binding reduction targets solely for developed Annex-I countries vs bottom-up voluntary Nationally Determined Contributions (NDCs) applicable to all countries	क्योटो (शीर्ष से नीचे बाध्यकारी) / पेरिस (स्वैच्छिक NDCs)
Basel Convention (1989)	**Rotterdam Convention (1998)**	Regulates transboundary movements and disposal of hazardous wastes vs enforces Prior Informed Consent (PIC) procedure for hazardous international trade chemicals	बासेल (खतरनाक अपशिष्ट आवागमन) / रॉटरडैम (पूर्व सूचित सहमति)
Stockholm Convention (2001)	**Minamata Convention (2013)**	Mandates global elimination and phase-out of Persistent Organic Pollutants (POPs - "Dirty Dozen") vs protects human health and environment from anthropogenic emissions of toxic mercury (Hg)	स्टॉकहोम (पीओपी कीटनाशक) / मिनामाता (पारा नियंत्रण)
Ramsar Convention (1971)	**CITES (1973)**	Intergovernmental treaty dedicated specifically to the conservation and wise use of wetlands of international importance vs treaty controlling illegal commercial wildlife trade	रामसर कन्वेंशन (आर्द्रभूमि) / साइट्स (वन्यजीव व्यापार)
Stockholm Conference (1972)	**Rio Earth Summit (1992)**	UN Conference on the Human Environment that led to creation of UNEP vs UN Conference on Environment & Development giving birth to Agenda 21, CBD, UNFCCC, and UNCCD	स्टॉकहोम सम्मेलन 1972 / रियो पृथ्वी शिखर सम्मेलन 1992

Treaty ↔ Issue Matrix

Treaty	Year	Issue
Ramsar	1971	Wetlands
CITES	1973	Wildlife trade
CMS (Bonn)	1979	Migratory species
Basel	1989	Hazardous waste
UNFCCC	1992	Climate framework
CBD	1992	Biodiversity
Kyoto	1997	Climate (Annex-I)
Rotterdam	1998	PIC chemicals
Stockholm	2001	POPs
Minamata	2013	Mercury
Paris	2015	Climate (NDCs)

Common Traps

Trap	Correct
Montreal = climate mitigation	**Ozone**; Kyoto/Paris = climate
Kyoto = save water	**Climate / GHG**
Stockholm Convention = 1972 conference	**POPs 2001** vs UNEP conference 1972
CITES = IUCN Red List	Trade rules vs risk assessment
CMS = CITES	Migration vs trade
Ramsar = forests/ozone	**Wetlands**
CITES born at Rio 1992	**1973 Washington**
Paris = Kyoto Annex-only model	**Universal NDCs**
Basel = wildlife trade	**Hazardous waste**
Minamata = all hazardous waste	**Mercury**

Topic 44 — Current Environmental Issues

Current Affairs (this topic)

- Primary living CA chapter — NbS, plastics, heat, Himalaya (हिमालय) cryosphere, LiFE.
- Cross-link facts to Climate / Biodiversity / Pollution without relocating treaty years.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Carbon Footprint	**Ecological Footprint**	Total amount of greenhouse gases (in tonnes of CO_2e) emitted directly/indirectly by an entity vs total biocapacity area (in global hectares, gha) required to sustain a lifestyle	कार्बन पदचिह्न (CO_2e उत्सर्जन) / पारिस्थितिक पदचिह्न (भूमि/संसाधन मांग)
Net Zero Emissions	**Gross Zero Emissions**	Balancing residual carbon emissions with equivalent carbon removals via sinks and DACCS vs completely eliminating all greenhouse gas emissions across all sectors without offsets	नेट जीरो (उत्सर्जन-अवशोषण संतुलन) / सकल शून्य (पूर्ण उत्सर्जन शून्यता)
Green Economy	**Circular Economy**	Broad macro-framework promoting low-carbon, resource-efficient, and socially inclusive growth vs industrial economic model eliminating waste through closed-loop recycling and repair	हरित अर्थव्यवस्था / चक्रीय अर्थव्यवस्था
Green Hydrogen	**Grey Hydrogen / Blue Hydrogen**	Produced by electrolysis of water powered entirely by renewable electricity vs produced from natural gas without carbon capture (Grey) or with carbon capture & storage (Blue)	हरित हाइड्रोजन (नवीकरणीय) / धूसर व नीला हाइड्रोजन
Primary Microplastics	**Secondary Microplastics**	Tiny plastic beads (<5mm) intentionally manufactured for cosmetics and industrial pellets vs microscopic plastic fragments resulting from physical weathering and breakdown of larger plastic debris	प्राथमिक माइक्रोप्लास्टिक (निर्मित मनके) / द्वितीयक माइक्रोप्लास्टिक (अपघटित टुकड़े)
Nature-based Solutions (NbS)	**Geoengineering**	Ecosystem protection, restoration, and sustainable management to address societal challenges (e.g. mangrove restoration) vs large-scale technological manipulation of Earth's systems (e.g. solar radiation management)	प्रकृति-आधारित समाधान (NbS) / भू-इंजीनियरिंग
Mission LiFE Genesis	**Mission LiFE Official Launch**	Conceptualized and introduced by PM Modi at COP26 Glasgow in **November 2021** vs officially launched jointly with UN Secretary-General António Guterres at Ekta Nagar (Kevadia), Gujarat in **October 2022**	मिशन लाइफ विचार (ग्लासगो 2021) / औपचारिक शुभारंभ (केवड़िया 2022)

44.1 Carbon Footprint

Scope	Type	Example
Scope 1	Direct	Factory (फैक्टरी) boiler
Scope 2	Purchased energy	Grid electricity
Scope 3	Other indirect	Supplier emissions, flights

44.2 Ecological Footprint

Metric	Unit	Rough India
Ecological footprint	gha	~1.0–1.2 / person

Metric	Unit	Rough India
Carbon footprint	tCO_2e	~2 / person

44.3 Net Zero Emissions

Country/bloc	Net zero year
EU / UK	2050
China	2060
India	**2070**

44.6 Circular Economy

Linear	Circular
Virgin in → dump out	Recycled/regenerative loops
Waste = disposal problem	Waste = design flaw

44.7 Green Hydrogen

Colour	Method
Green	Electrolysis + renewables
Grey	Natural gas reforming
Blue	Grey + CCS
Black/brown	Coal gasification

44.8 Microplastics

Type	Origin
Primary	Made small
Secondary	Fragmented from large items

44.9 Climate Refugees

Type	Example
Sudden-onset	Cyclone / flash flood displacement
Slow-onset	Drought / sea-level rise
Legal refugee (1951)	Persecution — not climate alone

Common Traps

Trap	Correct
LiFE idea at COP-25 Madrid	**COP26 Glasgow 2021**; launch **June 2022**
MISHTI initiated 2019	**Budget 2023–24**

Trap	Correct
India net zero 2050	**2070**
Net zero = emissions to absolute zero	Balance with **removals**
Emissions Gap = IPCC	**UNEP**
Ecological footprint in tCO_2e	Unit = **gha**
Microplastics <5 cm	**<5 mm**
Green H_2 = grey H_2	Grey = natural gas reforming
Green economy = circular identically	Circular = **material loops**; green = broader
Climate refugee = 1951 legal status	**Not** a formal Convention category
NbS = geoengineering	Ecosystem protect/manage/restore
MISHTI for Ganga basin inland	**Coastal mangroves**
Highest India per capita carbon footprint	High **total**, low **per capita**

Uttarakhand — 01 — Natural Resources & Climate Contribution

Current Affairs (this topic)

- Keep FSI / FRI / WII function traps and Chipko **1973 Reni** stable.
- Himalayan Mission and spring / landslide seasons refresh climate–resource CA.

Confused Pairs

Term	One-line difference
FSI	Forest cover / ISFR — Dehradun
FRI	Forestry research — Dehradun
WII	Wildlife research + training — Dehradun
Chipko	1973 Reni — forest protection movement
Weathering	In-situ rock breakdown
Erosion	Transport of weathered material
NEERI	Nagpur 1958 — CSIR env. engineering
Natural capital	Forests, water, minerals — not roads

1.4 Dehradun Institution Triangle

Institute	Job
FSI	Forest cover / ISFR
FRI	Forestry research
WII	Wildlife science + training

Uttarakhand — 02 — Biodiversity & Protected Areas

Current Affairs (this topic)

- Keep NP notification years and sanctuary–district pairs stable.
- Nanda Devi / Valley of Flowers (फूलों की घाटी) WHS + BR fauna (प्राणीजात) set recycles every year.

Quick Revision — Spine Only

Head	Key
Biogeographic zone	**Himalaya** (हिमालय) — high endemism; altitudinal belts
Forest richness A/R	A true; R (topography + climate) **explains** A
Nanda Devi BR	UNESCO WHS (यूनेस्को) (with Valley of Flowers); snow leopard, musk deer, blue sheep
Askot	**Pithoragarh** (पिथौरागढ़)
Binsar	**Almora** (अल्मोड़ा)
Govind	**Uttarkashi** (उत्तरकाशी)
Nandhaur	**Nainital** (नैनीताल)
Corbett NP	**1936**
Rajaji (राजाजी) NP	**1983**
Gangotri (गंगोत्री) NP	**1989**
Govind NP	**1990** (not 1989)

Confused Pairs

Pair	Key
Askot vs Binsar	Pithoragarh vs Almora
Govind 1989 vs 1990	**1990** keyed; Gangotri keeps **1989**
Nanda Devi fauna set	All three named species in
A true / R not explain	UKPCS keys **R explains A**
NP vs WLS vs BR	Strict NP / regulated WLS / BR zonation — different legal and management ideas

2.3 Sanctuary–district ratta

Sanctuary	District
Askot	Pithoragarh
Binsar	Almora
Govind	Uttarkashi
Nandhaur	Nainital

Common Traps

1. **Govind NP = 1989 → 1990**.
2. **Askot = Almora** → Pithoragarh.
3. **R does not explain forest richness** → keyed that it does.
4. **BR = only another NP name** → BR uses core–buffer–transition zonation.

Uttarakhand — 03 — Climate Vulnerability & Governance

Current Affairs (this topic)

- Keep SCCC **2011** and UAPCC department trap stable.
- Cloudburst / GLOF seasons refresh vulnerability without changing the institutional keys.

Quick Revision — Spine Only

Head	Key
National climate role	Carbon sink (कार्बन सिंक) · water tower · Himalayan Mission (NAPCC)
Vulnerability	Highly vulnerable mountain state
Documented effects	Water stress · biodiversity / habitat (वास स्थान) pressure
SCCC	State Council for Climate Change (जलवायु परिवर्तन) — **2011**
UAPCC prep	**Not** State Revenue Department — forest/environment (पर्यावरण) set-up
Hazards link	Landslides, cloudbursts, floods, glacial risk
National Panchamrit	COP26 Glasgow climate package — national climate chapters

Confused Pairs

Pair	Key
UAPCC by Revenue Dept	**False** — forest / environment set-up
SCCC 2011 vs 2009/2013	**2011** keyed
Only vulnerability true	2025 keys **both** vulnerability and water-stress/biodiversity statements
UK separate UNFCCC track	No — contributes via India's NDCs / ecosystem services

Common Traps

1. **UAPCC = Revenue Department** → false.
2. **Rejecting biodiversity impact of climate change** → keyed true with vulnerability.
3. **SCCC from statehood 2000 → 2011**.